



Chapter 11

Biological analysis

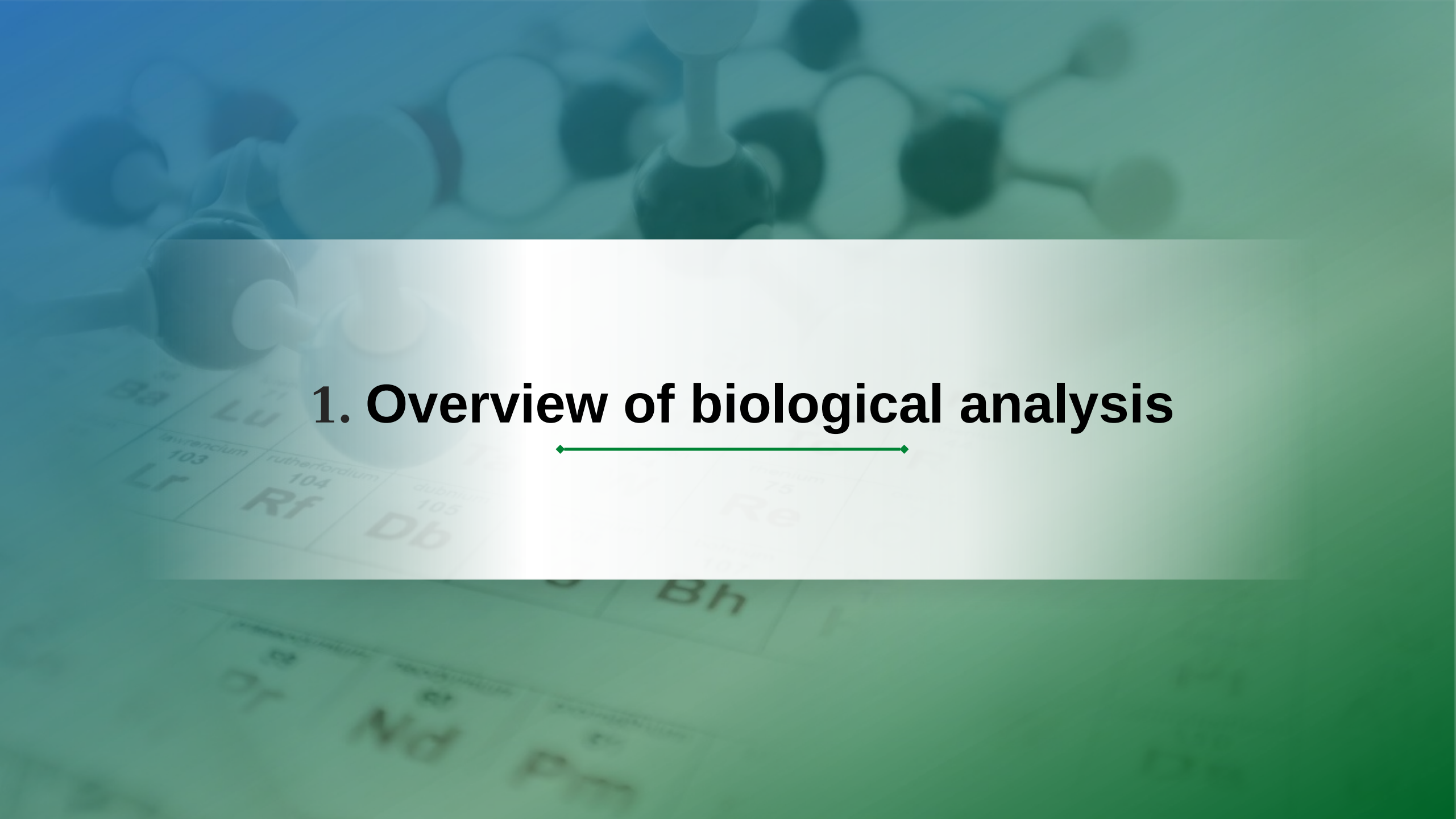


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2. Overview of biological analysis

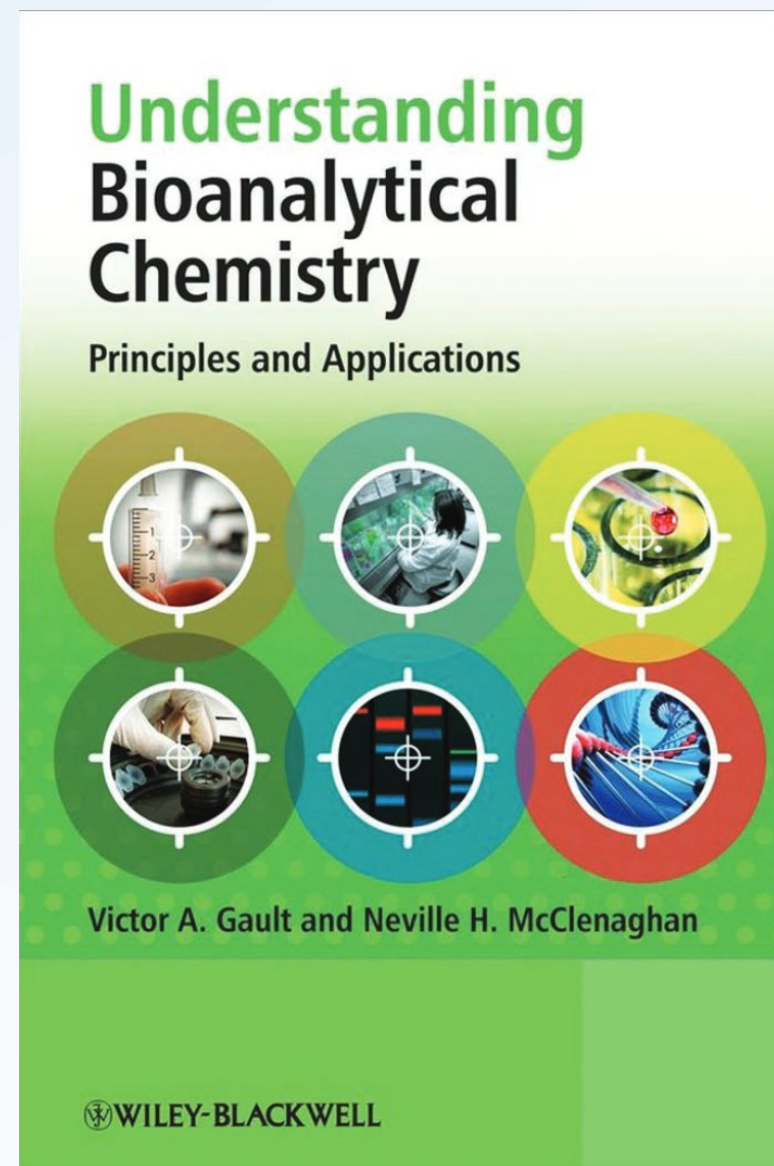
- Single-cell
- Structure biology
- Nanopore
- Bioinformatics
-



1. Overview of biological analysis

Bioanalytical chemistry

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Brief Introduction



- Overview of chemical and physical attributes of biomolecules
 - Atoms and elements
 - Bonding
 - Phases of matter
 - Physical and chemical properties
 - Studying physical and chemical properties of biomolecules

Charge, Density, Mass, Shape...

Bonding, solubility, structure...

- Classification of biomolecules

Organic / inorganic molecules

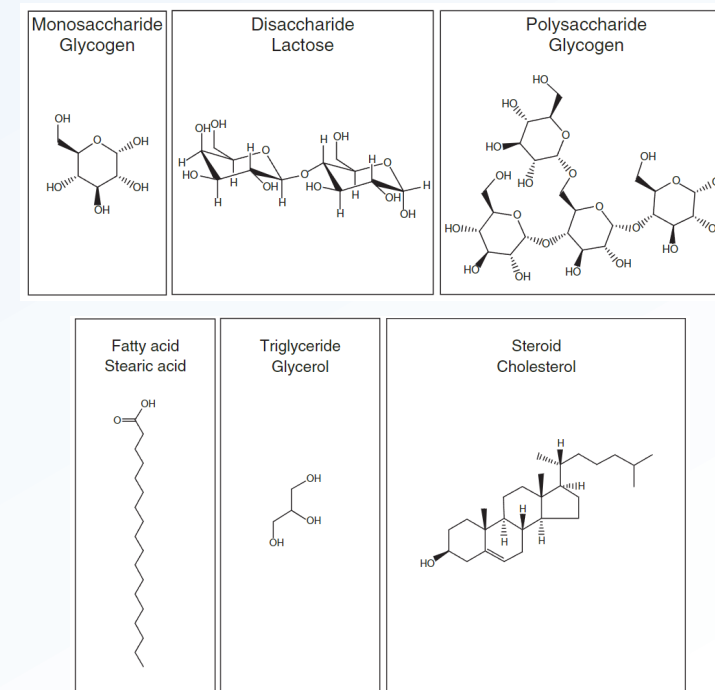
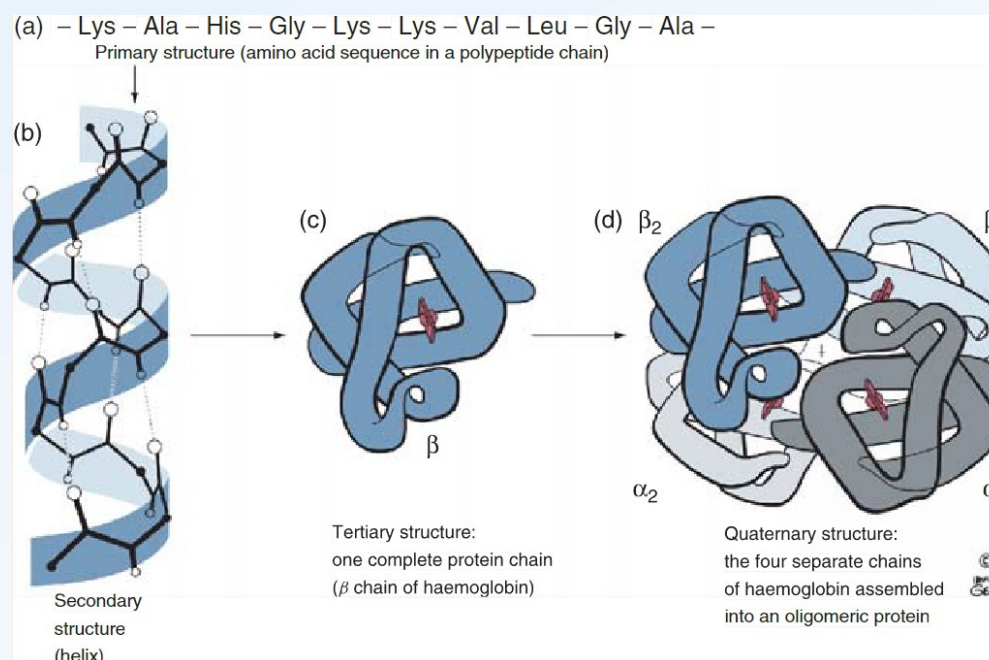
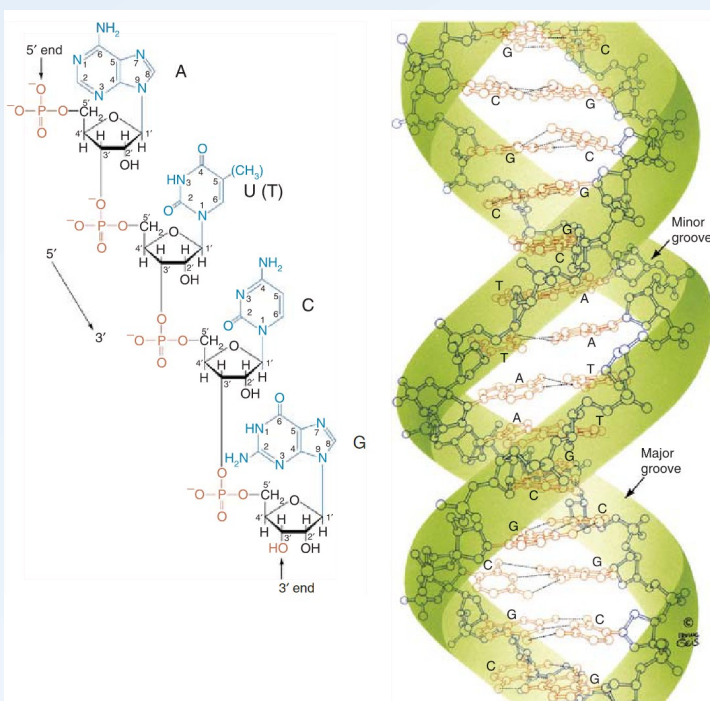
Small molecules: The term *small molecule* refers to a diverse range of substances including: lipids and derivatives; vitamins; hormones and neurotransmitters; and carbohydrates.

Monomers: The term *monomer* refers to compounds which act as building blocks to construct larger molecules called *polymers* and includes: amino acids; nucleotides; and monosaccharides.

Polymers: Constructed of repeating linked structural units or monomers, polymers (derived from the Greek words *polys* meaning many and *meros* meaning parts) include: peptides/oligopeptides/polypeptides/proteins; nucleic acids; and oligosaccharides/polysaccharides.

• Features and characteristics of major biomolecules

- Building biomolecules (elements)
- Constructing complex biomolecules (building block of different biomolecules)



- Structure–function relationships

The structure of a given biomolecule will confer certain functional attributes and, as such, is a key defining feature of that biomolecule. There are very many environmental influences that can impact on structure and/or function and, importantly, the stability of the biomolecule. Furthermore, this information is also important when considering bioanalysis, as the outcome of an analytical process or procedure, and indeed the stability of a biomolecule, is dependent on numerous physical and chemical factors including pH, temperature and solvent concentration/polarity.

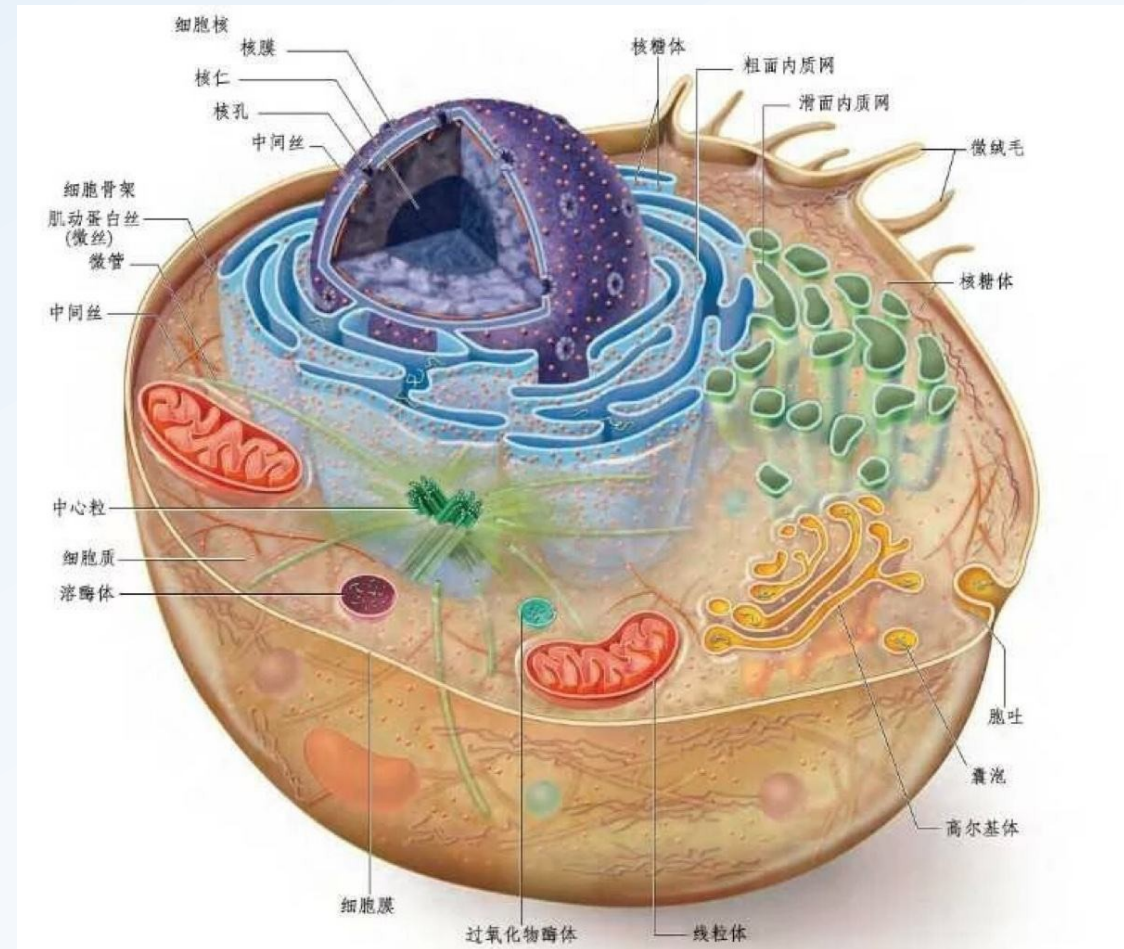
QSAR

It is also important to map the chemical structure of biomolecules with certain well-defined processes, which may relate to biological activity or chemical reactivity. Medicinal chemists place particular importance on understanding *structure–activity relationships* (SARs) of biomolecules in order to bioengineer modified biomolecules with enhanced activity (*potency*), for example by changing amino acid composition of peptides or insertion/addition of chemical groups. This approach has given new and exciting insights into chemical groups that affect biological processes, and allowed complex mathematical modelling of *quantitative structure–function relationships* (QSARs). This has inherent difficulties, as certain features such as *post-translational modification* of proteins may depend on multiple factors, and thus not all related biomolecules have similar activities (so-called SAR paradox). Historically, one of the first simple examples of QSAR was to predict boiling points on the basis of the number of carbon atoms in organic compounds; more modern applications of QSAR are in drug design and discovery, discussed in more detail in Chapter 12.

- Significance of biomolecules in nature and science

- Nucleic acid
- Protein/peptides
- Lipid
- Carbon hydrates

Biomarkers





Analysis and quantification of biomolecules

- Importance of accurate determination of biomolecules

Without the ability to accurately determine the presence and amount of any given biomolecule in nature, we would not be able to understand the contribution and function of these important chemical entities. Since the first basic tests available to distinguish chemical elements, numerous methods have been developed to measure the amounts and types of large and small biomolecules. Individual biomolecules are principally distinguished on the basis of their chemical and physical properties, for example molecular mass. In fact, molecular mass is analogous to a 'molecular fingerprint', enabling scientists to specifically distinguish one biomolecule from another using, for example, mass spectrometry techniques that will be considered later.

Clinical, environment, drugs....

- Major methods to detect and quantify biomolecules

Table 2.1 Basic overview of key principles underlying major bioanalytical methods

Method/instrument	Physical and chemical attributes
Ion-selective electrode	pH and/or ionic charge
Centrifugation	Shape, volume and/or density
Spectroscopy	Energy, molecular structure
NMR	Charge and magnetic properties (e.g. spin/resonance)
Mass spectrometry	Mass
Chromatography	Mass, size (volume/density), shape, charge
Electrophoresis	Mass, charge, shape

- Understanding mass, weight, volume and density

Mass and weight

Mass is the quantity of matter in a given object and is independent of gravity.

Weight is the force of gravity on a given object and is calculated as mass multiplied by gravity.

$$\text{Weight} = \text{mass} \times \text{acceleration due to gravity}$$

While an object will always have a given mass, its weight will vary depending on gravity. So an object with the same mass will have a different weight on Earth compared with the Moon or a weightless environment (such as in space).



Volume and density

Volume is the space that mass in a given object occupies.

Density is a comparison of an object's mass to its volume.

$$\text{Density} = \text{mass/volume}$$

It is no more correct to assume that a small object will have a low mass, than it is to assume that an object with a large volume has a high mass. Density, however, gives an indication of 'how big' something actually is, as each substance has a unique density. The above formula dictates that objects of high density have a lot of mass occupying a relatively small volume.

Example: The densities of gold and lead are 0.0193 g l^{-1} and 0.0113 g l^{-1} , so if you have a 200 g bar of each, which will occupy the most volume?

Substituting these values into the above formula:

Gold: $0.0193 = 200/\text{volume}$

Therefore, cross-multiplying, $\text{volume} \times 0.0193 = 200$

So, $\text{volume} = 200/0.0193 = 10\,362.691$

Lead: $0.0113 = 200/\text{volume}$

Therefore, cross-multiplying, $\text{volume} \times 0.0113 = 200$

So, $\text{volume} = 200/0.0113 = 17\,699.111$

From the above you will see that lead occupies a considerably greater volume than gold.

- Understanding moles and molarity

The Mole

The **mole (mol)** is the SI unit of a quantity of a given molecule.

Origin of the term mole

The term *mole* is the English version of the German word for molecular weight ('mol', short for *molekulargewicht*, rather than referring to the burrowing mammal). Chemists who coined this term around 200 years ago were trying to quantify the chemical composition of gases, solids and liquids. For this they needed a universal unit of measurement so they could connect the mass of an atom (or molecule) with the number of atoms or molecules in any given compound. They decided to use Avogadro's number.

What is a mole?

A mole is a term that represents a number; just like the word 'dozen' represents '12', the word 'mole' represents 6.02×10^{23} (or 602 216 900 000 000 000

000 000). The term *mole*, while initially confusing, is extremely useful. To give an example: if we were to count the number of water molecules in a single drop it would be around 10 trillion (10 000 000 000 000), so rightly or wrongly chemists argue that it is easier to quantify such numbers in moles and thus cut down on the number of zeros!

Why 6.02×10^{23} ?

This number is 'Avogadro's number', named after the Italian physics professor who proposed that equal volumes of different gases at the same temperature contain the same number of molecules. In its simplest form, 1 mol of any substance contains 6.02×10^{23} atoms or molecules of that substance, and as this relationship was first put forward by Avogadro they named this number after him.

Concept of the mole

A defined mass of an element (its atomic weight) contains an exact number of atoms, that is, Avogadro's number. Therefore, for any given molecule, one mole of the substance has a mass (in grams) equal (numerically) to the atomic mass of the molecule. One mole is the number of atoms in exactly 12 thousandths of a kilogram (i.e. 12 g) of carbon (^{12}C), which is the most abundant isotope of carbon.

Example: A single molecule of silver has an atomic mass of 107.8682; therefore 1 mol of silver weighs 107.8682 g.

Molarity

The **molarity** of a solution is the number of moles of dissolved substance (solute) in a given volume of liquid (solution).

Importance of molarity

Molarity is a fundamental way of expressing the concentration of a compound in solution. A one molar solution (denoted 1 M) contains one mole of a given substance in a litre of a solution. In other words, 1 M is equivalent to 1 mole per litre (1 mol l^{-1}), and is a convenient means of representing the concentration of substances in all liquids and body fluids including blood plasma, urine and cerebrospinal fluid.

- Understanding solubility and dilutions

Solubility

The **solubility** is the amount of a particular substance (solute) that can dissolve in a given solvent (giving a saturated solution) at a particular temperature.

Given this, solubility is critically dependent upon two main factors, namely, type/properties of **solvent** (e.g. organic or inorganic) and **temperature** (e.g. 12 °C or 37 °C, sub-physiological and physiological temperature).

Solvent: Water is the most abundant solvent, and while a substance may be soluble in water it may not be soluble in other solvents (hence the popular terms water-soluble or fat-soluble).

Temperature: In general, increasing the temperature increases the solubility up to the *point of saturation*, which is when no more substance can be dissolved in that particular volume of solvent, and excess solute may be visible in suspension.

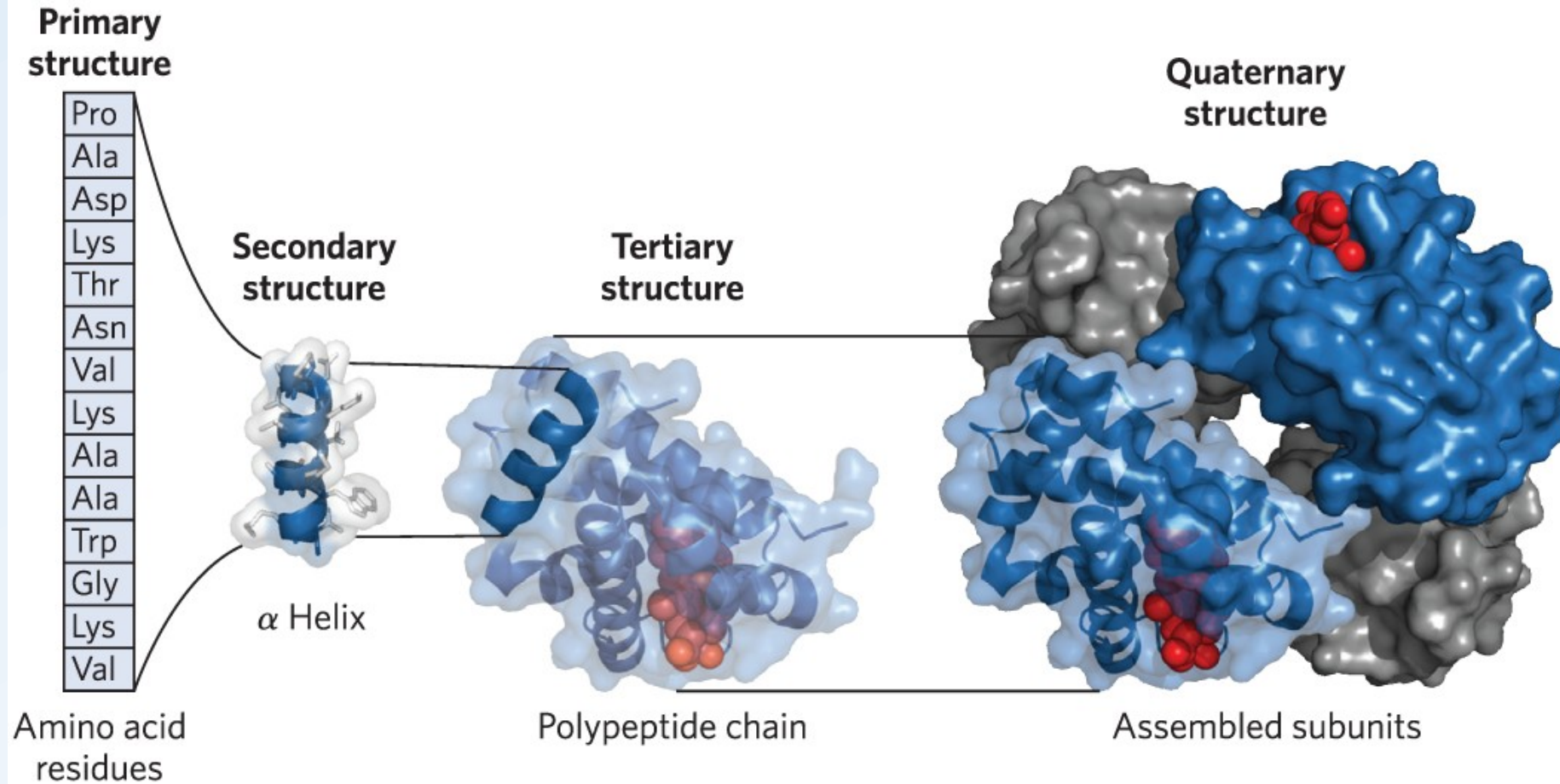
Solubility can be expressed in several ways, the most common being molarity (as moles of solute per litre) and % solubility, which is the number of grams of solute dissolved in 100 g (or 100 dm³) of solvent.

To confuse matters, the solubility of a substance is often defined on its ability to 'go into solution' at 25 °C (i.e. around room temperature). On the basis of this definition, whereas soluble substances can form a 0.10 to 1 M solution at 25 °C, insoluble substances cannot form a 0.10 to 1 M solution at 25 °C.

Dilution

Dilution is the process of making a weaker solution, which has a lower concentration of a given solute, and is fundamental when preparing solutions.

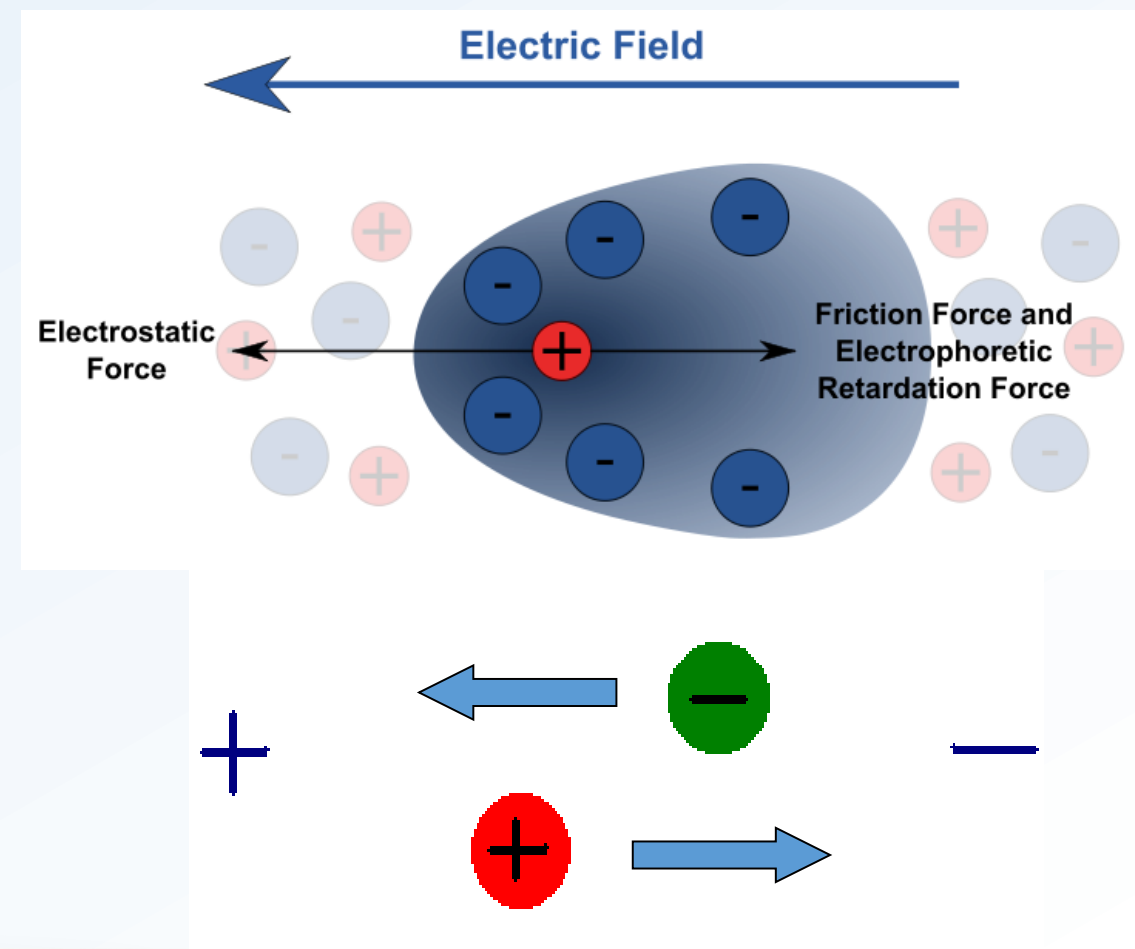
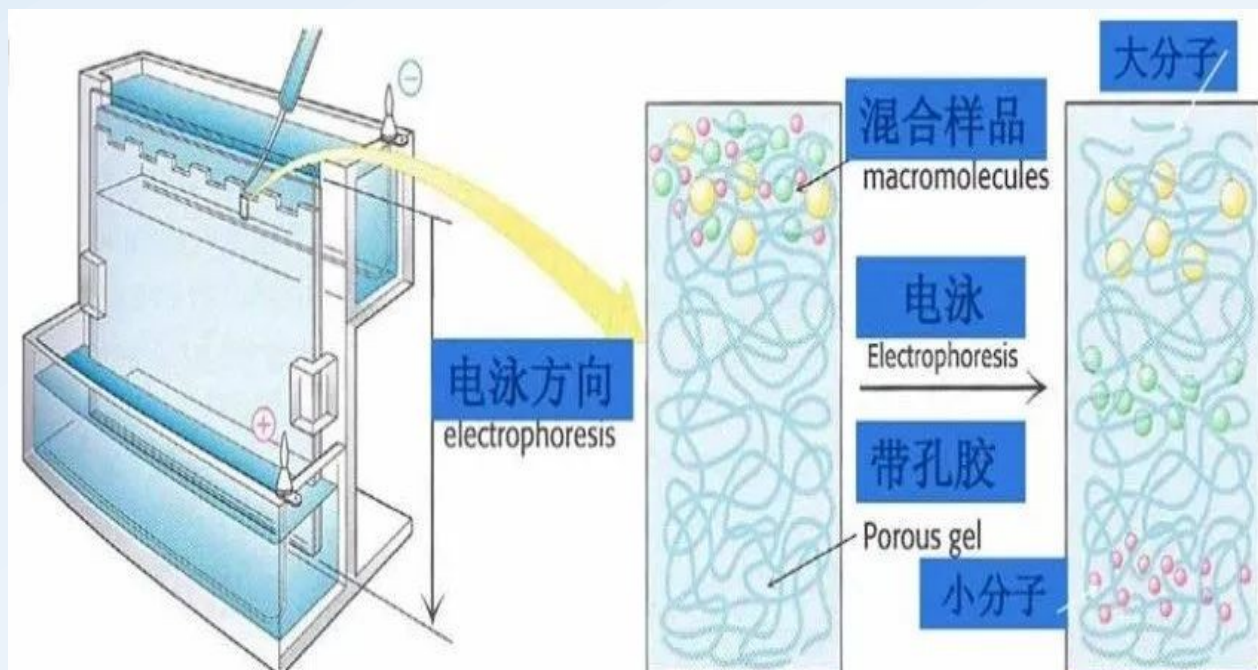
So-called **stock solutions** are solutions typically 10, 100 or 1000 times more concentrated than that ultimately required in the final solution. Stock solutions are particularly useful when the same ingredients are required in multiple test solutions, when various concentrations of these ingredients are required or simply for storage purposes. A useful example is fruit-flavoured drink concentrates (syrops or cordials) that are mixed/diluted with water to taste; if these drinks were pre-diluted they would typically fill hundreds of bottles. The same principle applies to the preparation and storage of chemical concentrates for preparation of laboratory reagents, solutions and buffers.

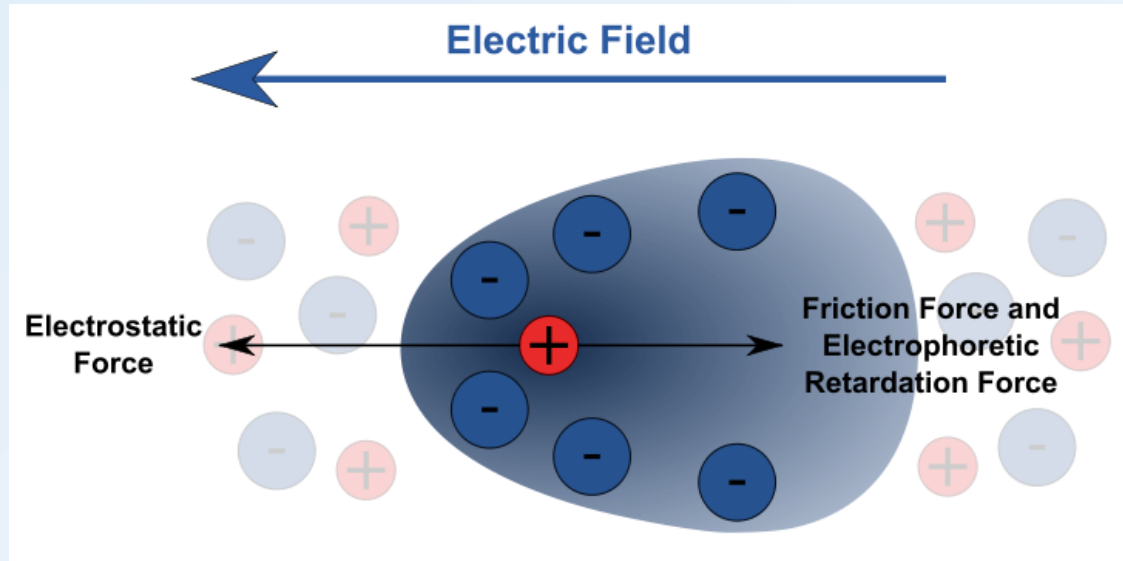


Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021
W. H. Freeman and Company

— SDS-page







$$F = EQ \quad f = 6\pi r\eta v$$

$$F = f$$

$$6\pi r\eta v = EQ$$

$$\longrightarrow \frac{v}{E} = \frac{Q}{6\pi r\eta}$$

F
Q
E
f

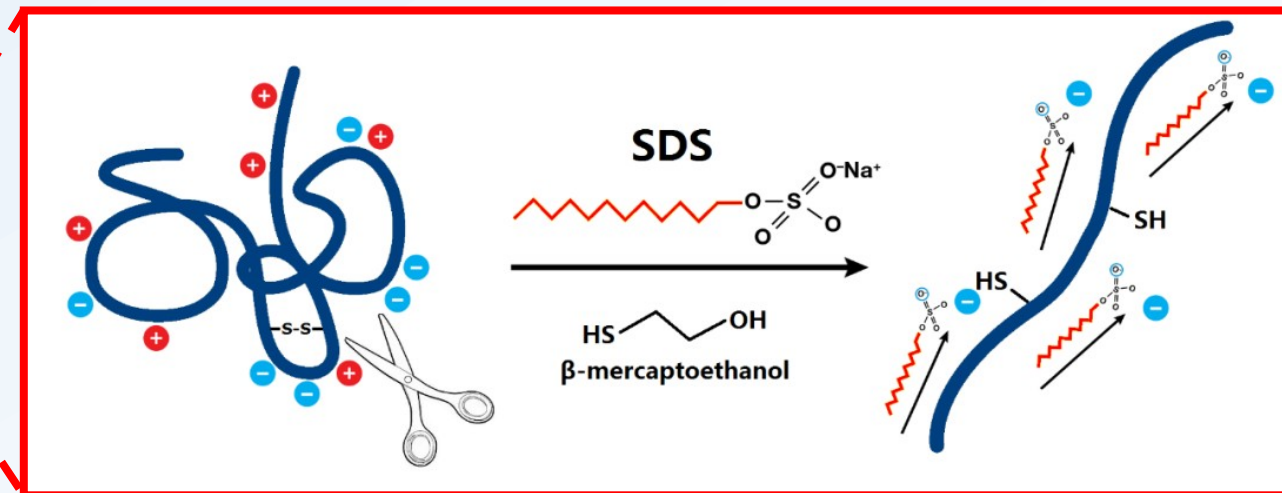
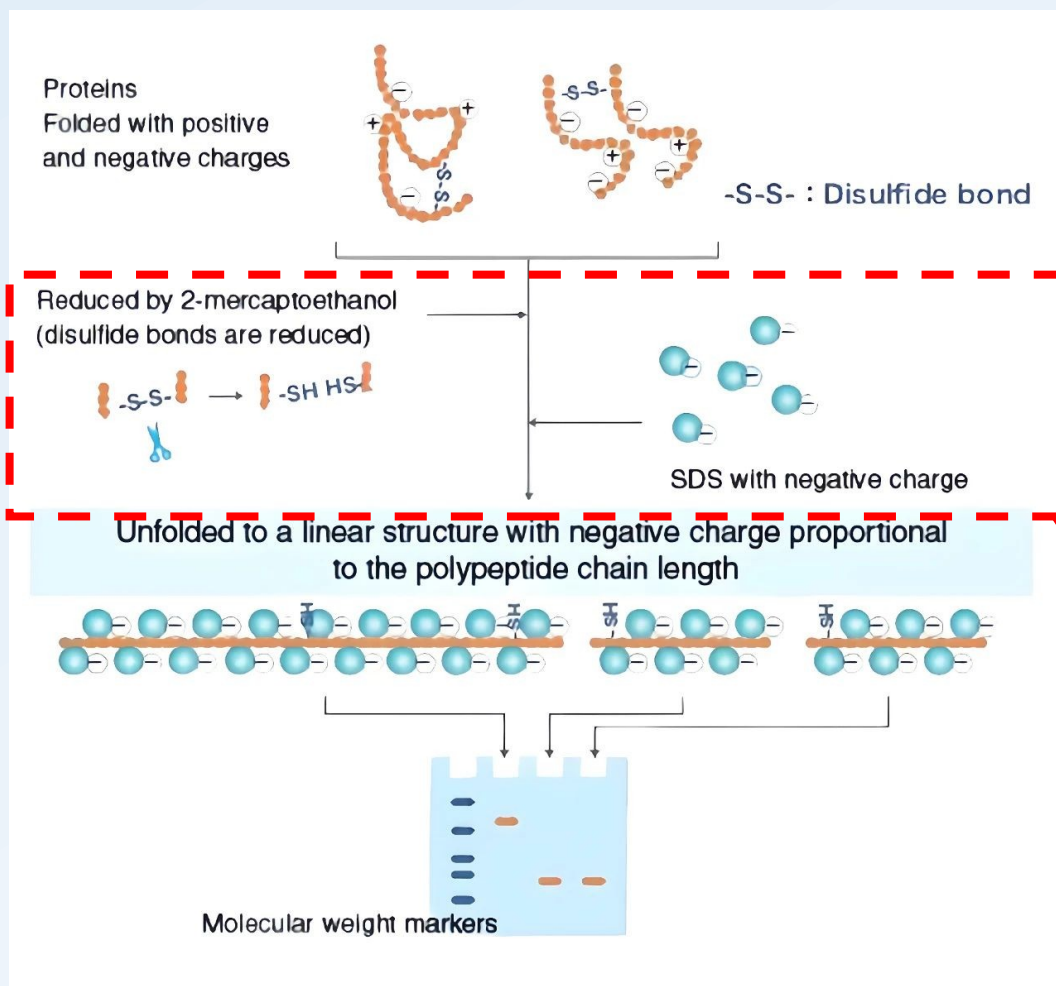
r
η
v

v/E

u

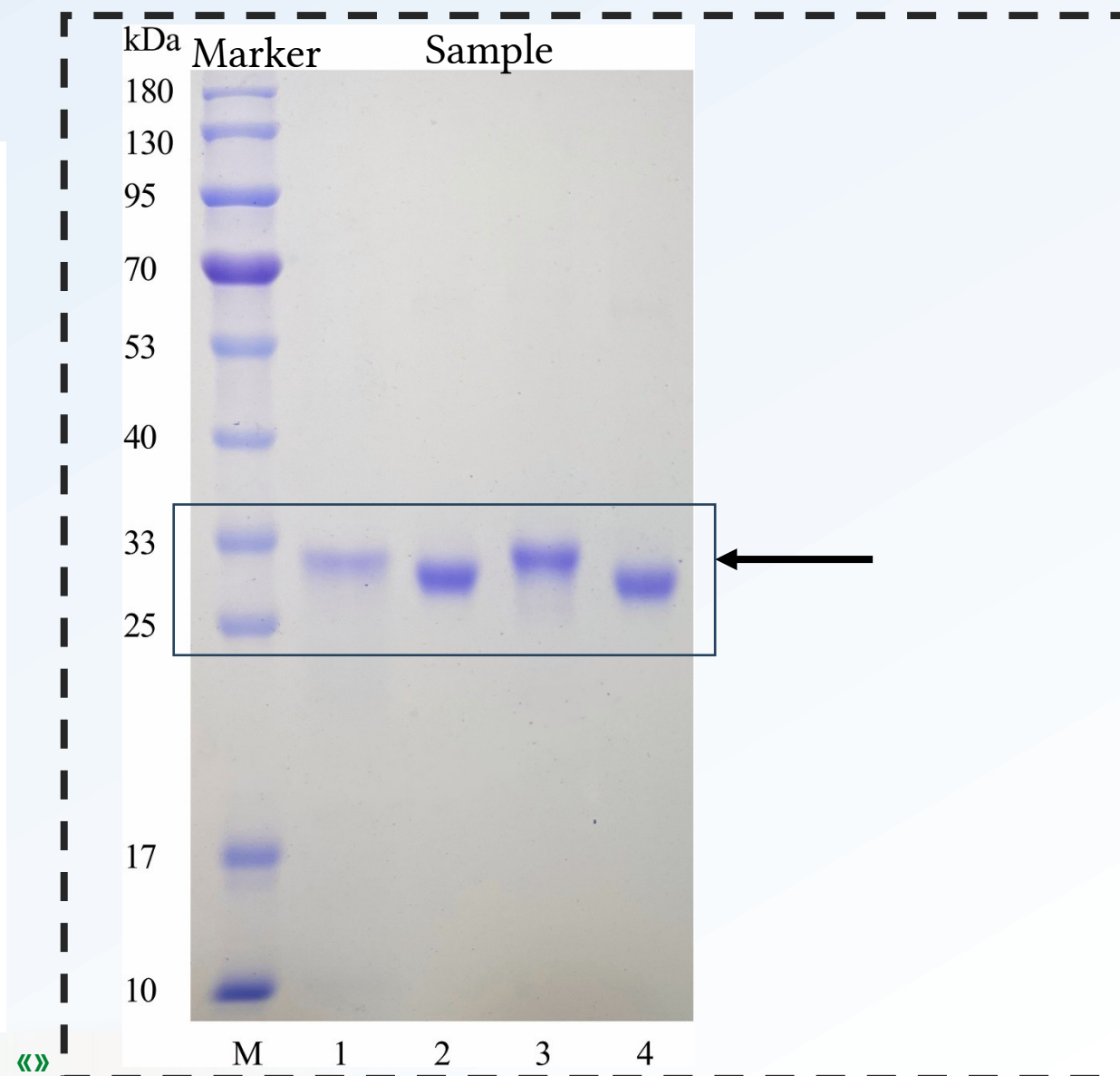
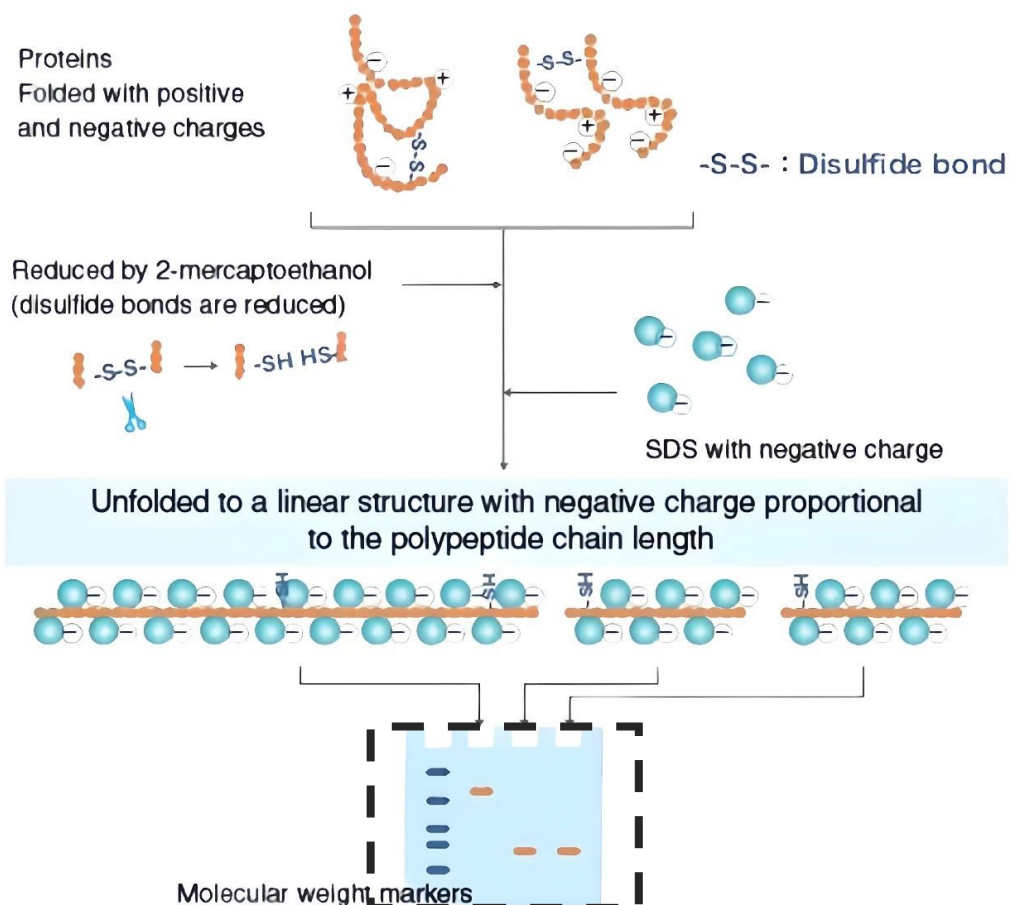
(mobility)

Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis, SDS-PAGE



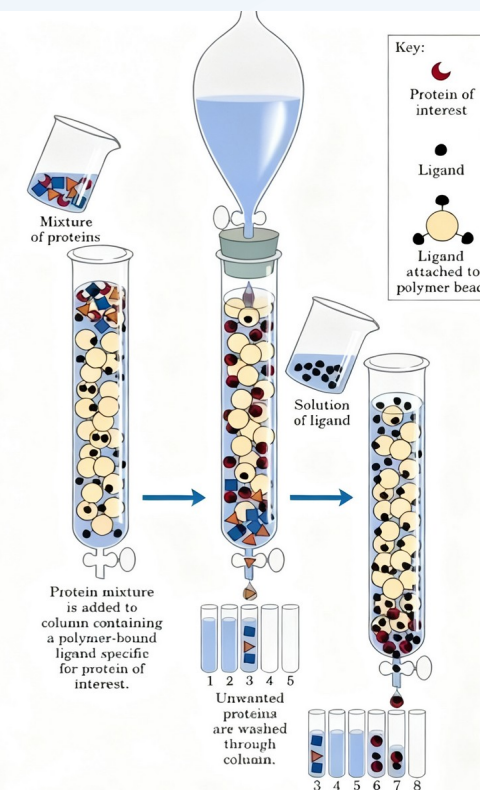
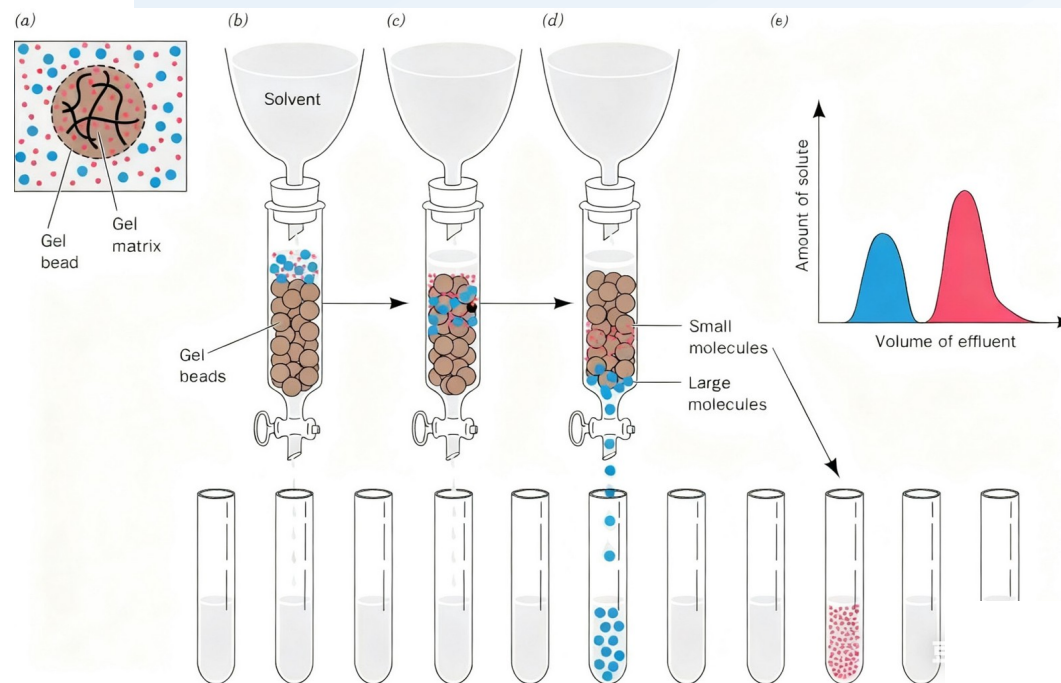
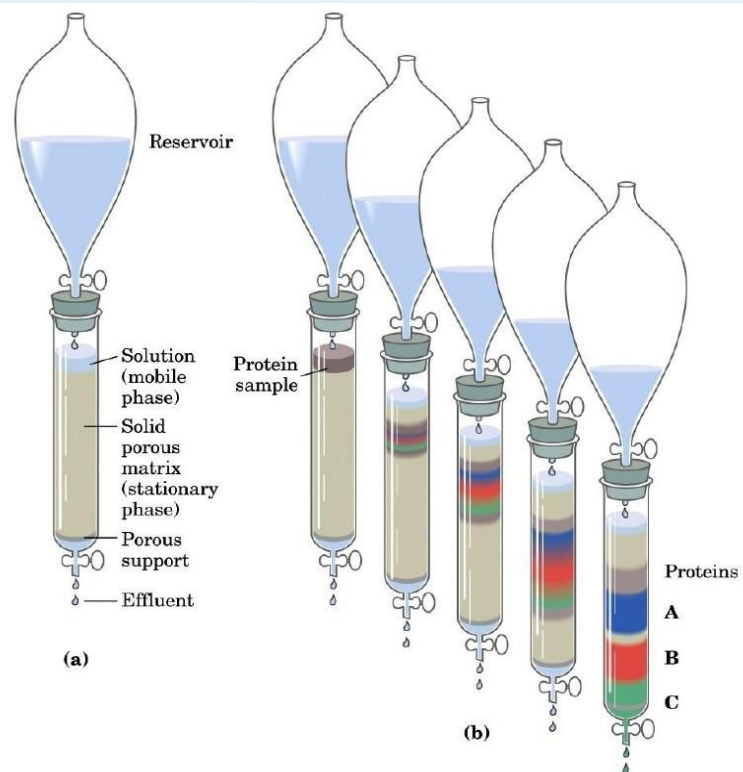
SDS-PAGE

Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis, SDS-PAGE



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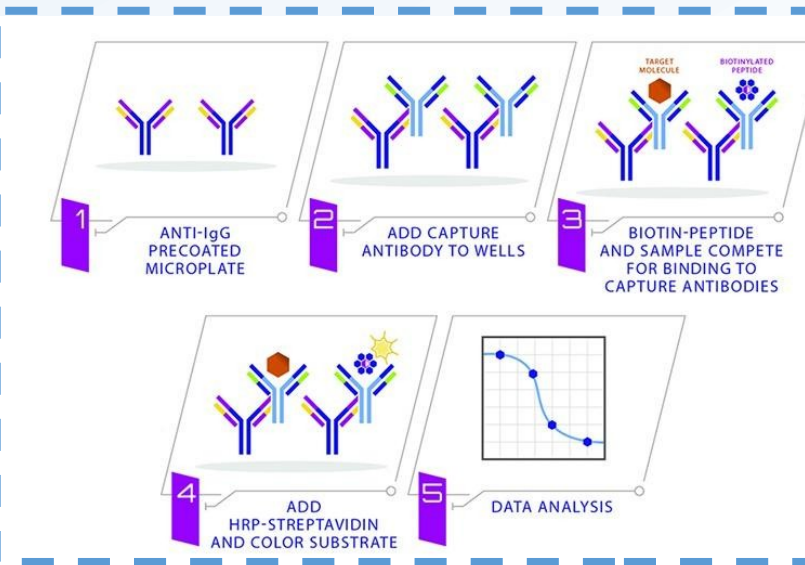
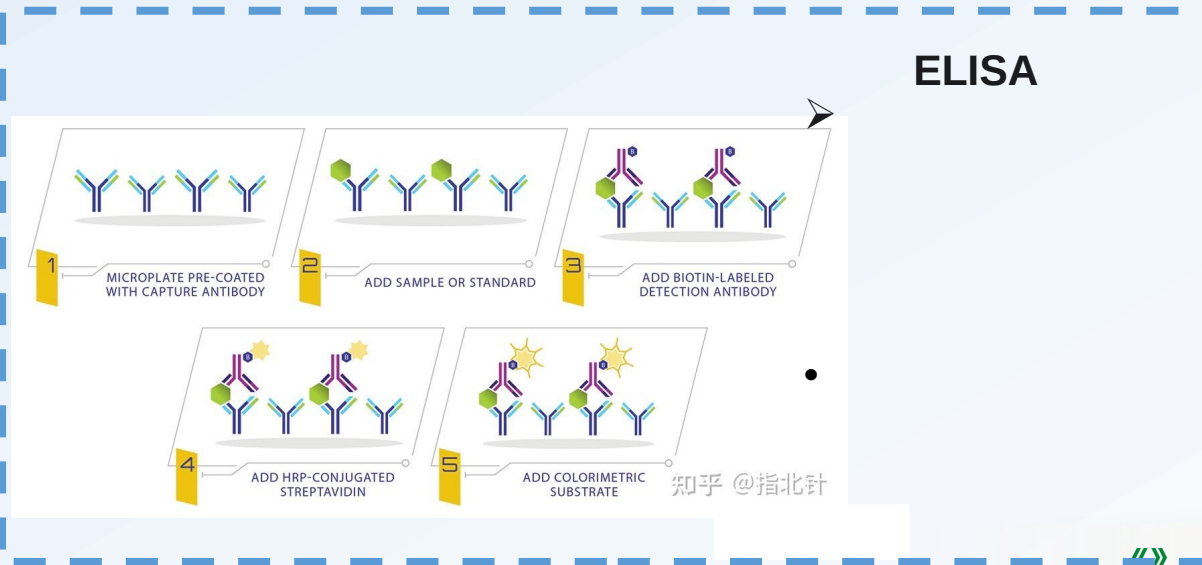
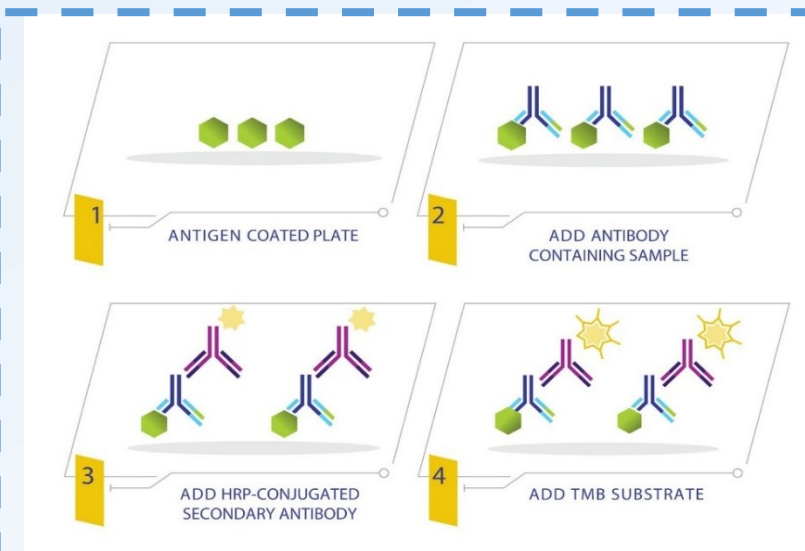
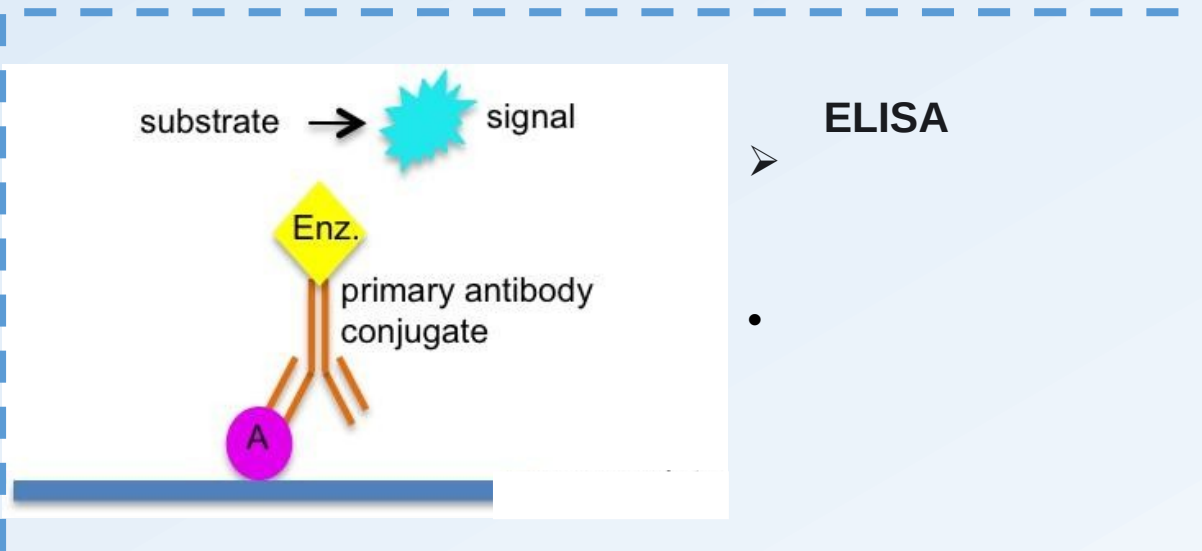




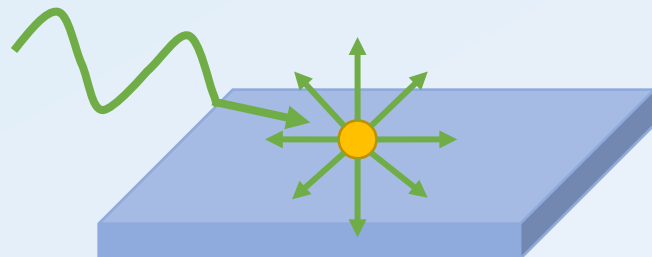
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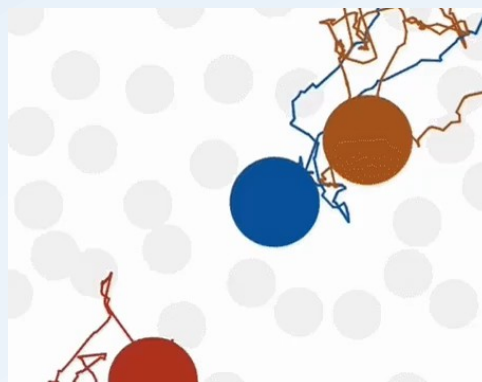


$$\alpha = 3\epsilon_m V \frac{\epsilon_p - \epsilon_m}{\epsilon_p + 2\epsilon_m}$$

$$\sigma_s = \frac{8}{3\pi^2} \frac{|\alpha|^2}{\lambda^4}$$

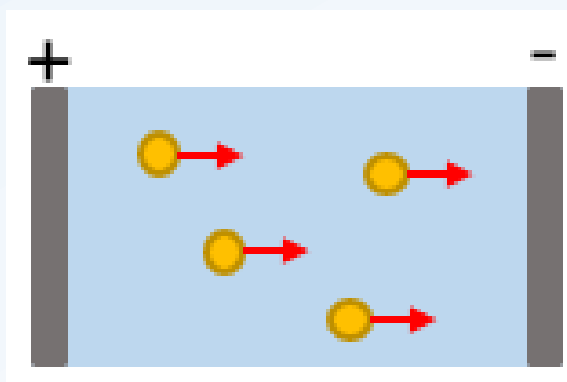
$$I_{scat} \propto \sigma_s \propto \alpha^2 \propto V^2$$

Brownian □ □



<https://zhuanlan.zhihu.com/p/564870325>

□ □



$$\text{Stokes-Einstein公式: } D = \frac{k_B T}{3\pi\eta R}$$

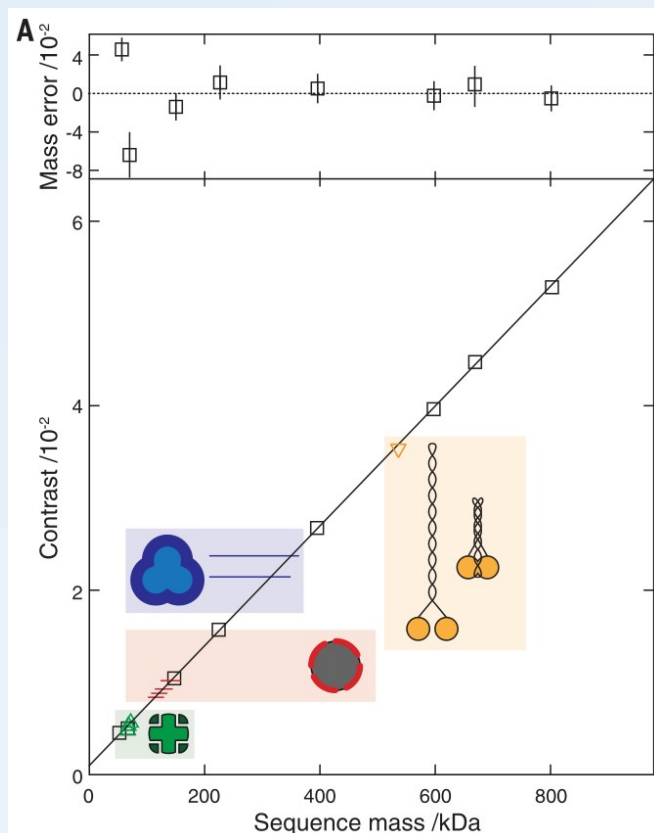
平移扩散系数 D $\longrightarrow$ 粒径分布 R

Langmuir, 2015, 31, 3-12.

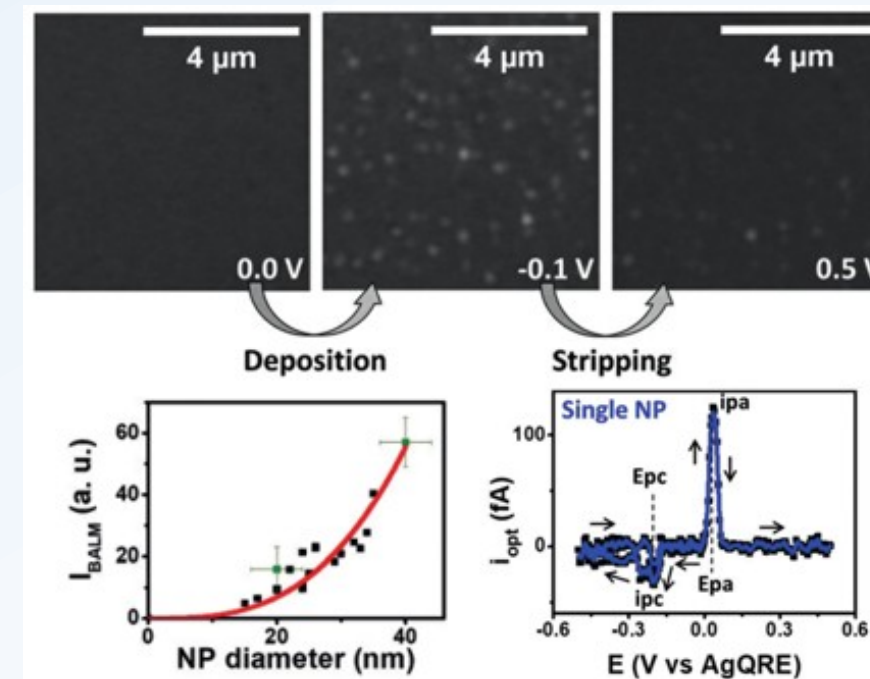
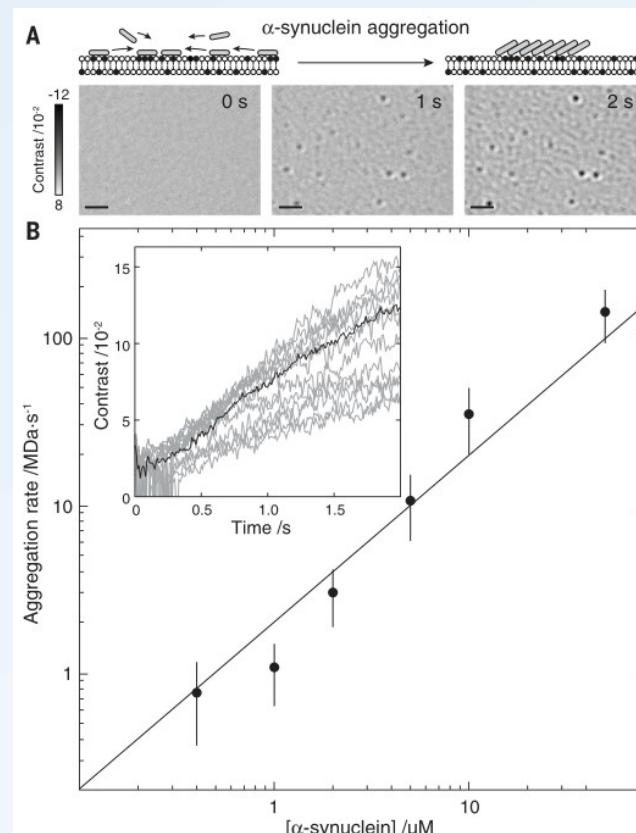
$$\text{Henry公式: } \mu_e = \frac{2\epsilon\zeta f(Ka)}{3\eta}$$

$f(Ka)$: Henry函数
 双电层厚度 $\ll$ 粒径, $f(Ka) \approx 1.5$
 双电层厚度 $\gg$ 粒径, $f(Ka) \approx 1$

电泳迁移率 μ_e $\longrightarrow$ Zeta电位 ζ



Young G, Hundt N, Cole D, et al. Science, 2018, 360(6387): 423-427.

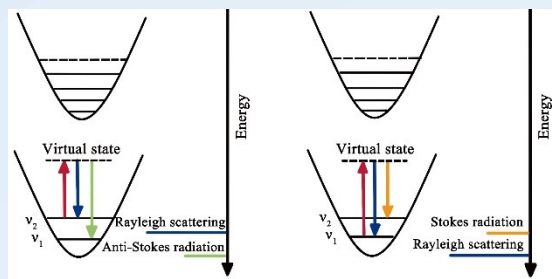


Lemineur J F, Noël J M, Ausserré D, et al. Angewandte Chemie, 2018, 130(37): 12174-12178.

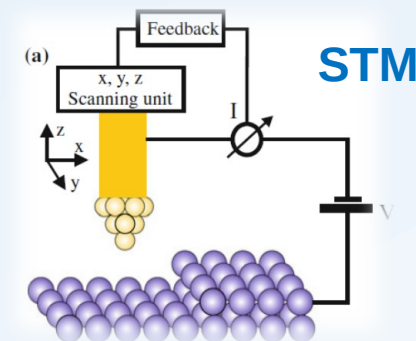
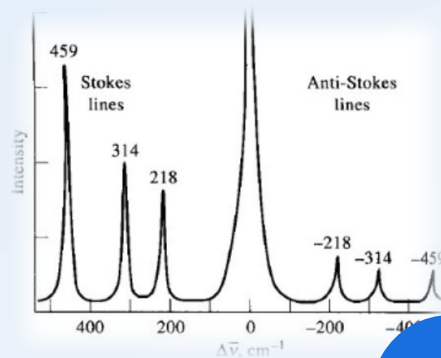
$$\text{Contrast} \propto V = \frac{m}{\rho} \xrightarrow{\text{蛋白质的密度}\rho\text{基本相同}} \text{Contrast} \propto m$$

《》

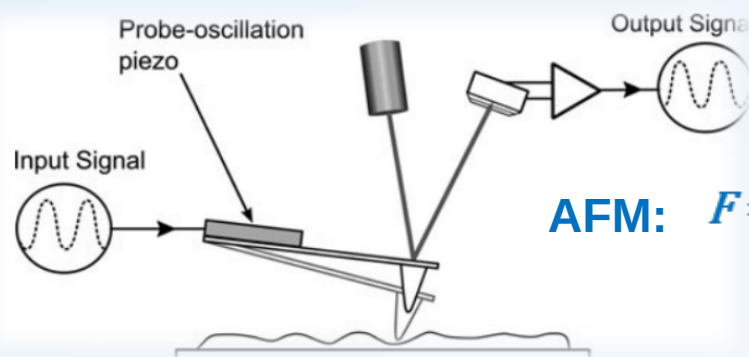
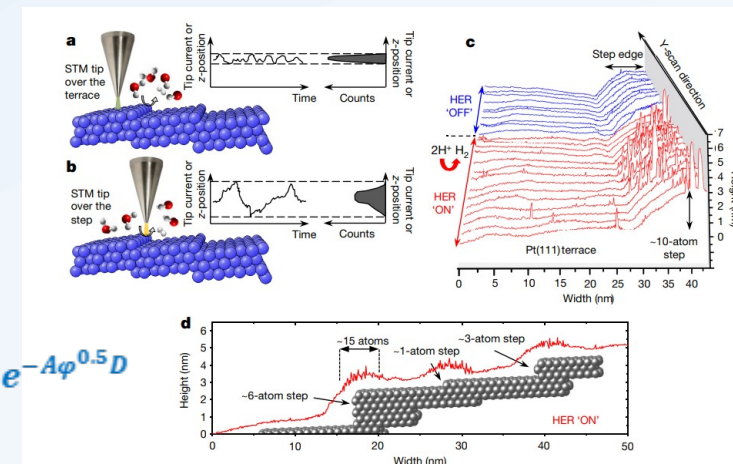
$$i_{opt} = -\frac{F\rho}{M} \frac{dV_{NP}}{dt} = -k \frac{F\rho}{M} \frac{dI_{BALM}}{dt}$$



$$T = \frac{h\nu_i}{k_B} \ln \left[\frac{I_S}{I_{AS}} \cdot \frac{\nu + \nu_i^4}{\nu - \nu_i^4} \right]$$



$$I_{\text{tunneling}} = V_b e^{-A\phi^{0.5}D}$$



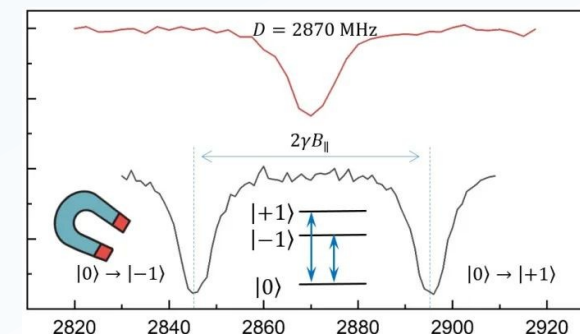
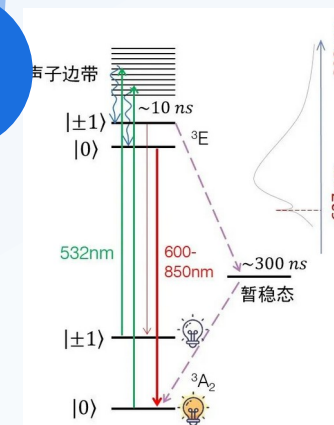
$$\text{AFM: } F = -k \cdot D$$

$$F_{\text{loc}} = \Delta h \cdot k_t = \Delta T \cdot \alpha_s \cdot k_t = E_i \cdot k_s \cdot \alpha_s \cdot k_t$$

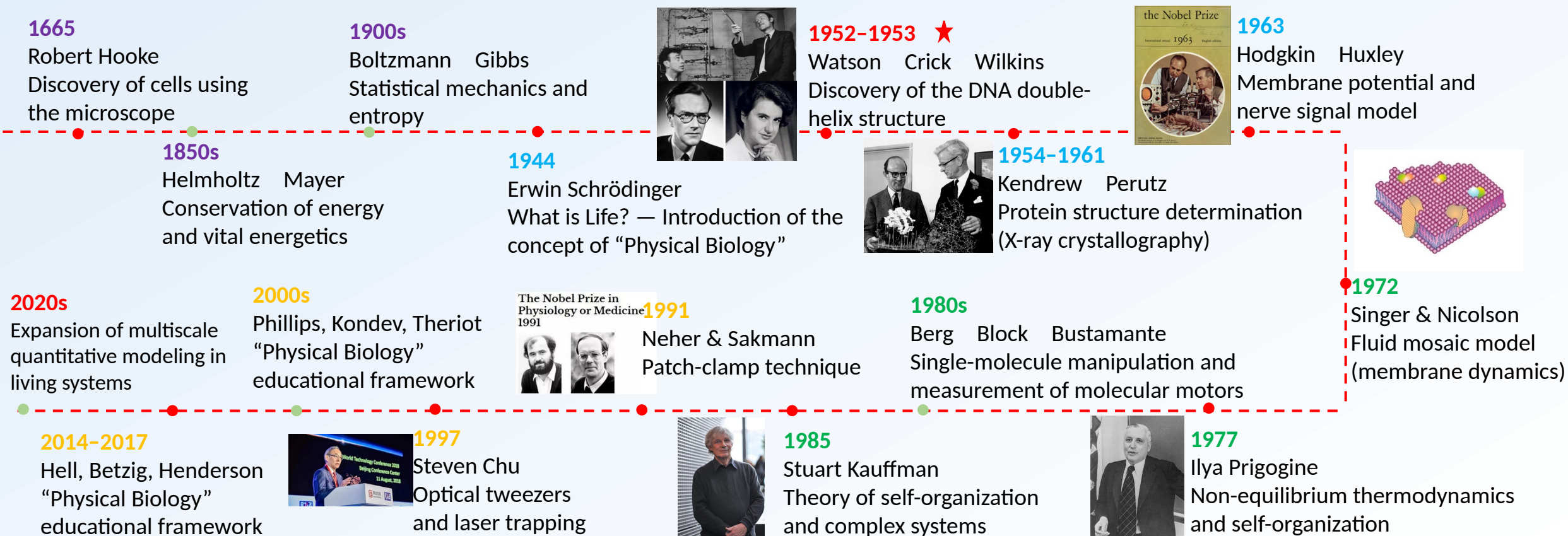
PFIR

$$F_{\text{vdW}} = -\frac{\pi^2 C \rho_{\text{tip}} \rho_{\text{sample}} R_{\text{tip}}}{6d^2}$$

《》



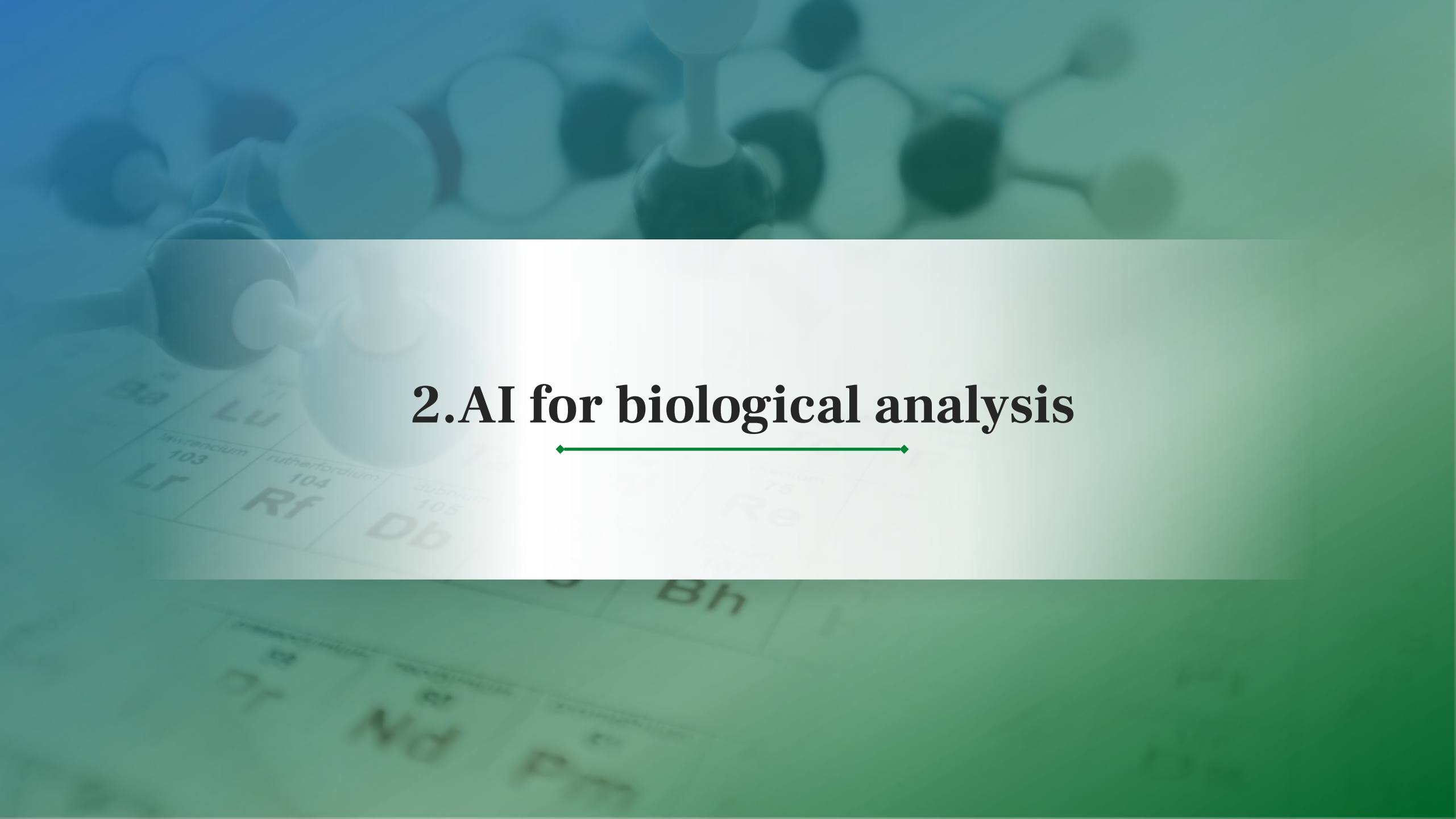
Optically Detected Magnetic Resonance NV







1	m	$1/299,792,458$
2	kg	$(h = 6.62607015 \times 10^{-34}, \text{J} \cdot \text{s})$
3	s	$9,192,631,770^{-133}$
4	A	$(e = 1.602176634 \times 10^{-19}, \text{C})$
5	K	$(k = 1.380649 \times 10^{-23}, \text{J/K})$
6	mol	$(N_A = 6.02214076 \times 10^{23})$
7	cd	$540 \times 10^{12} \text{ Hz}$



2.AI for biological analysis



Input data	Example prediction tasks	Recommended models	Challenges
Gene sequence	DNA accessibility ¹⁴	1D CNNs	Repetitive regions in genome
	3D genome organization ⁵⁸	RNNs	Sparse regions of interest
	Enhancer–promoter interactions ⁴⁰	Transformers	Very long sequences
Protein sequence	Protein structure ^{23,55}	2D CNNs and residual networks using co-variation data	Metagenome data stored in many places and therefore hard to access
	Protein function ¹³²	Multilayer perceptrons with windowing	Data leakage (from homology) can make validation difficult
	Protein–protein interaction ¹³³	Transformers	
Protein 3D structure	Protein model refinement ¹³⁴	GCNs using molecular graph	Lack of data, particularly on protein complexes
	Protein model quality assessment ¹³⁵	3D CNNs using coordinates	Lack of data on disordered proteins
	Change in stability upon mutation ¹³⁶	Traditional methods using structural features Clustering	
Gene expression	Intergenic interactions or co-expression ¹³⁷	Clustering	Unclear link between co-expression and function
	Organization of transcription machinery ¹³⁸	CNNs Autoencoders	High dimensionality High noise
Mass spectrometry	Detecting peaks in spectra ¹³⁹	CNNs using spectral data	Lack of standardized benchmarks ¹⁴¹
	Metabolite annotation ¹⁴⁰	Traditional methods using derived features	Normalization ^a required between different datasets
Images	Medical image recognition ^{24,62}	2D CNNs and residual networks	Systematic differences in data collection affect prediction
	Cryo-EM image reconstruction ^{60,142}	Autoencoders	Hard to obtain large datasets of consistent data
	RNA-sequencing profiles ¹⁴³	Traditional methods using image features	
Molecular structure	Antibiotic activity ⁷³	GCNs using molecular graph	Experimental data available for only a tiny fraction of possible small molecules
	Drug toxicity ⁵⁴	Traditional methods or multilayer perceptrons using molecular properties	
	Protein–ligand docking ³⁹	RNNs using text-based representations of molecular structure such as SMILES	
	Novel drug generation ¹⁴⁴	Autoencoders	
Protein–protein interaction network	Polypharmacology side effects ⁷⁷	GCNs	Interaction networks can be incomplete
	Protein function ¹⁴⁵	Graph embedding	Cellular location affects whether proteins interact High number of possible combinations

Recommendations for the use of machine learning strategies for different biological data types.

Contents

1. Overview of biological analysis

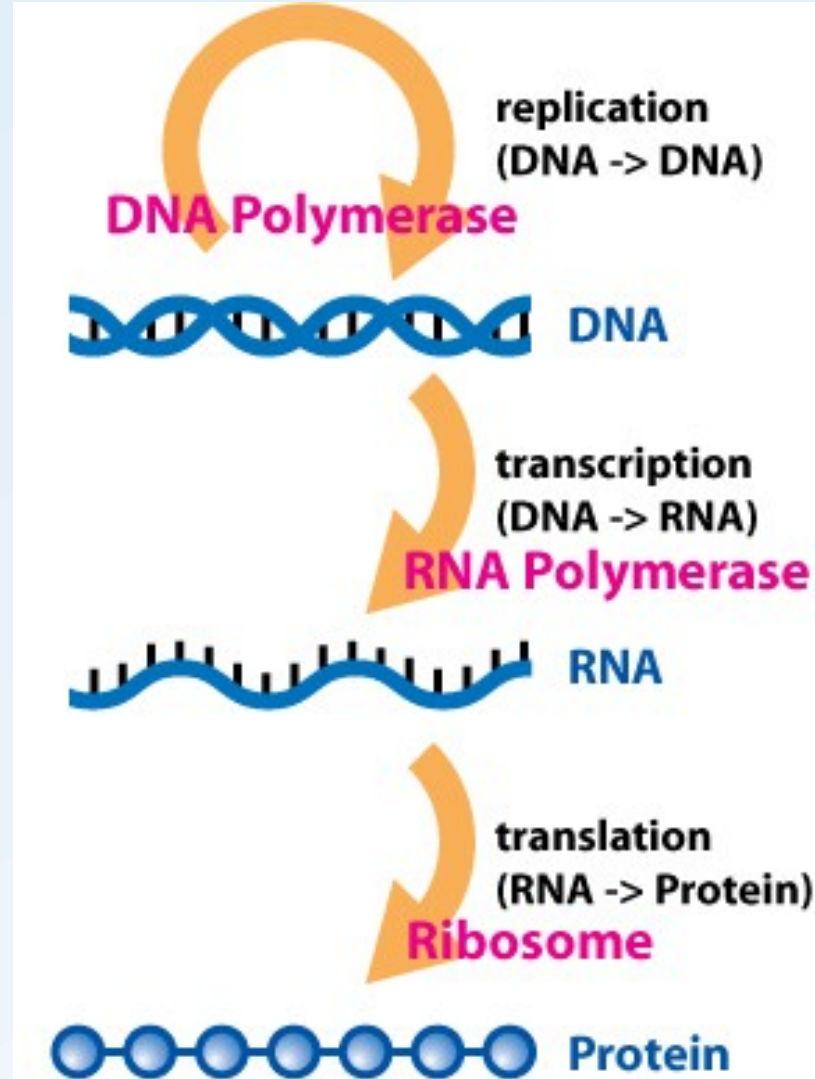
2. **AI for biological analysis**

- **Nanopore**
- **Structure biology**
- **Single-cell**
- **Bioinformatics**
- **.....**

AI for biological analysis



◆ Genetic central dogma ()



The central dogma of molecular biology explains the flow of **genetic information** within a cell

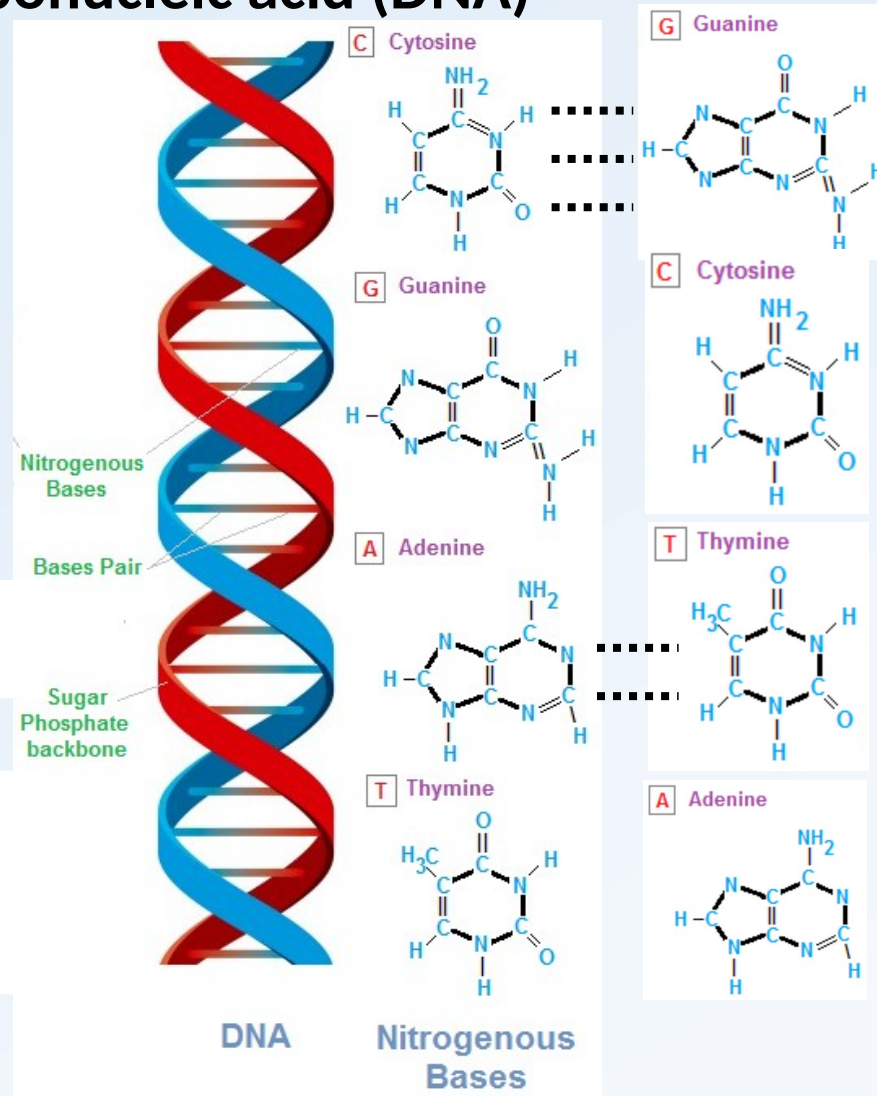
- DNA codes for RNA via the process of transcription (occurs within the nucleus)
- RNA codes for protein via the process of translation (occurs at the ribosomes)

This information flow was always considered to be uni-directional:
DNA → RNA → Protein

AI for biological analysis



◆ Deoxyribonucleic acid (DNA)



The Nobel Prize in Physiology or Medicine 1962



Photo from the Nobel Foundation archive.
Francis Harry Compton Crick
Prize share: 1/3



Photo from the Nobel Foundation archive.
James Dewey Watson
Prize share: 1/3



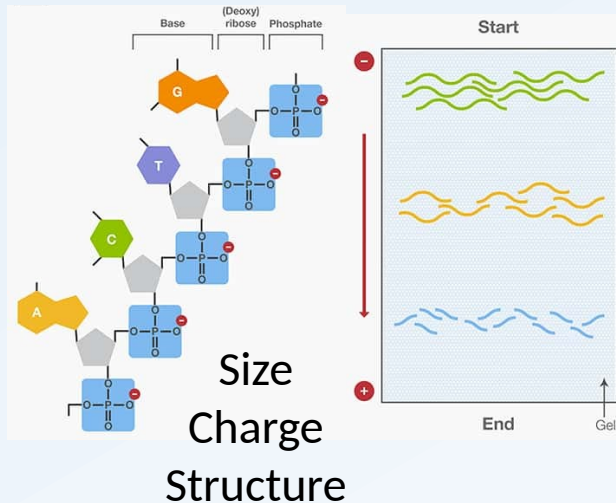
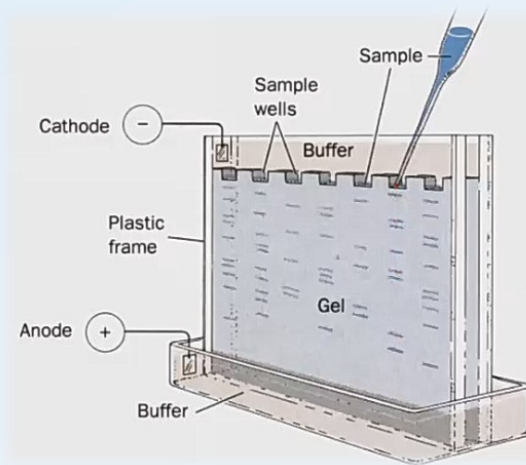
Photo from the Nobel Foundation archive.
Maurice Hugh Frederick Wilkins
Prize share: 1/3

DNA

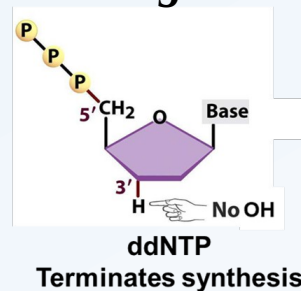
AI for biological analysis



◆ First-generation sequencing Gel Electrophoresis

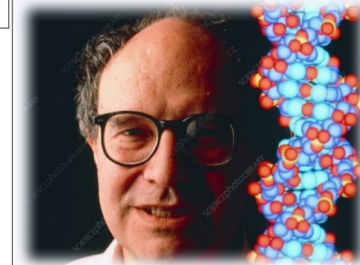
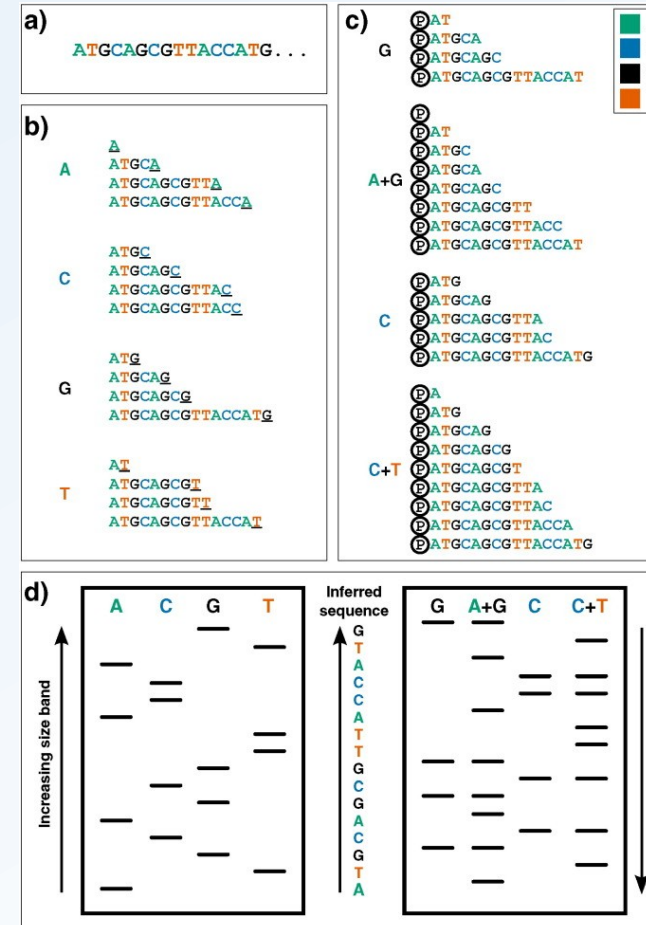


Frederick Sanger



"chain-termination"

chemical sequencing



Walter Gilbert

AI for biological analysis



◆ First-generation sequencing

The Nobel Prize in Chemistry 1980



Photo from the Nobel Foundation archive.

Paul Berg

Prize share: 1/2



Photo from the Nobel Foundation archive.

Walter Gilbert

Prize share: 1/4

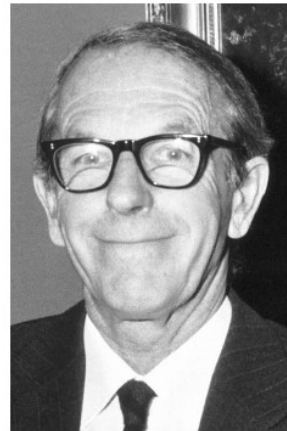


Photo from the Nobel Foundation archive.

Frederick Sanger

Prize share: 1/4

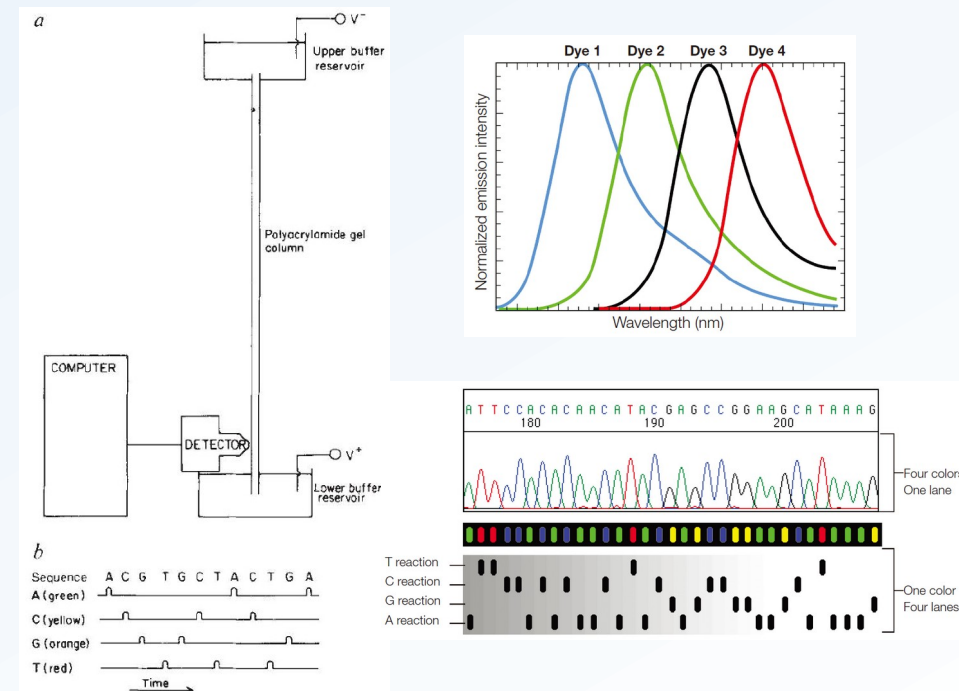
The Nobel Prize in Chemistry 1980 was divided, one half awarded to Paul Berg "for his fundamental studies of the biochemistry of nucleic acids, with particular regard to recombinant-DNA", the other half jointly to Walter Gilbert and Frederick Sanger "for their contributions concerning the determination of **base sequences in nucleic acids**"

Fluorescence detection in automated DNA sequence analysis

Lloyd M. Smith, Jane Z. Sanders, Robert J. Kaiser, Peter Hughes, Chris Dodd,
Charles R. Connell*, Cheryl Heiner*, Stephen B. H. Kent & Leroy E. Hood

Division of Biology, California Institute of Technology, Pasadena, California 91125, USA
* Applied Biosystems, Inc., Foster City, California 94404, USA

We have developed a method for the partial automation of DNA sequence analysis. Fluorescence detection of the DNA fragments is accomplished by means of a fluorophore covalently attached to the oligonucleotide primer used in enzymatic DNA sequence analysis. A different coloured fluorophore is used for each of the reactions specific for the bases A, C, G and T. The reaction mixtures are combined and co-electrophoresed down a single polyacrylamide gel tube, the separated fluorescent bands of DNA are detected near the bottom of the tube, and the sequence information is acquired directly by computer.



Hood, L. E., et al, **Nature**, 1986, 321(6071), 674.

AI for biological analysis



◆ Sanger sequencing

Automated DNA sequencing workflow, *continued*



1. Prepare the instrument.
2. Set up the plate record in Data Collection Software:
 - Results group
 - Instrument protocol
 - Analysis protocol
3. Load samples.
4. Perform the run.

Applied Biosystems instruments:



3730/3730xl analyzer



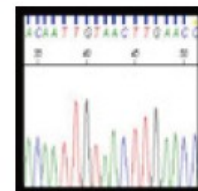
3500 analyzer



3130/3130xl analyzer



310 analyzer



AB1 file

AB Applied Biosystems

ThermoFisher
SCIENTIFIC

AI for biological analysis



◆ Human Genome Project)

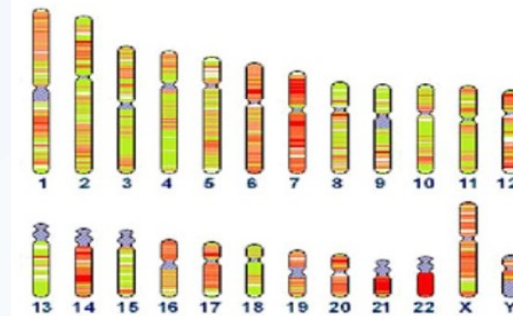


Human genome
30 0000 0000 base pairs

From 1990 to 2003

13 years, \$30 0000 0000

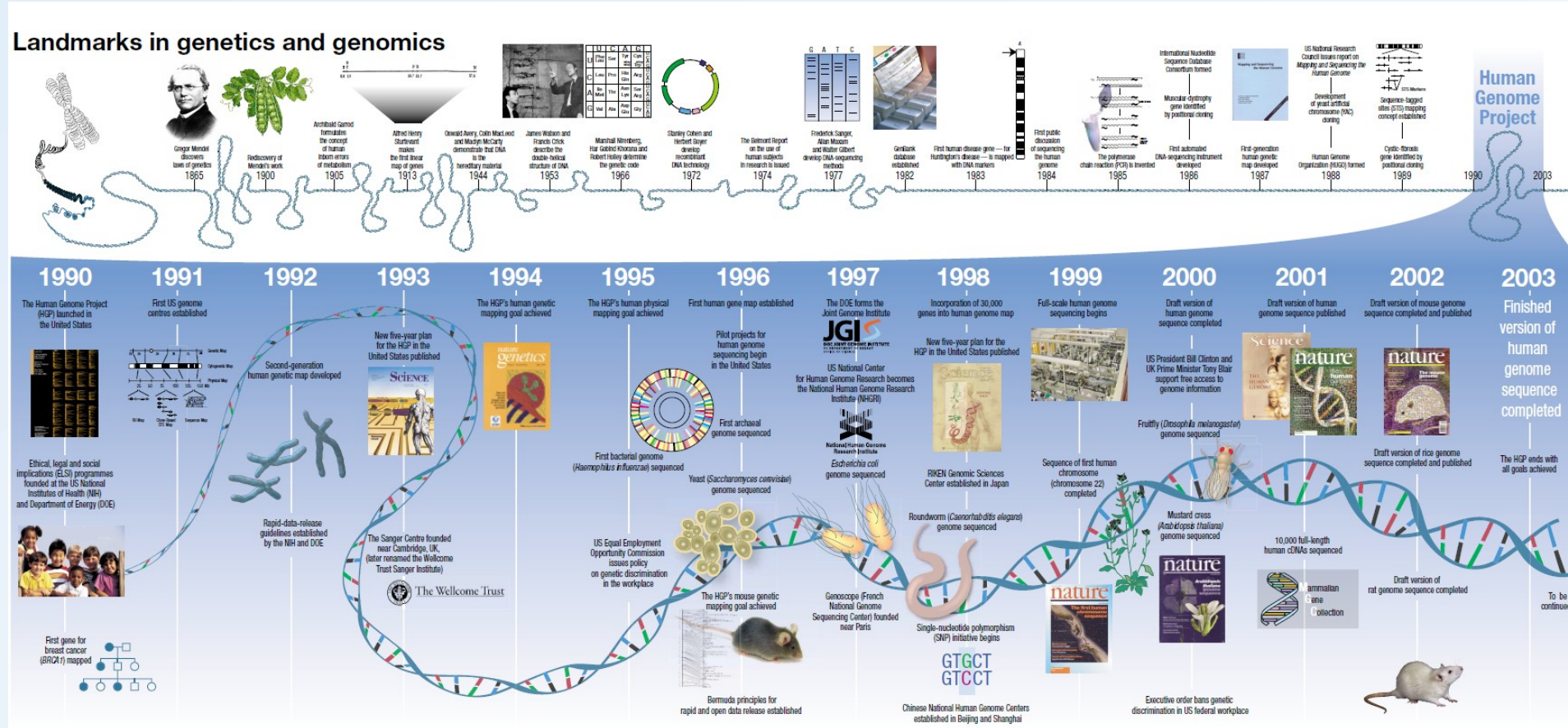
\$1 per base



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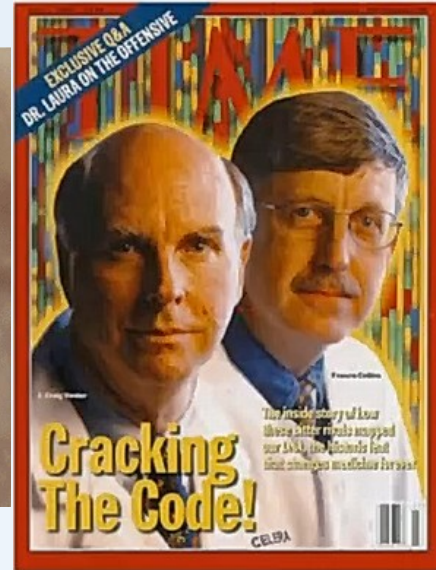
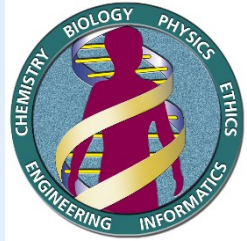
◆ Human Genome Project



AI for biological analysis



◆ Human Genome Project



J. Craig Venter

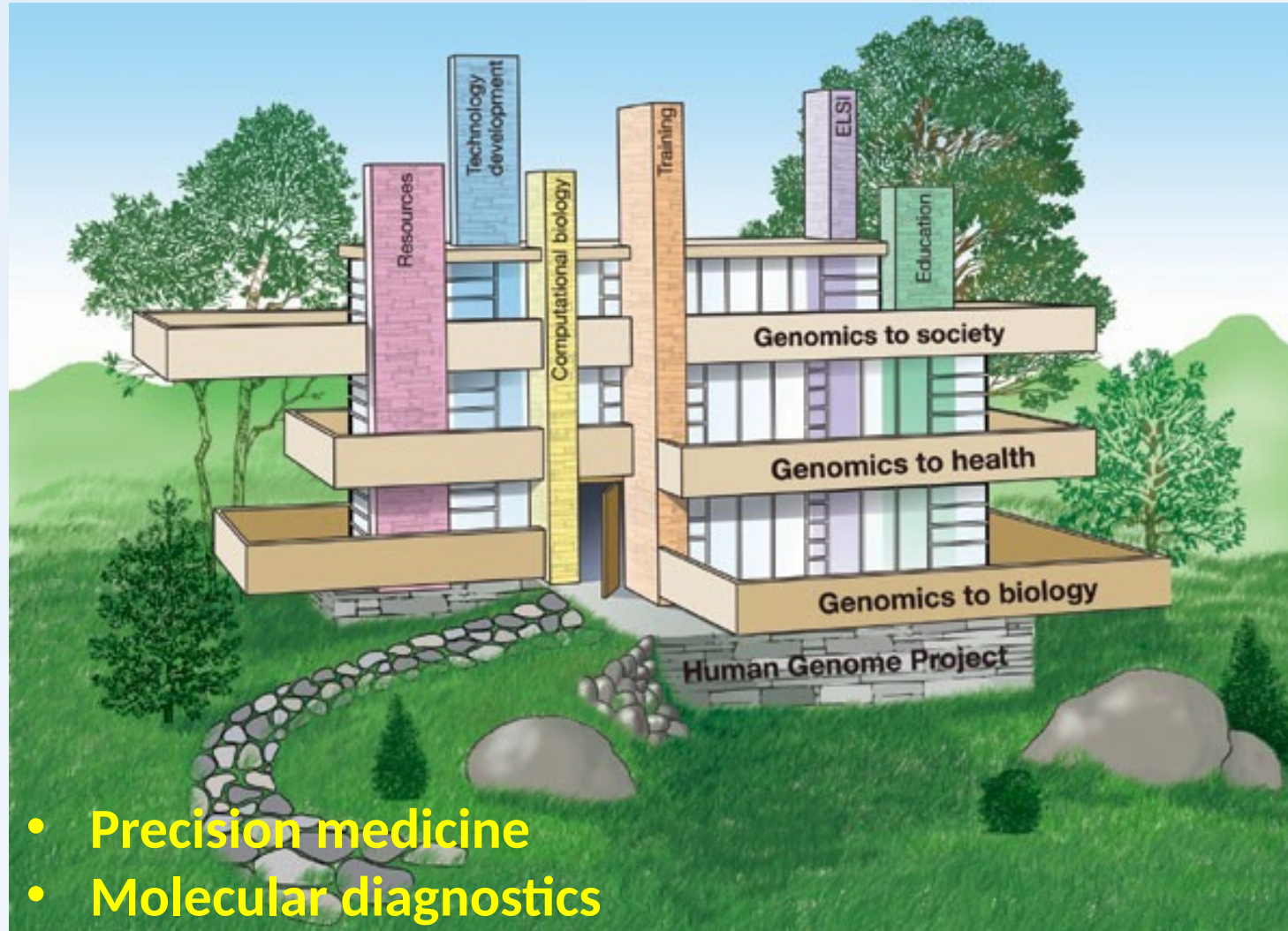
Francis Collins



AI for biological analysis



◆ The future of genomics research



- Precision medicine
- Molecular diagnostics

AI for biological analysis



◆ Second-generation DNA sequencing

Next-generation sequencing (NGS)

Disadvantage of the first

Low sequencing throughput

- **High-throughput sequencing**

	MiSeq	HiSeq	NovaSeq	Sanger
Reads (millions)	30	3,000	13,000	0.0004
Gigabases/day	7	500	4000	0.001



Thermo fisher-Ion Torrent



Illumina-Hiseq

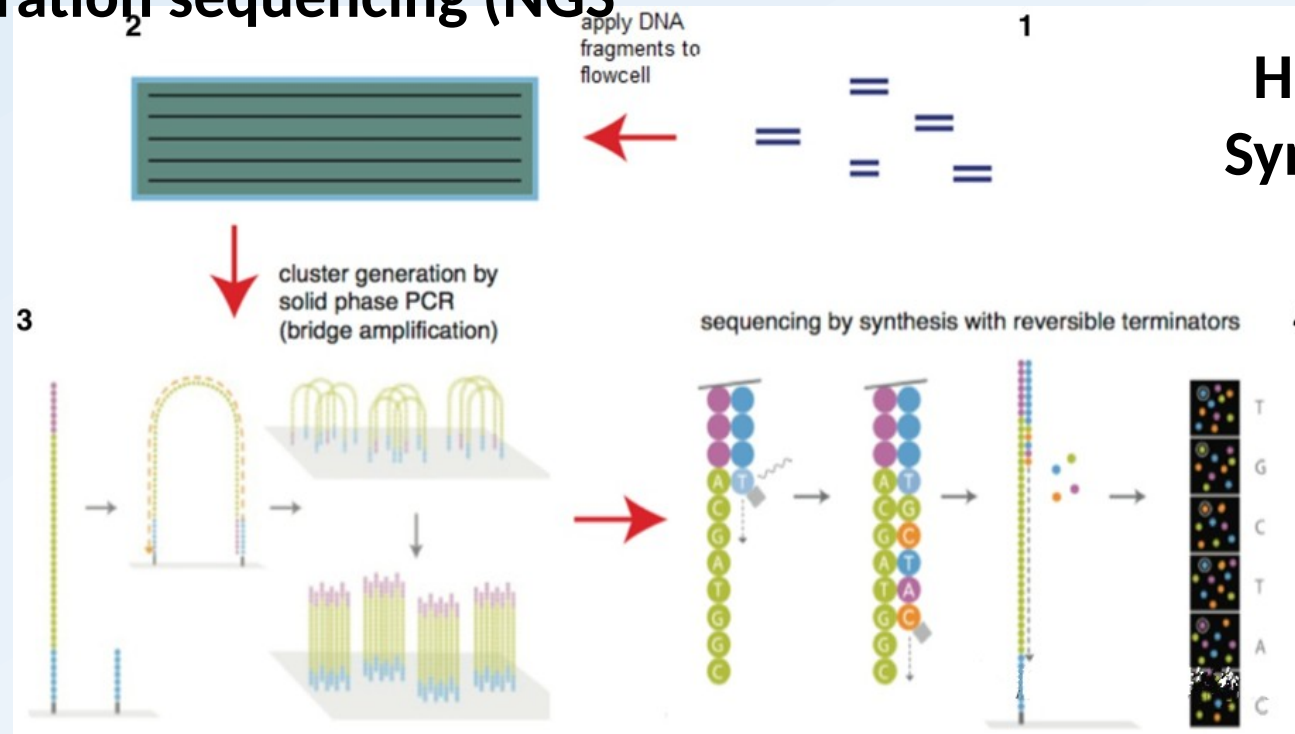


BGI()-MGISEQ

AI for biological analysis

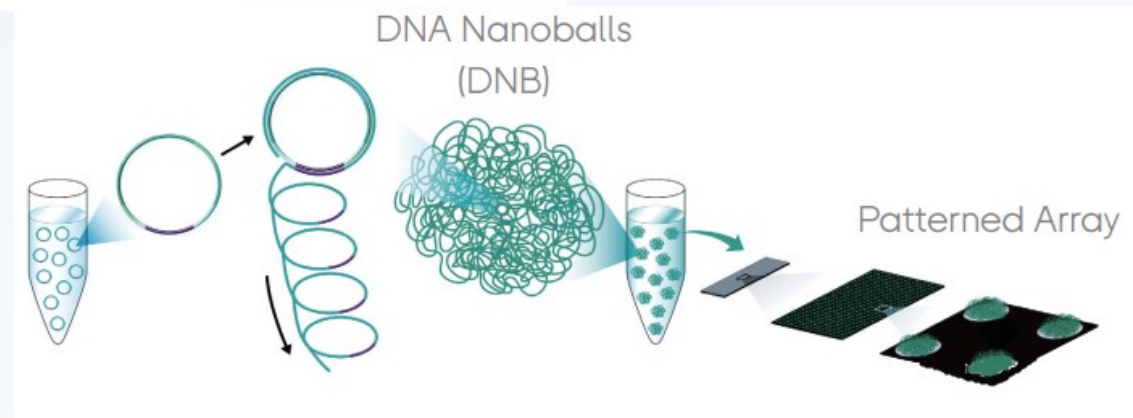


◆ Next-generation sequencing (NGS)



HiSeq (Illumina) :
Synthesis-cleaning-
photography

MGISEQ



AI for biological analysis



◆ BGI () DNA nanoball

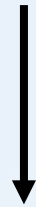
AI for biological analysis



◆ Third-generation DNA sequencing

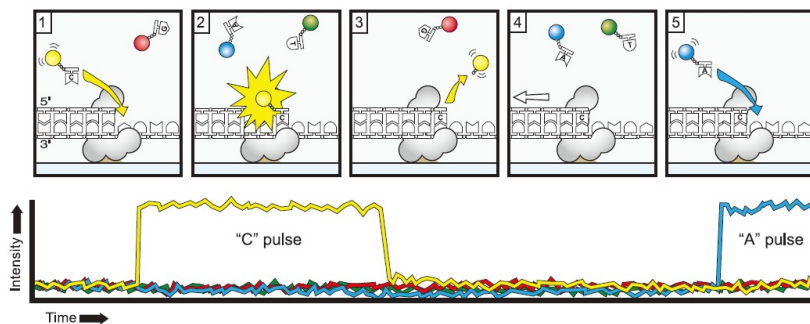
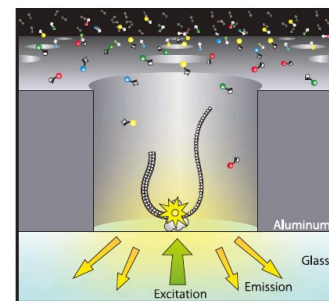
➤ Disadvantage of NGS

Since **PCR amplification** is utilized in library construction, PCR preference and a certain probability of **base mismatch** will be generated

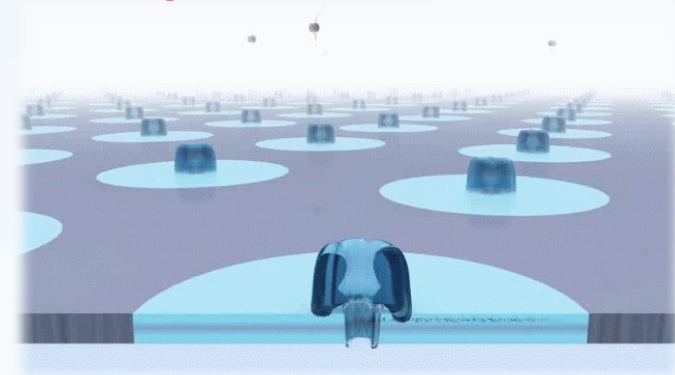


- Long sequencing time
- Short read length
- Poor end quality

PacBio & Oxford Nanopore Technologies



✓ Single molecule



Bayley H., et al, **Nat. Nanotechnol.**, 2009, 4.4, 265.

Eisenstein M., **Nat. Biotechnol.**, 2012, 30(4):295.

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Summary

	GO1	GO2	GO3
Accuracy	High	Medium	Low
Sequencing length	700-1000bp	100-150bp	∞
Cost	High	Low	High
Throughput	Low	High	High
Time	Long	Long	Short
PCR	Yes	Yes	No, direct detection
Application	Genotyping	Genetic diseases	Macrogenome

Accuracy → Throughput → Time



AI for biological analysis



◆ Nanopore technique

Nanoscale dimensions
chemical environment

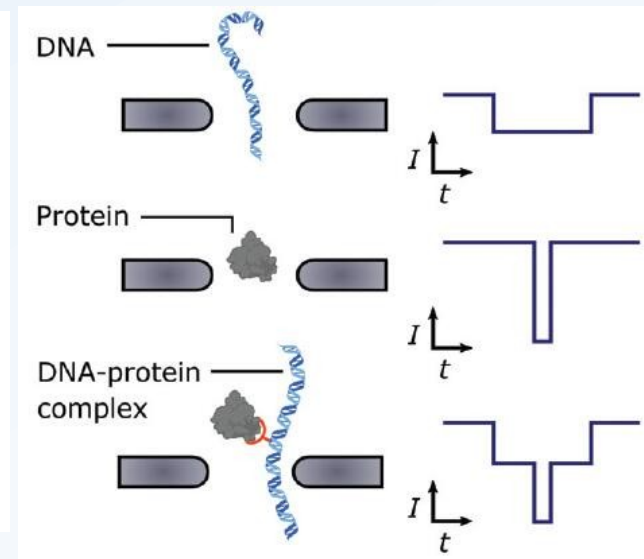
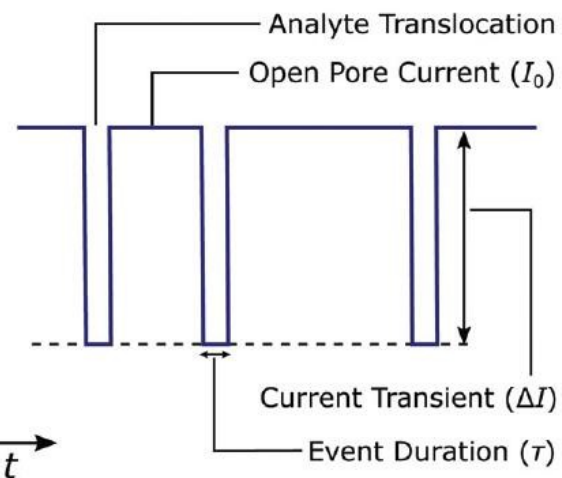
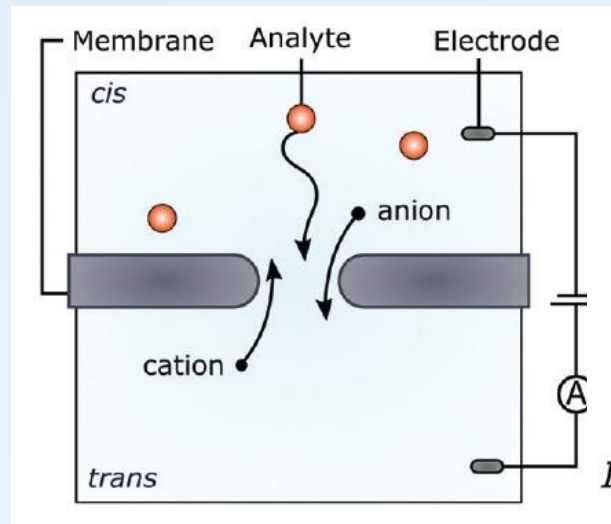


Single
molecule

Confinement
space

Chemical, Catalytic
and Biological
effects

Nanopore



◆ Temporal resolution

◆ Spatial resolution

J. B. Edel, et al, Chem. Soc. Rev., 2021, 50, 4974.

J. B. Edel, et al, Nat. Rev. Mater., 2020: 1.

2026/9/11

Amit Meller, et.al, Chem. Soc. Rev., 2018, 47(23): 8512.

Yi-Tao Long, et.al, Chem. Eur. J., 2018, 24, 13064.



AI for biological analysis

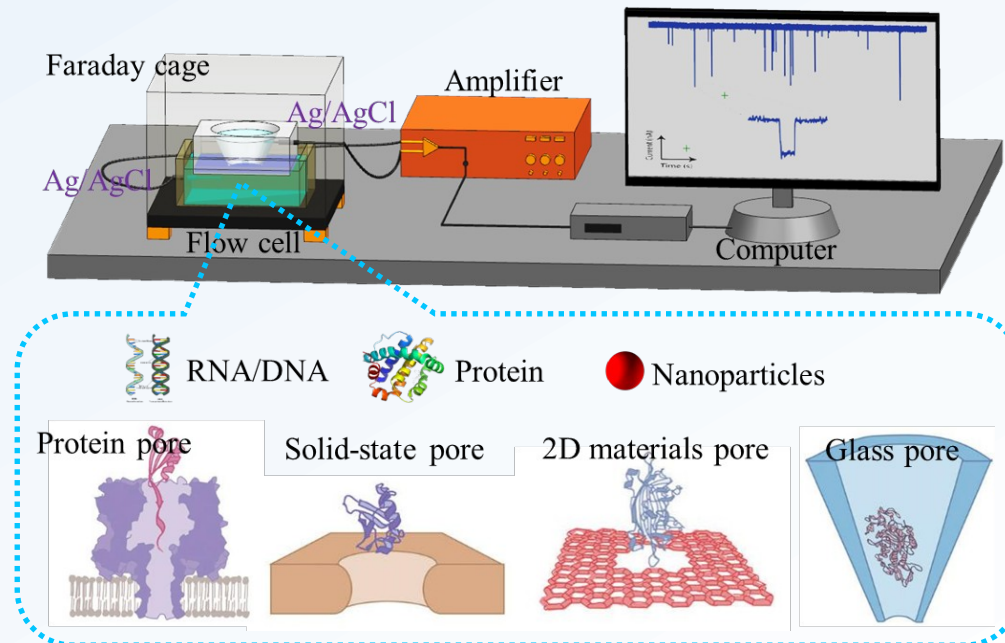
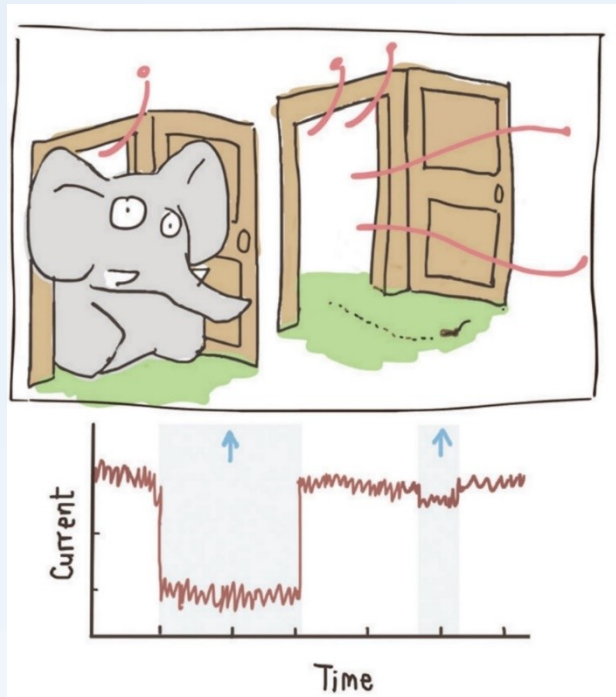


◆ Nanopore and device

The key point of nanopore detection

- The relationship between **nanopore size** and analyte
- The **velocity** of the analyte translocation the nanopore
- Signal current **amplification** at the pA level

- ✓ **Sensing area**
- ✓ **Translocation speed**



Meller A., et.al, **Chem. Soc. Rev.**, 2018, 47(23): 8512.

AI for biological analysis



◆ Biological nanopore

Biological Nanopore	Structure Side View	Top View	Critical Dimension	Types of Analytes
α -HL			1.4 nm ¹⁰⁸	small molecules, RNA, ssDNA, dsDNA, proteins
OmpG			1.3 nm ¹¹⁸	small molecules, proteins
MspA			1.2 nm ¹¹⁰	ssDNA, dsDNA
AeL			1.0 nm ¹¹¹	ssDNA, proteins
Phi29 Motor			3.6 nm ¹¹⁴	small molecules, ssDNA, dsDNA, proteins
ClyA			3.3 nm ¹¹⁶	dsDNA, proteins



Prof. Hagan Bayley

<https://www.bayleygroup.co.uk/>

University of Oxford



Prof. Jens H Gundlach

<http://faculty.washington.edu/gundlach/>

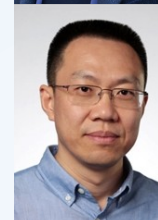
University of Washington



Prof. Shuo Huang ()

<https://hysz.nju.edu.cn/bionano/main.htm>

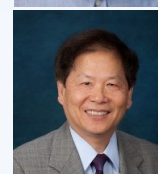
Nanjing University



Prof. Yi-tao Long ()

<https://hysz.nju.edu.cn/ytlong/main.psp>

Nanjing University



Prof. Peixuan Guo

<https://rnanano.osu.edu/Guo/peixuanguo.html>

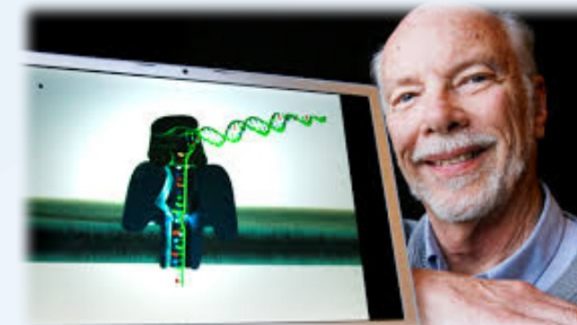
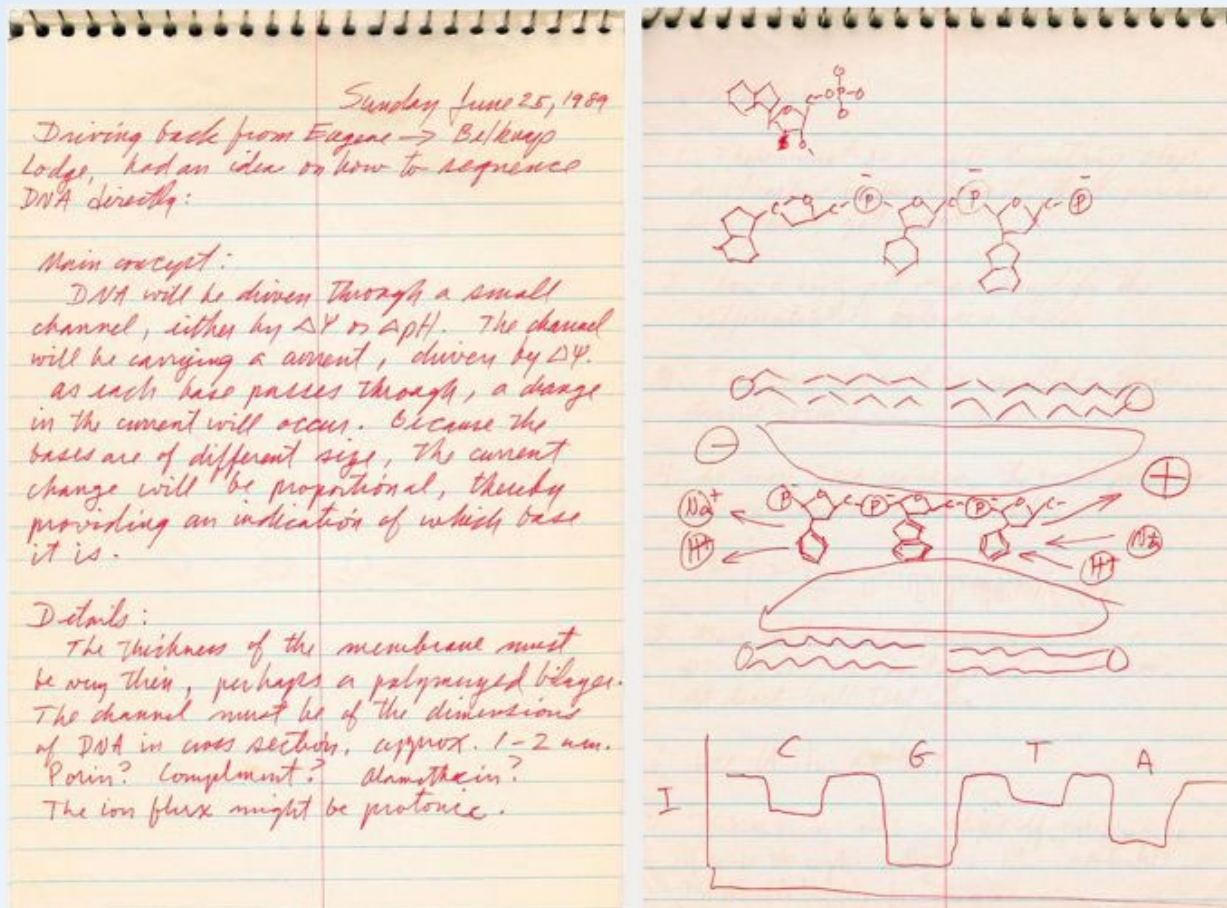
The Ohio State University

AI for biological analysis

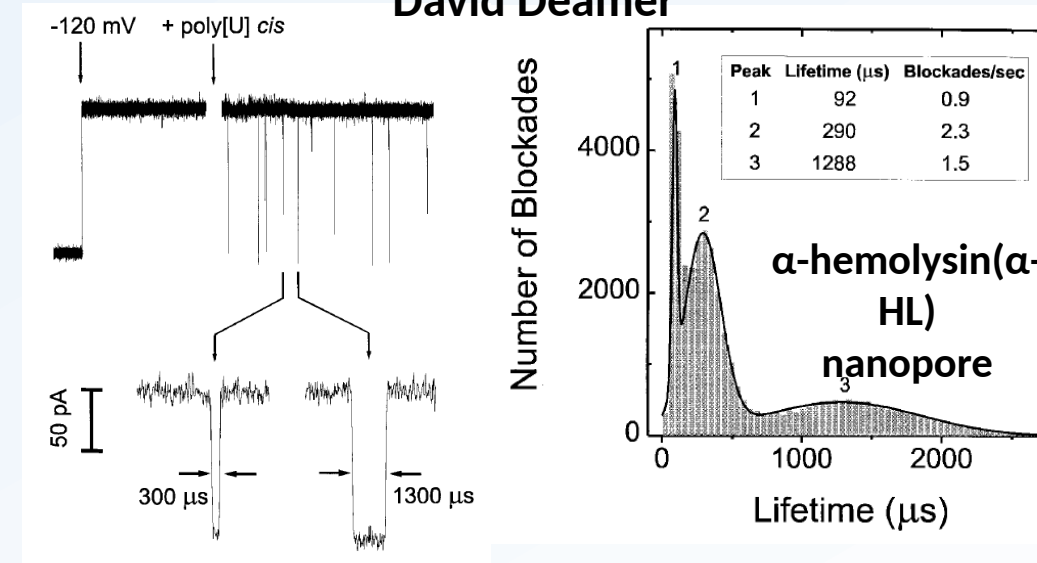


◆ Detection of nucleic acids

David Deamer's notebook (dated June 25, 1989)



David Deamer



D. W. Deamer, et al, **PNAS.**, 1996, 93, 13770.

Channel blocking methods can be used to measure the length of polynucleotides. The method could theoretically enable **direct, high-speed detection of base sequences** in single

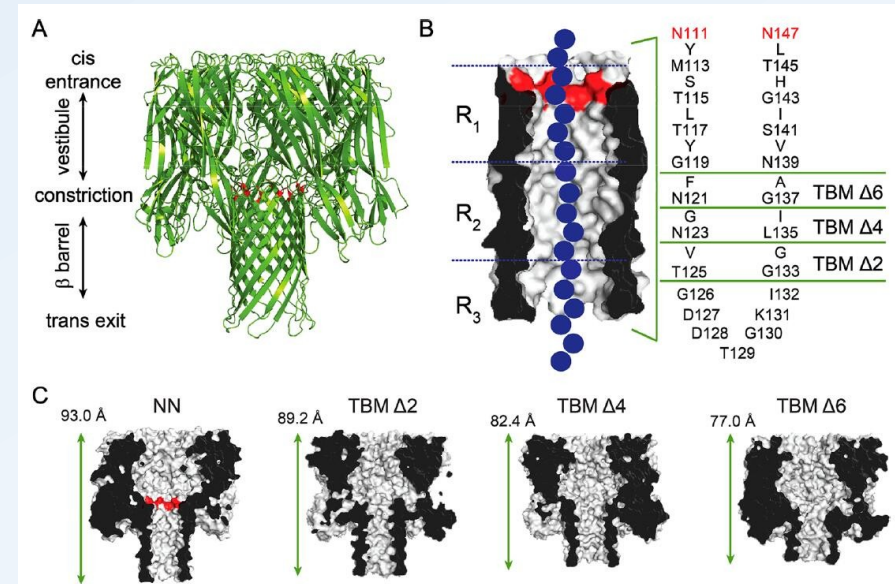
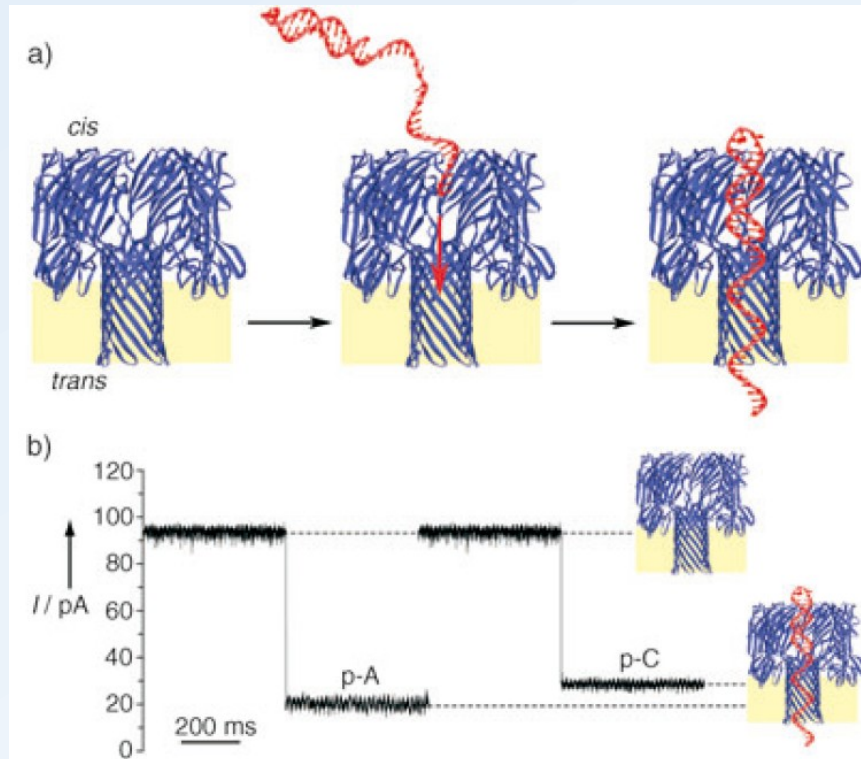
2026/9/11 DNA or RNA molecules

AI for biological analysis

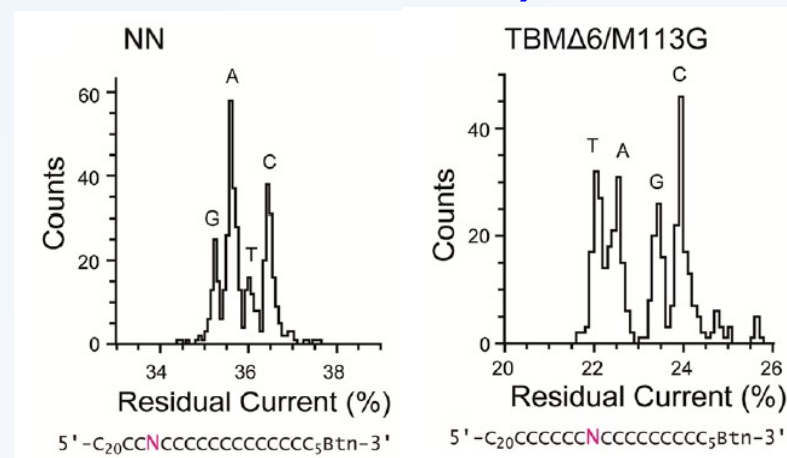


◆ Detection of nucleic acids

Recognizing a Single Base in an Individual DNA Strand



The barrel is shortened by ~16 Å



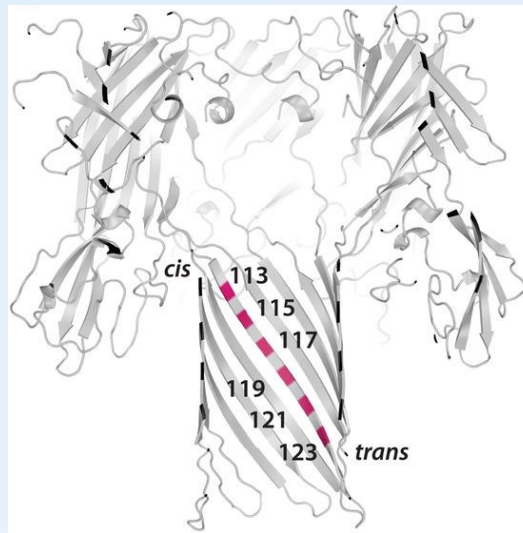
Bayley, H., et al, *Angew. Chem. Int. Ed.*, 2005, 117(9): 1425.
2026/9/11

Bayley H., et al, *ACS Nano*, 2015, 9(8): 7895.

AI for biological analysis

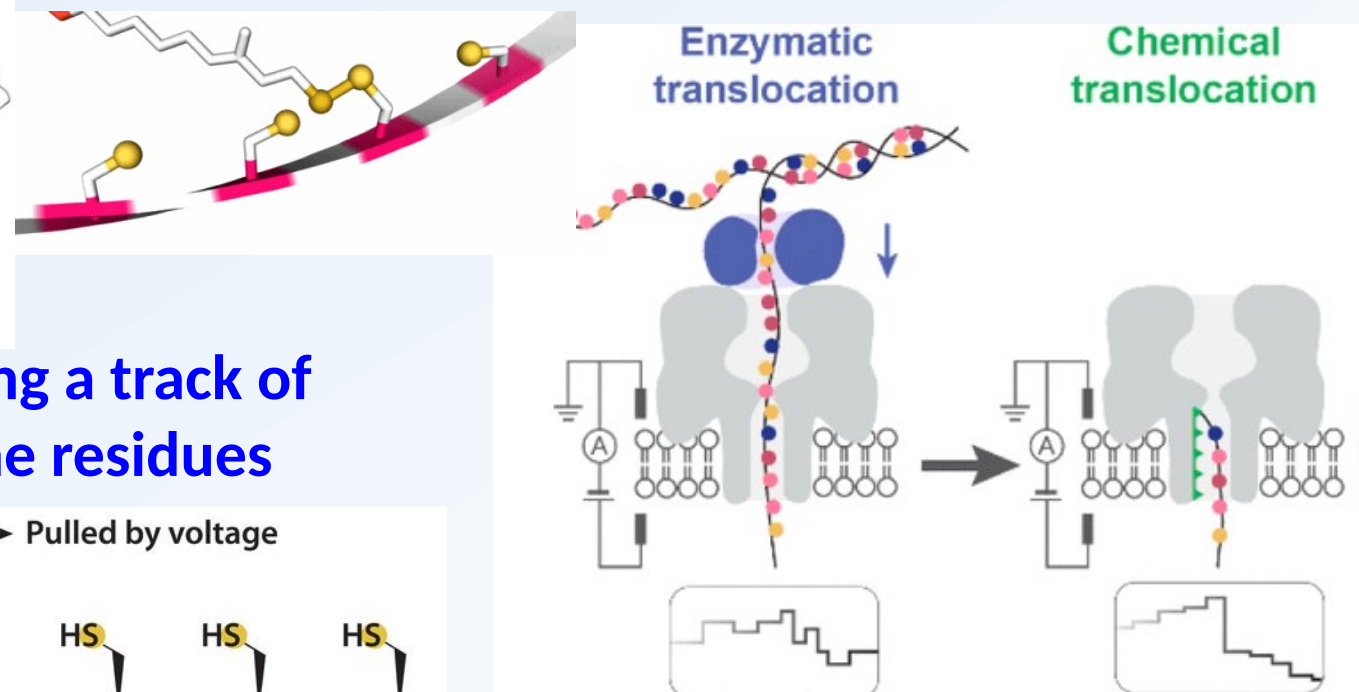
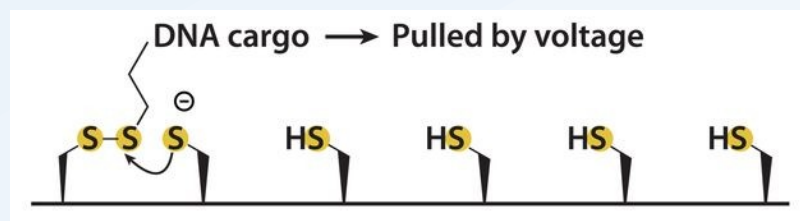


◆ Directional control of a processive molecular hopper



Multi-step directional movement via thiol-disulfide exchange reaction in electric field

Hop along a track of cysteine residues



A multi-step, long-distance directional movement along a given orbit

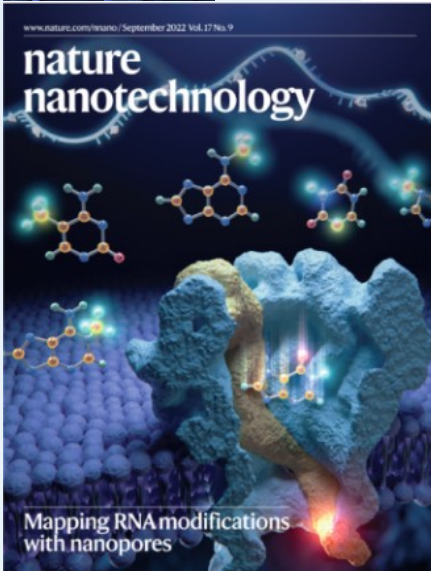
Bayley H., et al, *Science*, 2018, 361, 908.

Bayley H., et al, *J. Am. Chem. Soc.*, 2021, 143, 18181.

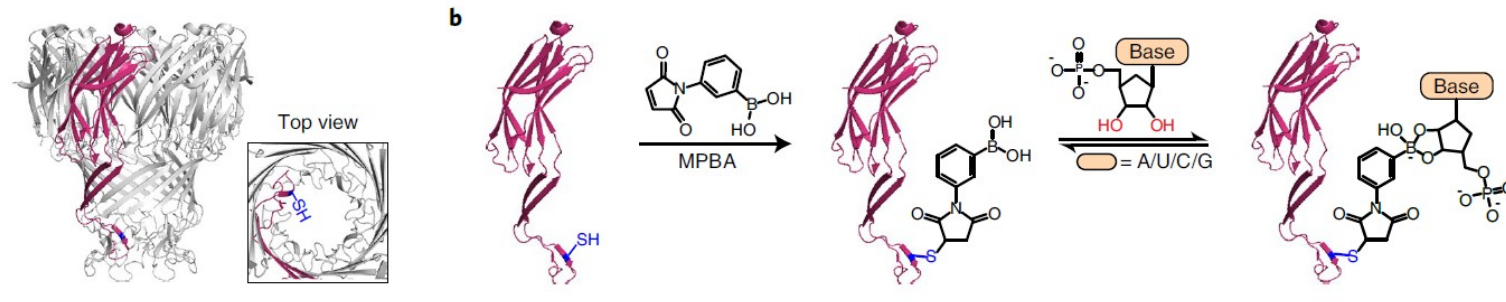
AI for biological analysis



◆ Direct identification of RNA modifications

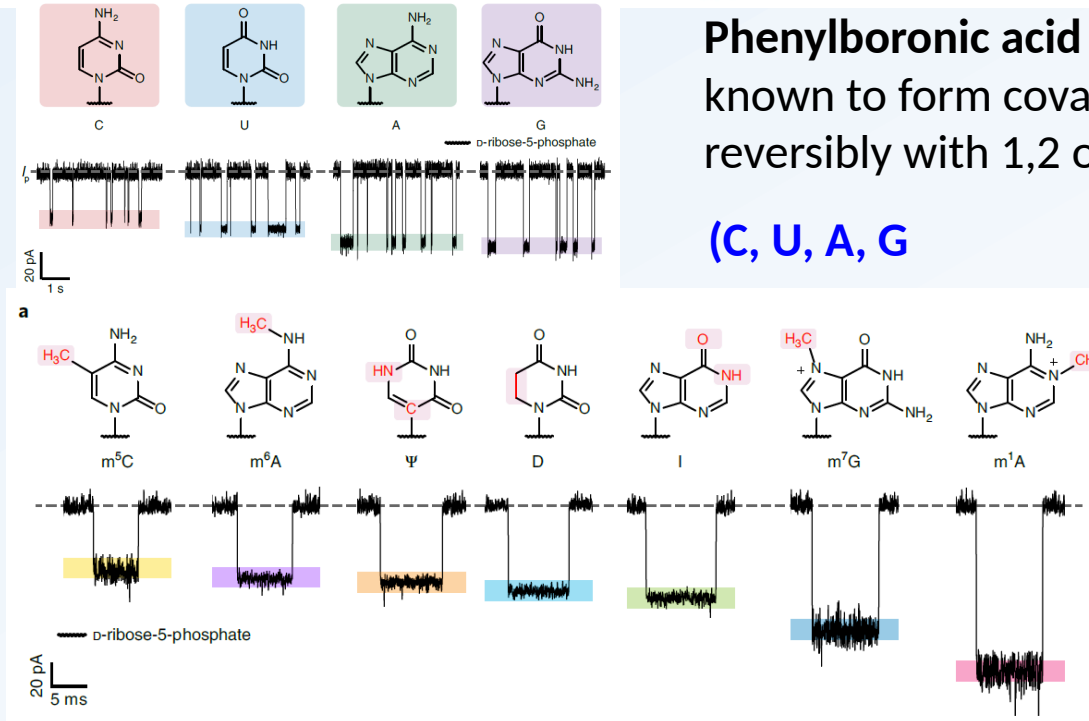


Shuo Huang et al, Nat. Nanotechnol., 2022: 1-8.



Phenylboronic acid (PBA) is known to form covalent bonds reversibly with 1,2 or 1,3-diols

(C, U, A, G)



5- (m⁵C) N6 - (m⁶A) N7 - (m⁷G) N1 - (m¹A) (I) (ψ) (D)

2026/9/11

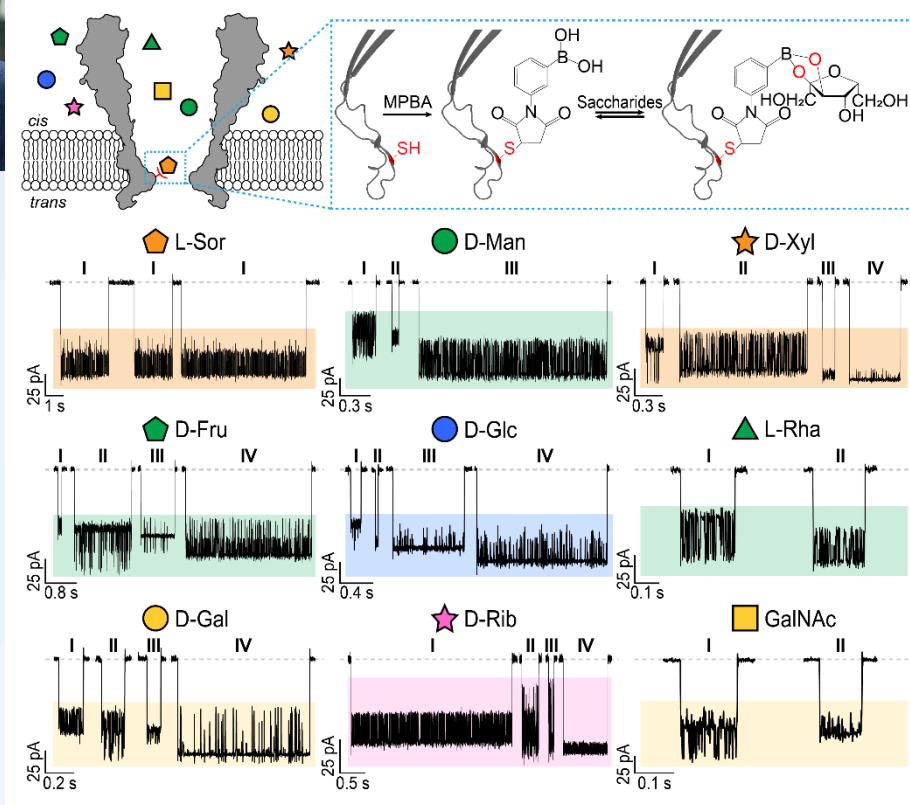
AI for biological analysis



◆ A Nanopore-Based Saccharide Sensor



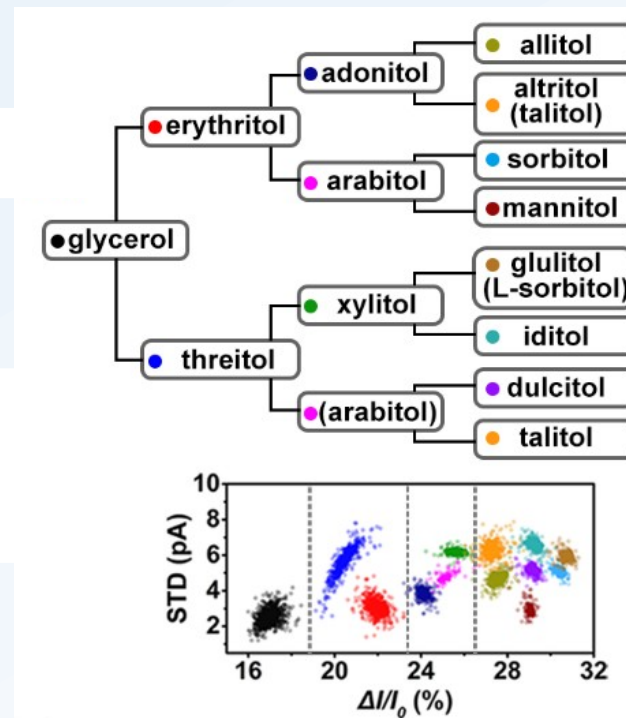
Saccharide structures are however **complicated and similar**, setting a technical hurdle for direct identification



D- N- -D- L- D- D- L-

Shuo Huang et al, *Angew. Chem. Int. Ed.* 2022: e202203769.

2026/9/11



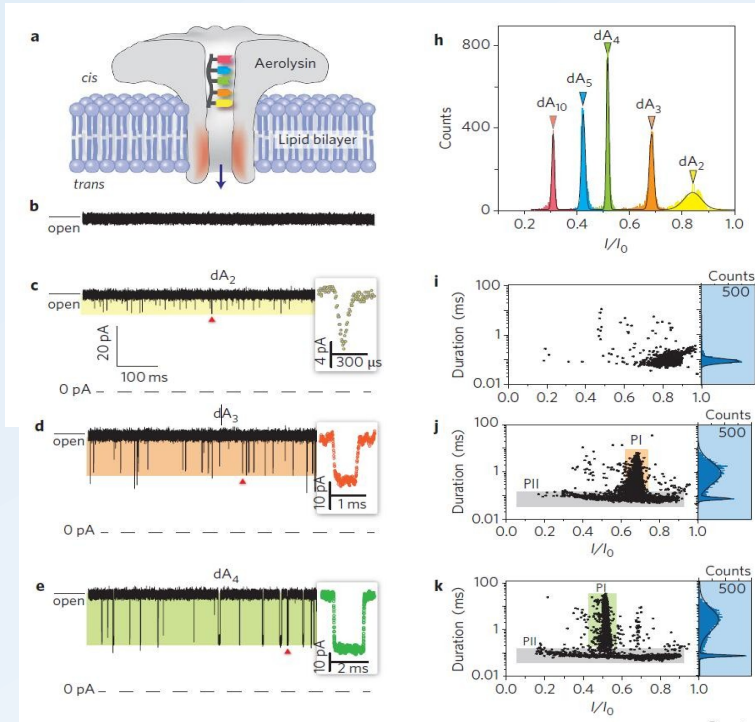
Direct distinguishing of alditols corresponding to all 13 monosaccharide aldoses()

Shuo Huang et al, *JACS*, 2022, 144(30): 13717.

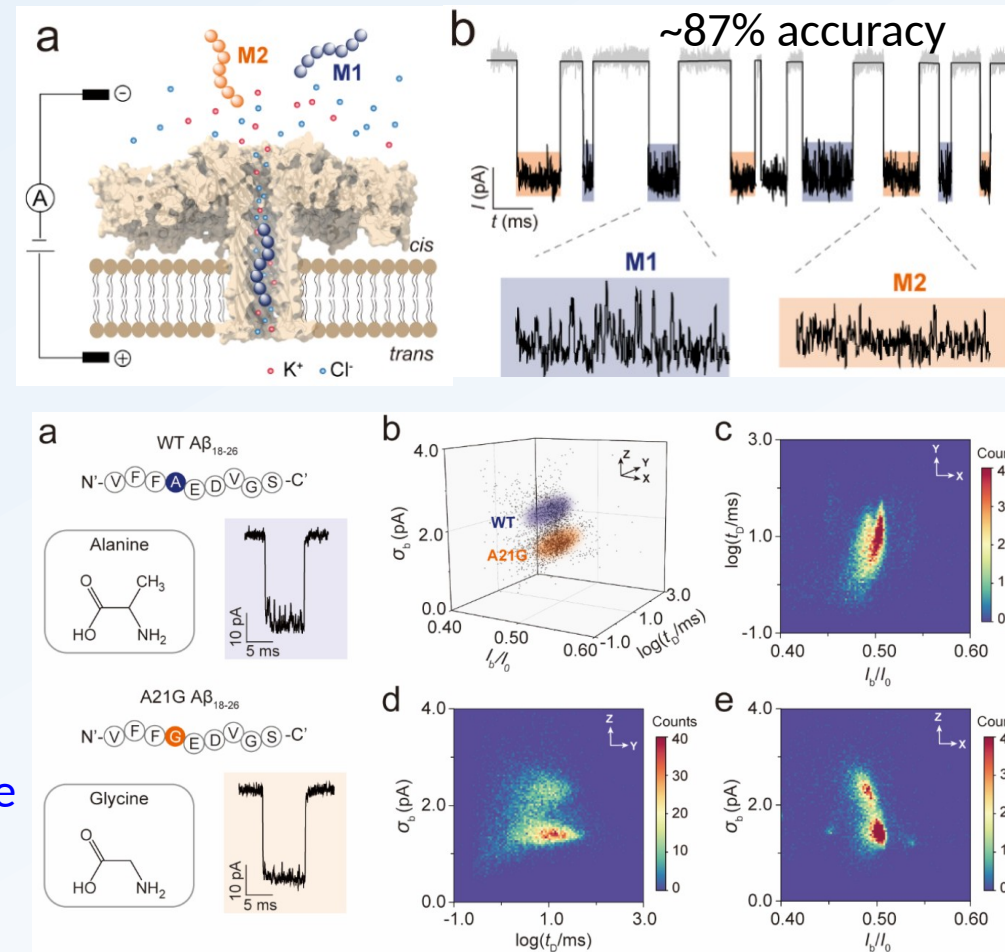
AI for biological analysis



◆ Wild-type aerolysin nanopore



A β peptides with familial point mutations



- The slow translocation of DNA in AeL pores (~2–10 nt/ms) is about **3 orders of magnitude slower**.
- The small constriction of the pore and the highly charged β -barrel walls.

Y-T. Long, et al, *Nat. Nanotechnol.*, 2016, 11, 713.
2026/9/11

Y-T. Long, et al, *Angew. Chem. Int. Ed.* 2022: 10.1002/anie.202209970

◆ Recognition of amino acid using AeL nanopore

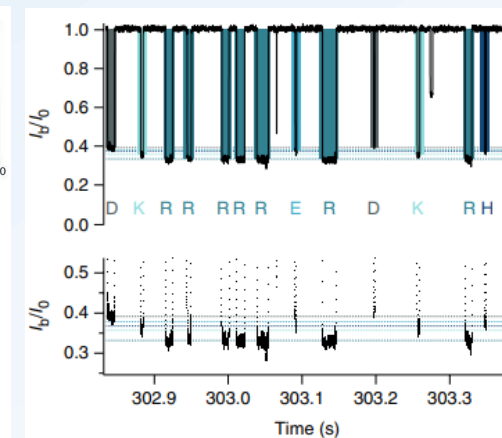
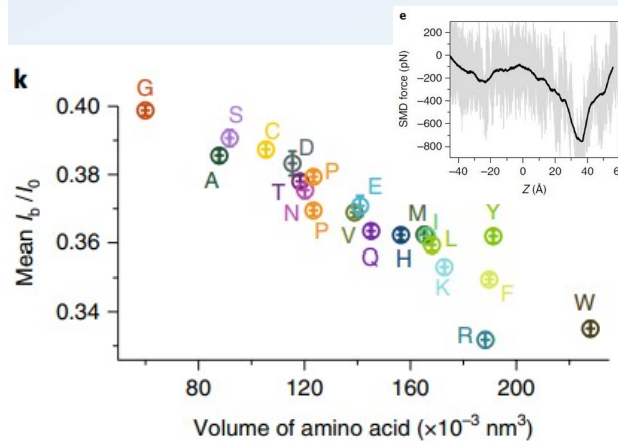
a

b

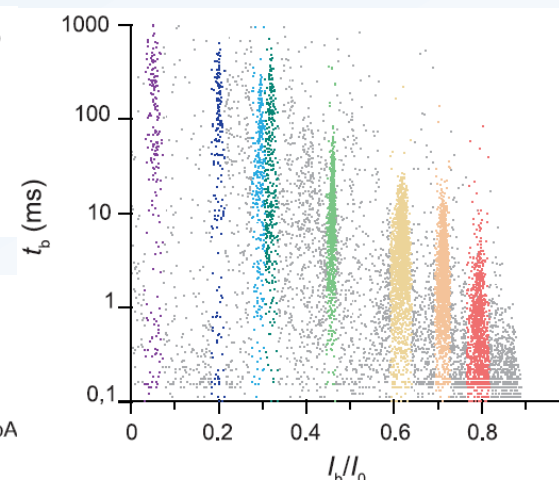
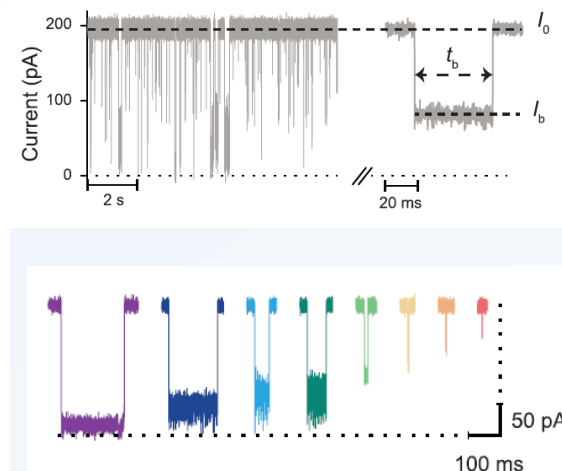
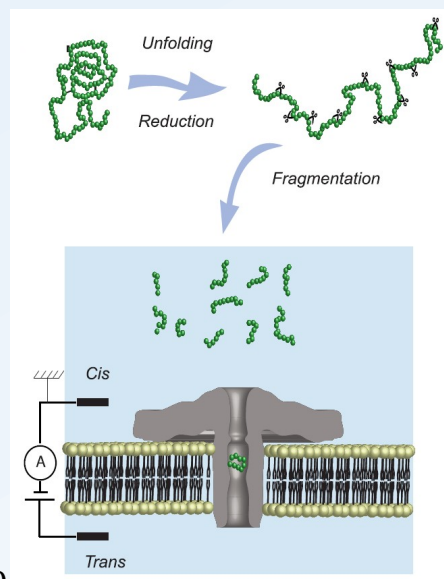
Cis

Trans

X_{R7}



Abdelghani O., et al, Nat. Biotechnol., 2020, 38, 176.



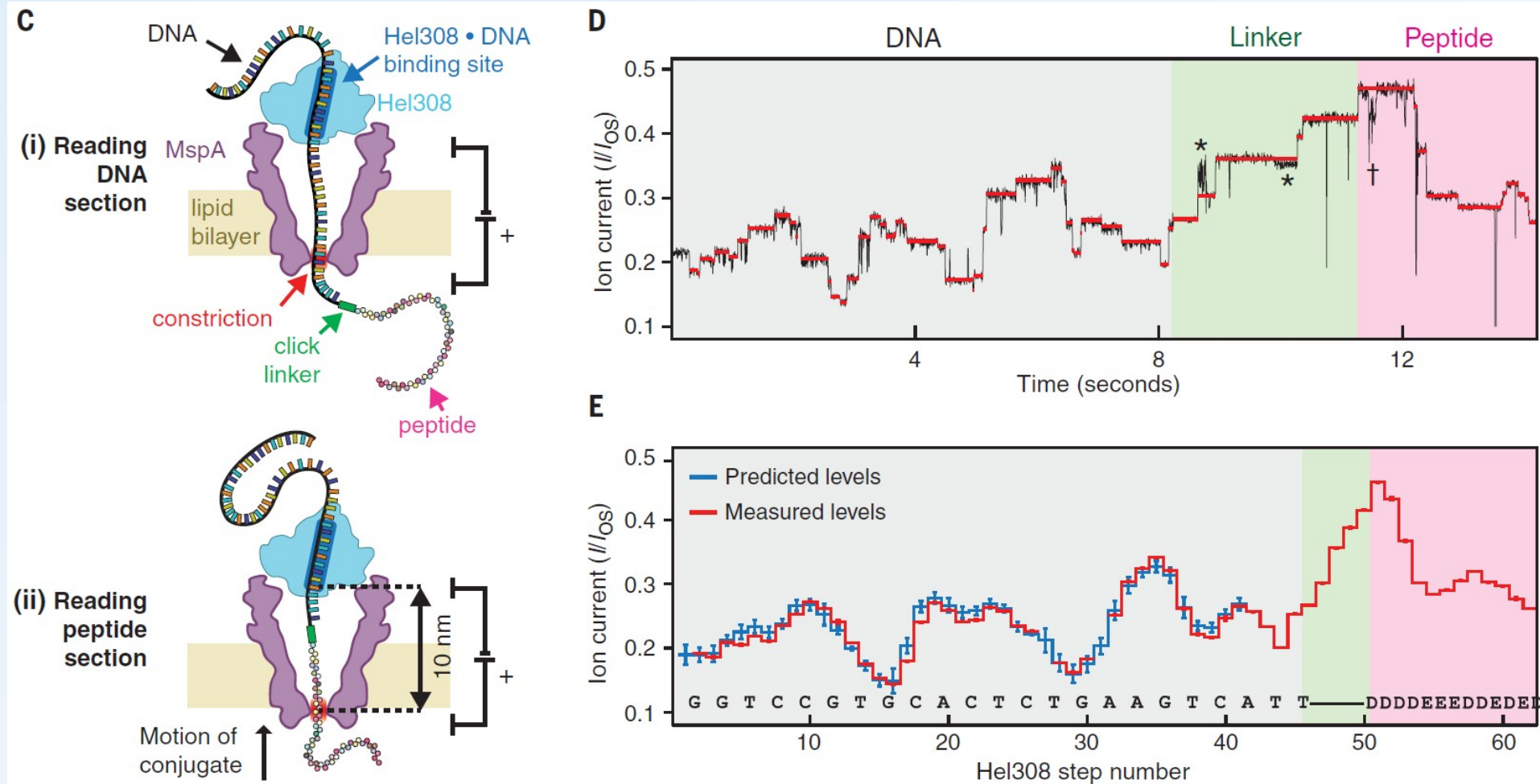
Protein fingerprinting at the single-molecule level

Oukhaled, A., et al, JACS, 2022, 144(6): 2716.

AI for biological analysis



◆ Single-amino acid resolution using nanopores

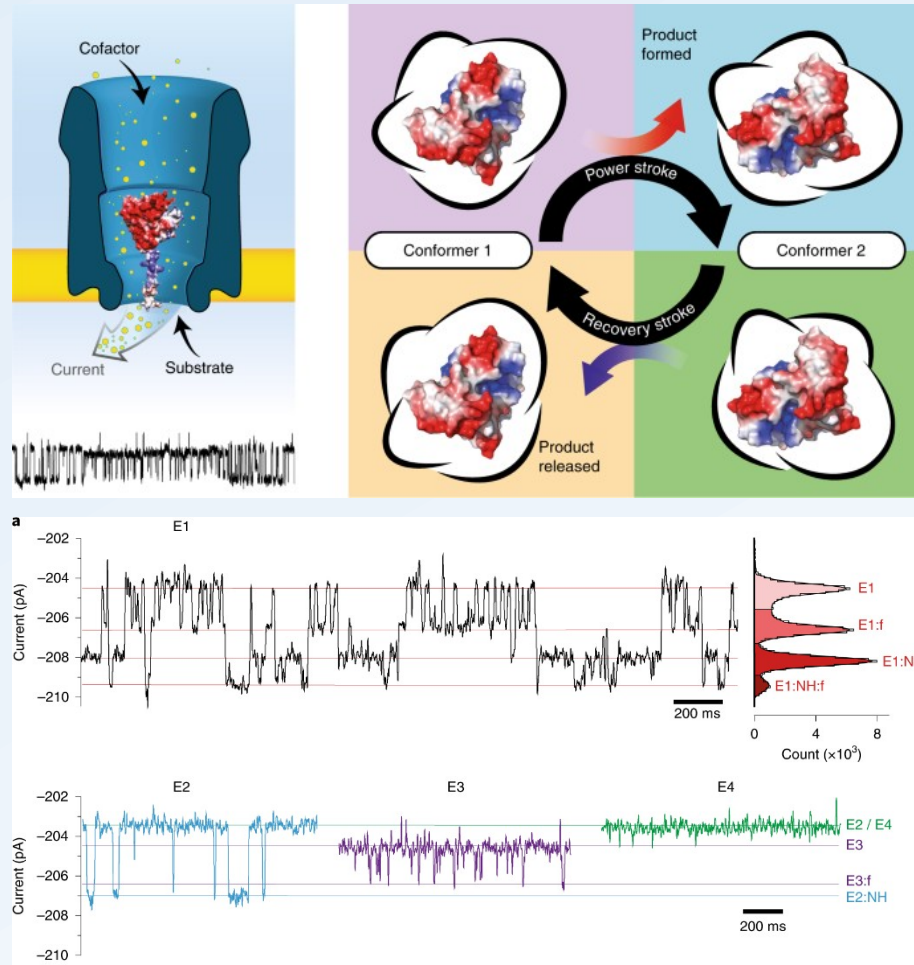


AI for biological analysis

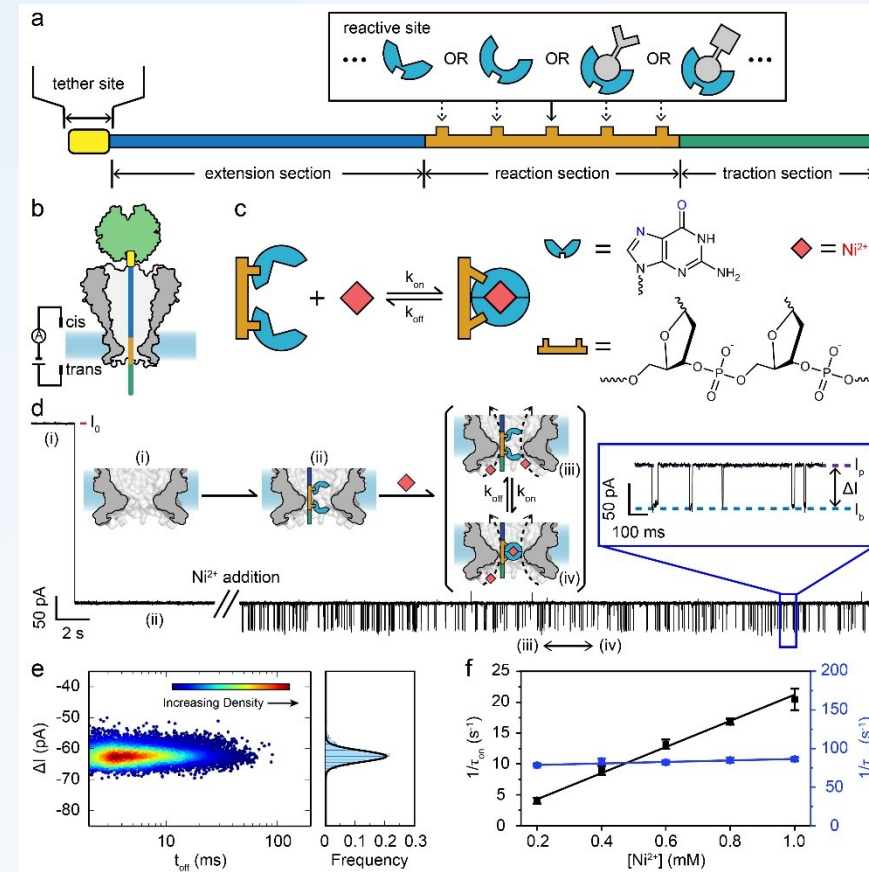


◆ Single-molecule dynamics

ligand-induced conformational changes of dihydrofolate reductase (DHFR)



Programmable Nano-Reactors for Stochastic Sensing



20 monomolecular chemical reactions

Contents

1. Overview of biological analysis

2. **AI for biological analysis**

- Nanopore
- **Structure biology**
- **Bioinformatics**
- Single-cell
-

4. Data preprocessing

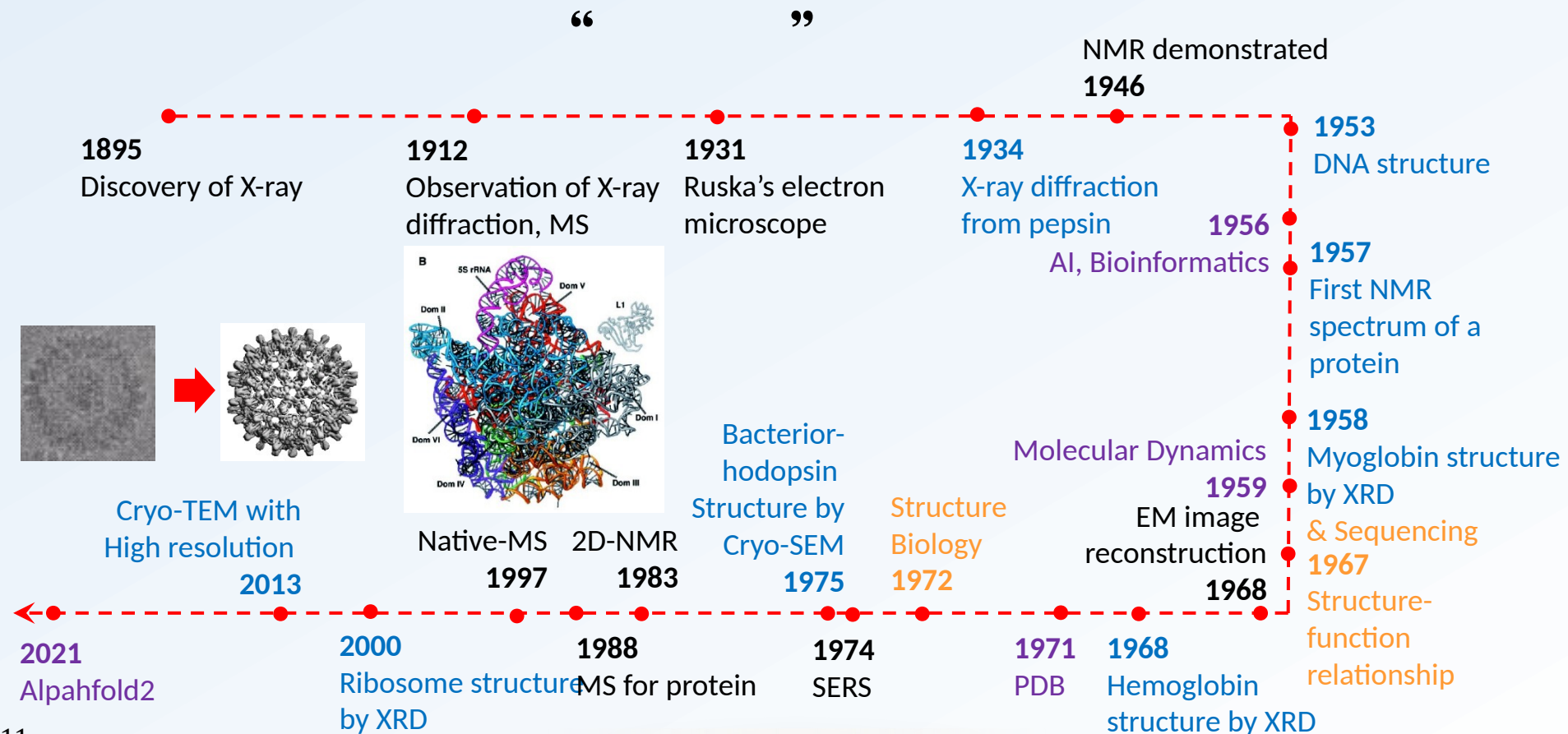
5. Code

AI for biological analysis



◆ Introduction of structure biology

Structural biology is the study of how biological molecules are built. Using a variety of imaging techniques, scientists view molecules in **three dimensions** to see how they are assembled, how they function, and how they interact.

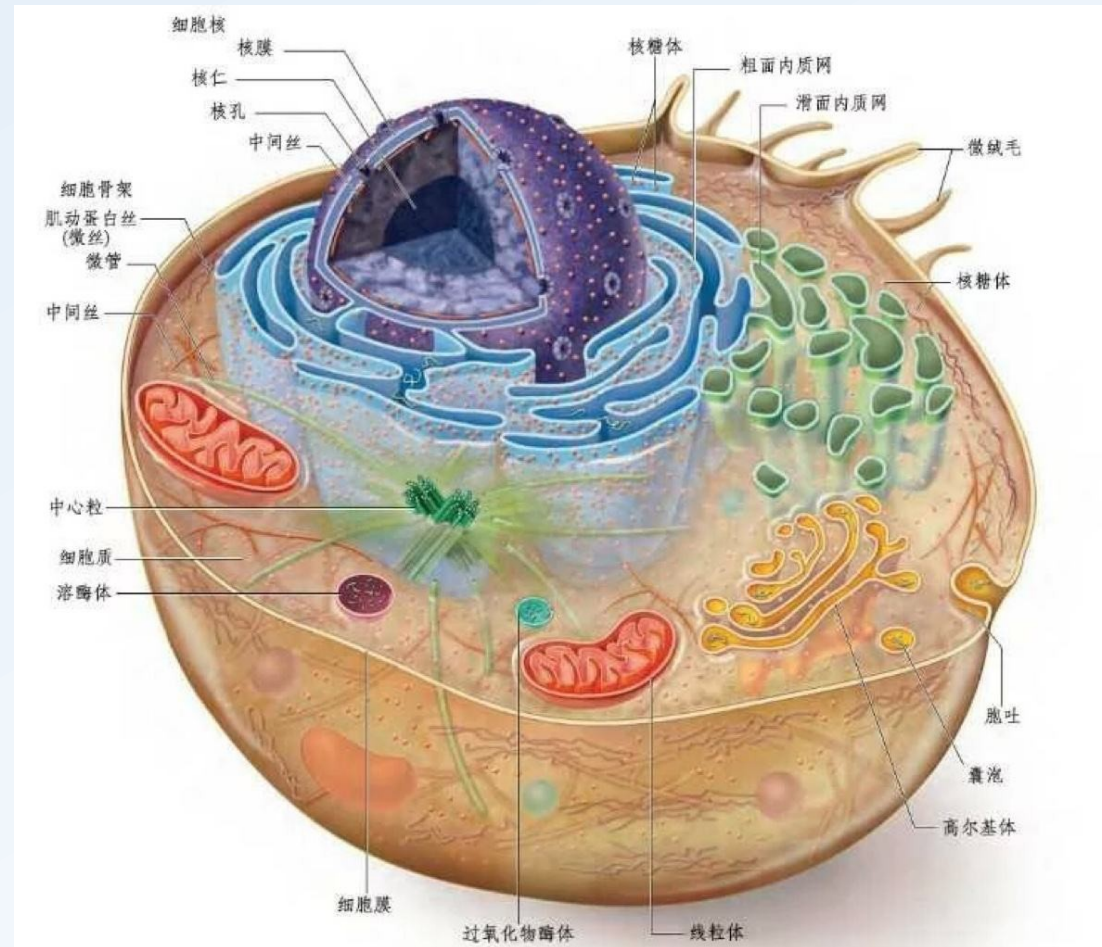


AI for biological analysis



◆ Introduction of structure biology: Molecules

Protein, Nucleic Acid, Lipid, Polysaccharide

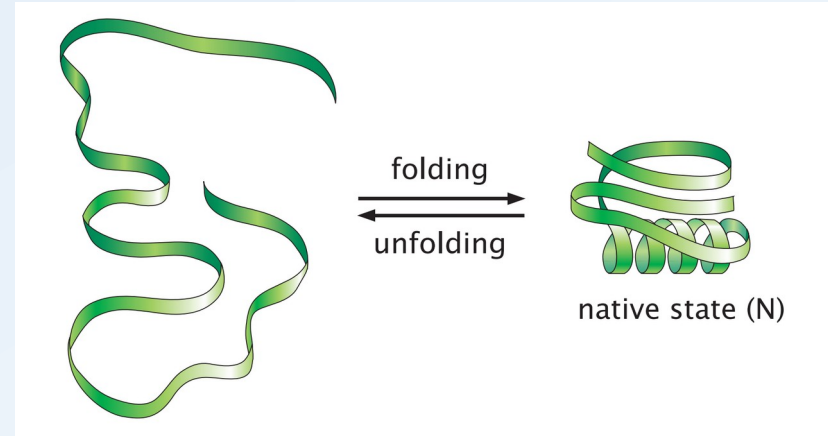


AI for biological analysis



◆ Introduction of structure biology: Molecules

Protein



The central paradigm of protein science has long been that the amino acid sequence dictates the 3D structure, which, in turn, dictates how the protein performs its function:

Sequence → Structure → Function



The biological mechanism of action of a protein can depend not only on its structure, but also on its *dynamics*. So, a more accurate statement of the protein paradigm is:

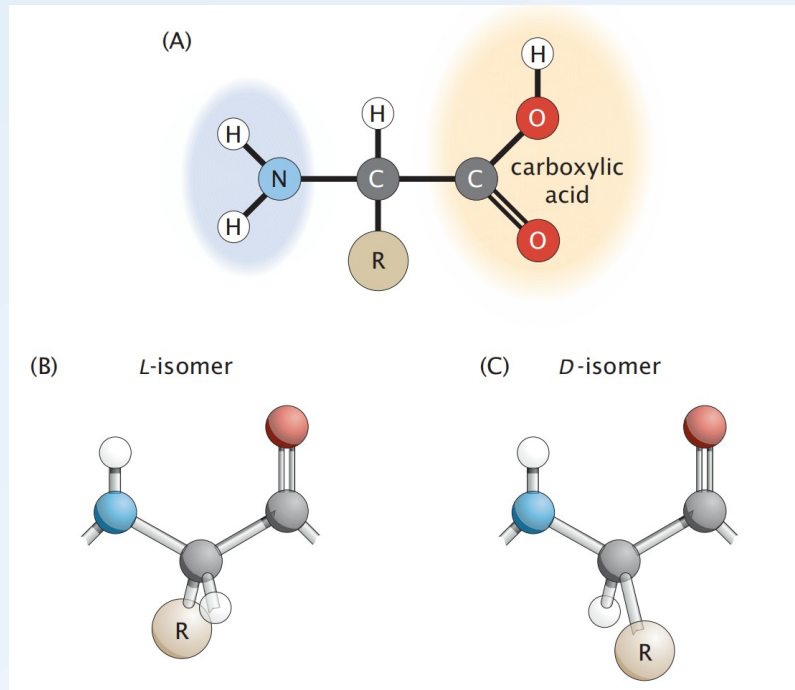
Sequence → Structures → Motions → Function

AI for biological analysis

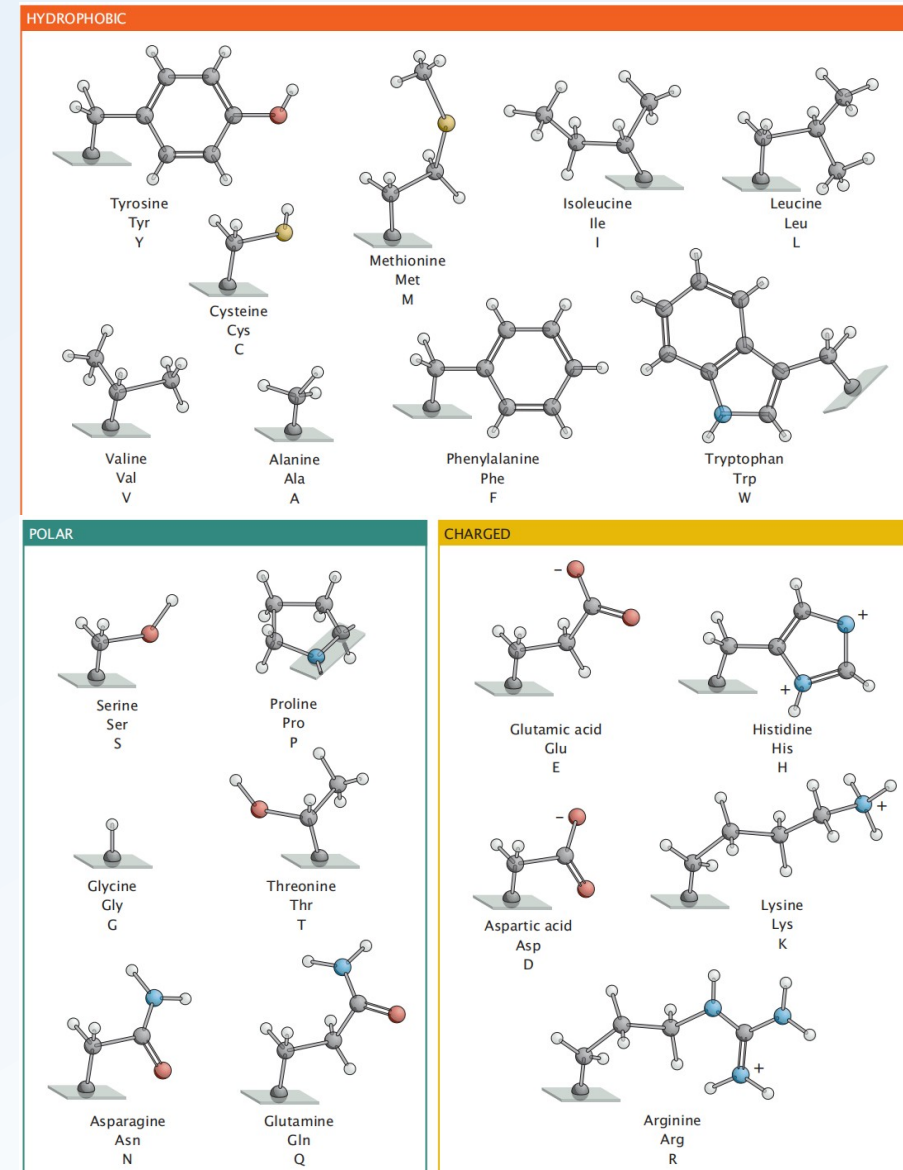


◆ Introduction of structure biology: Molecules

Sequence:



Amino acids are the repeat units of proteins;
each has a different chemical character:
e.g. charged, polar, nonpolar.....

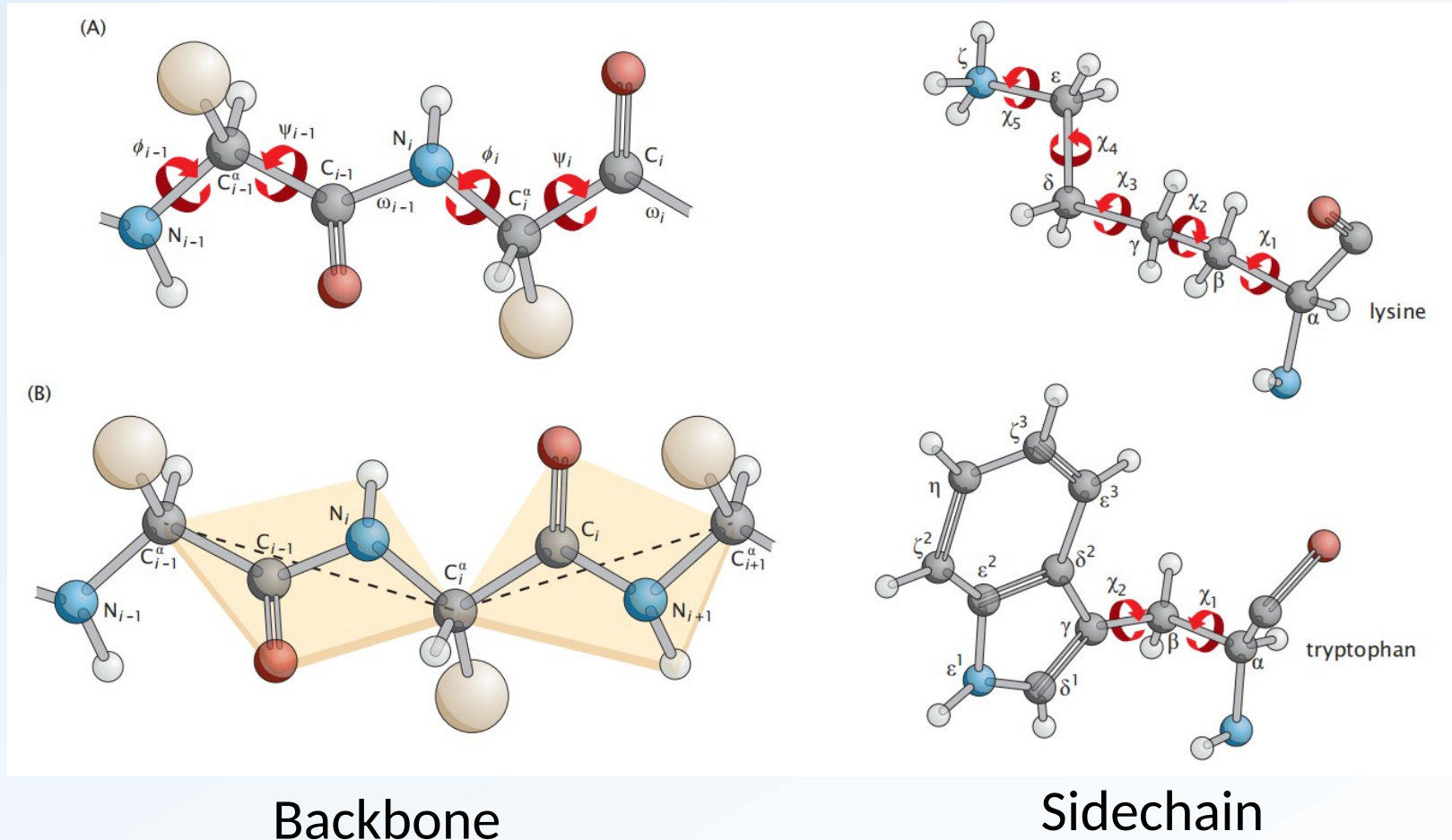


AI for biological analysis



◆ Introduction of structure biology: Molecules

Amino acids are linked together through peptide bonds.

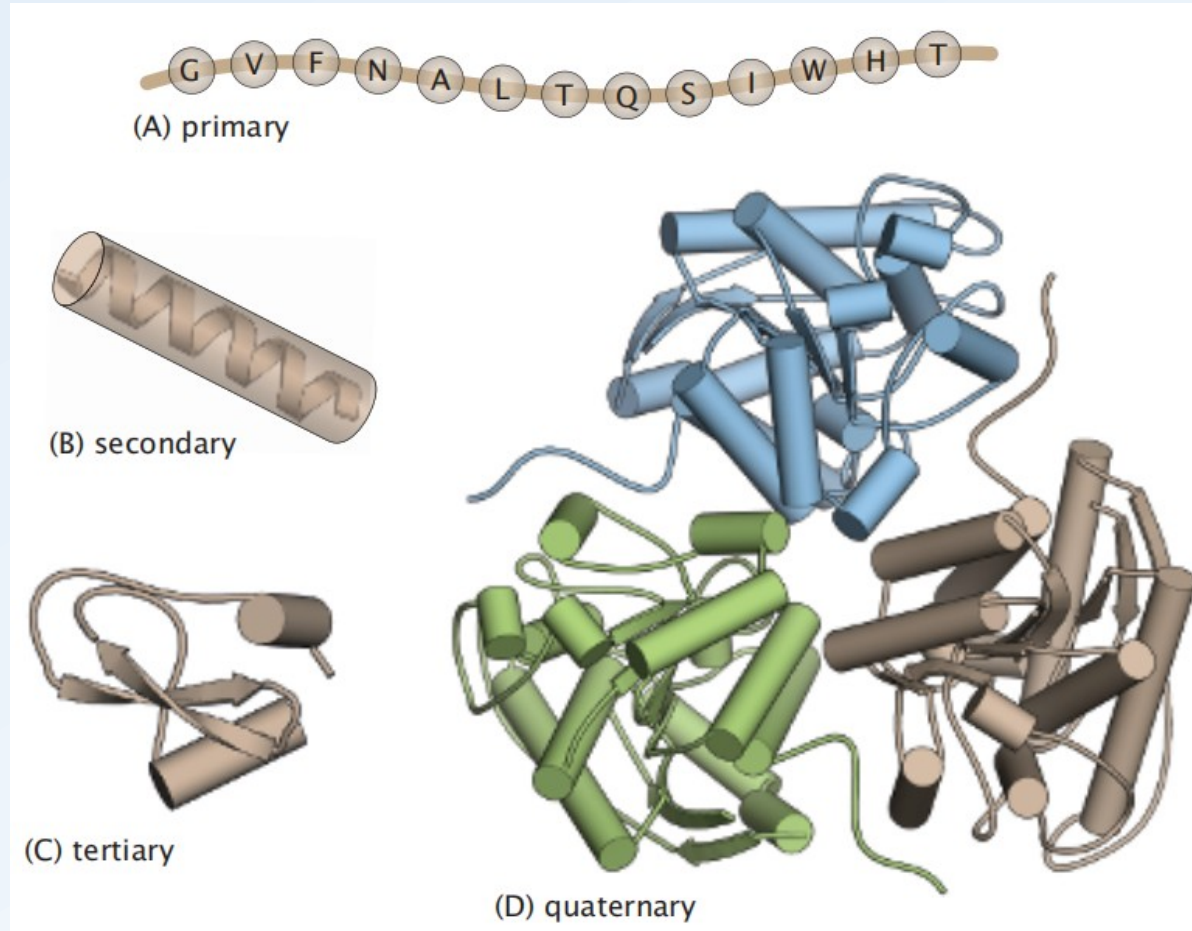


AI for biological analysis



◆ Introduction of structure biology: Molecules

Four hierarchical levels of protein structure:



α -helices, β -sheets, π -helix,
polyproline II helix, *turns and loops*

Quaternary Structures: Higher-Order Structures Result from Noncovalent Assemblies of Multiple Chains

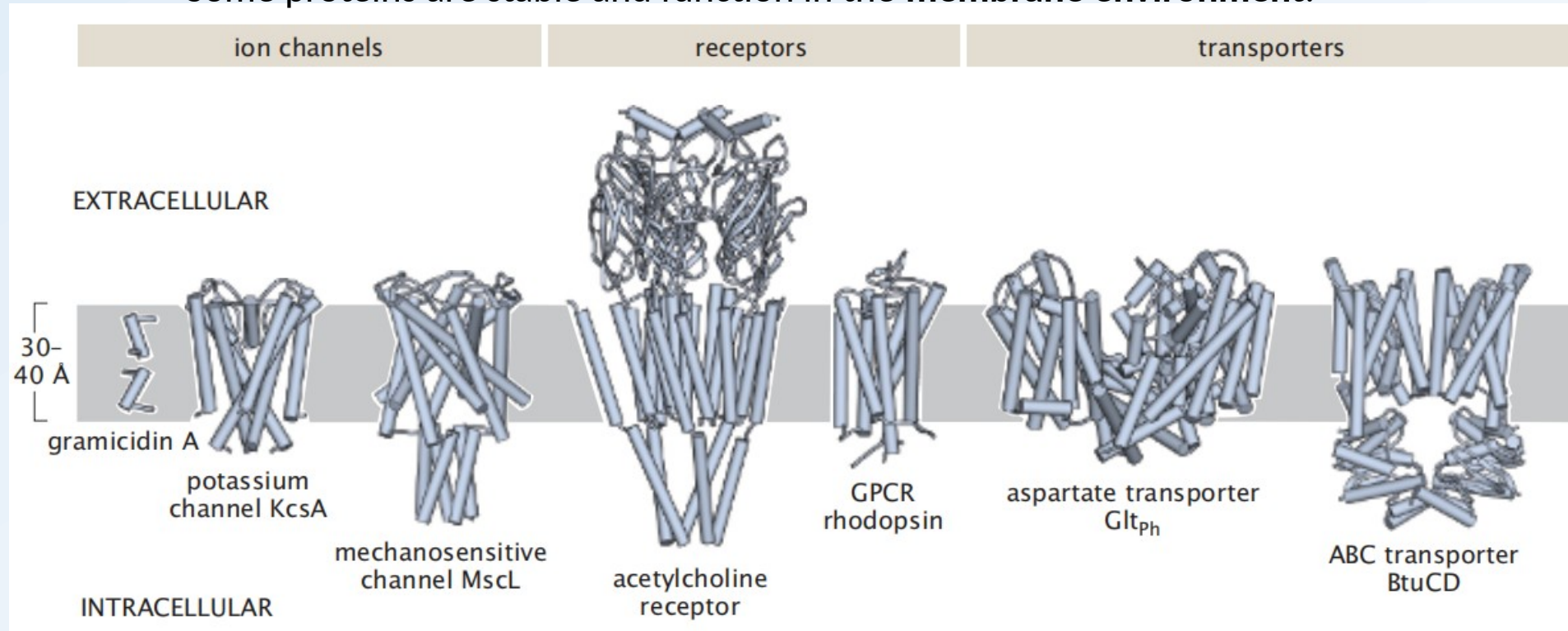
2026/9/11

AI for biological analysis



◆ Introduction of structure biology: Molecules

Some proteins are stable and function in the **membrane environment**.



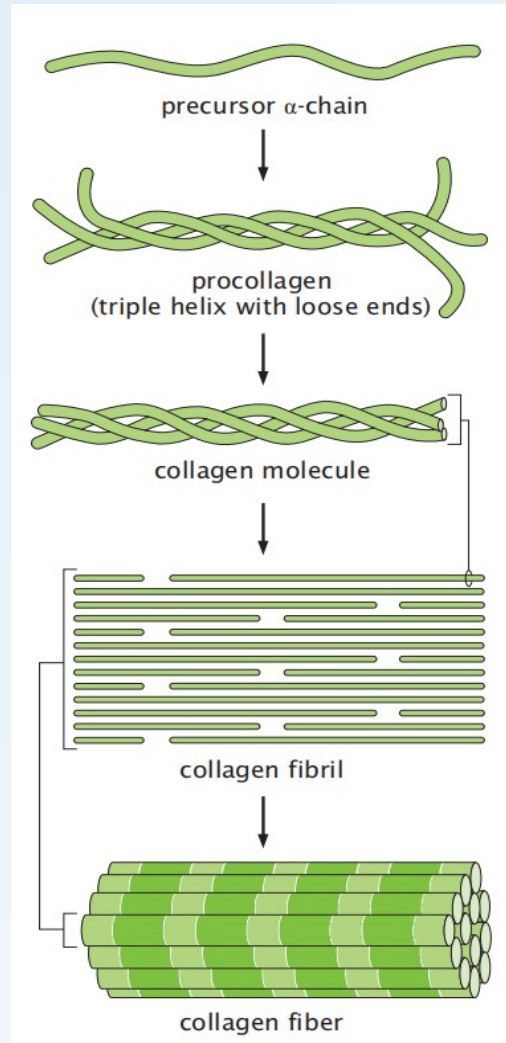
A membrane protein usually has three regions: transmembrane (TM), extracellular, and intracellular. The membrane is typically a lipid bilayer, which is a sandwich of two layers of lipid molecules.

AI for biological analysis

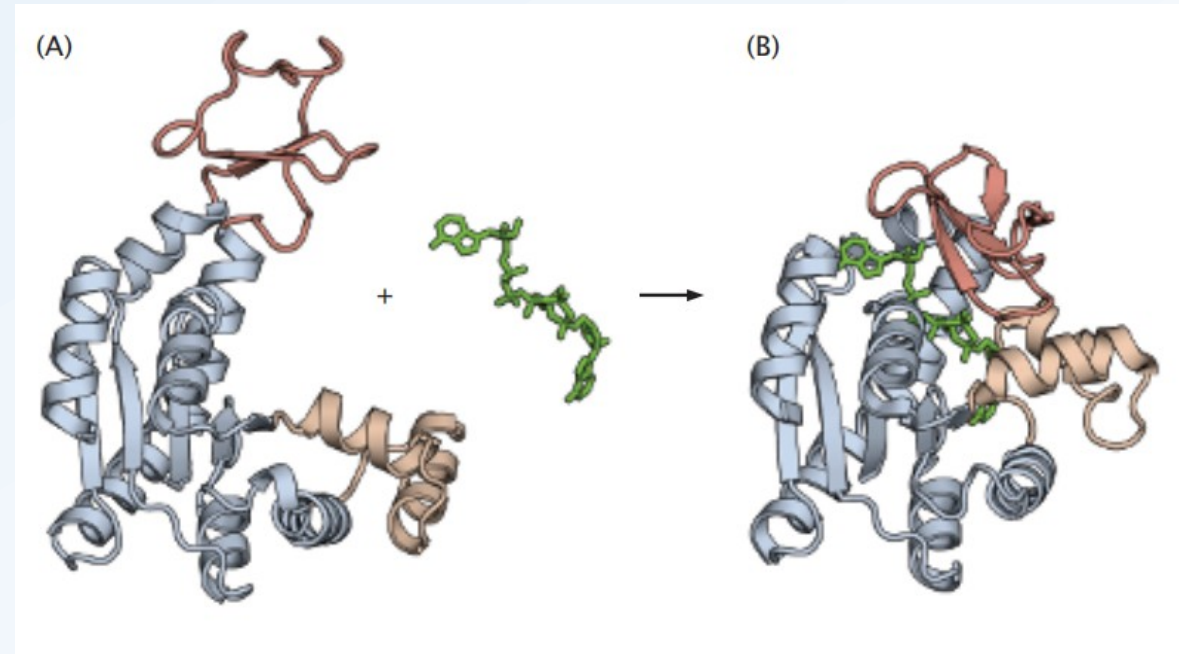


◆ Introduction of structure biology: Molecules

Some proteins have fibrous structures:



Native proteins are conformational ensembles.



AI for biological analysis

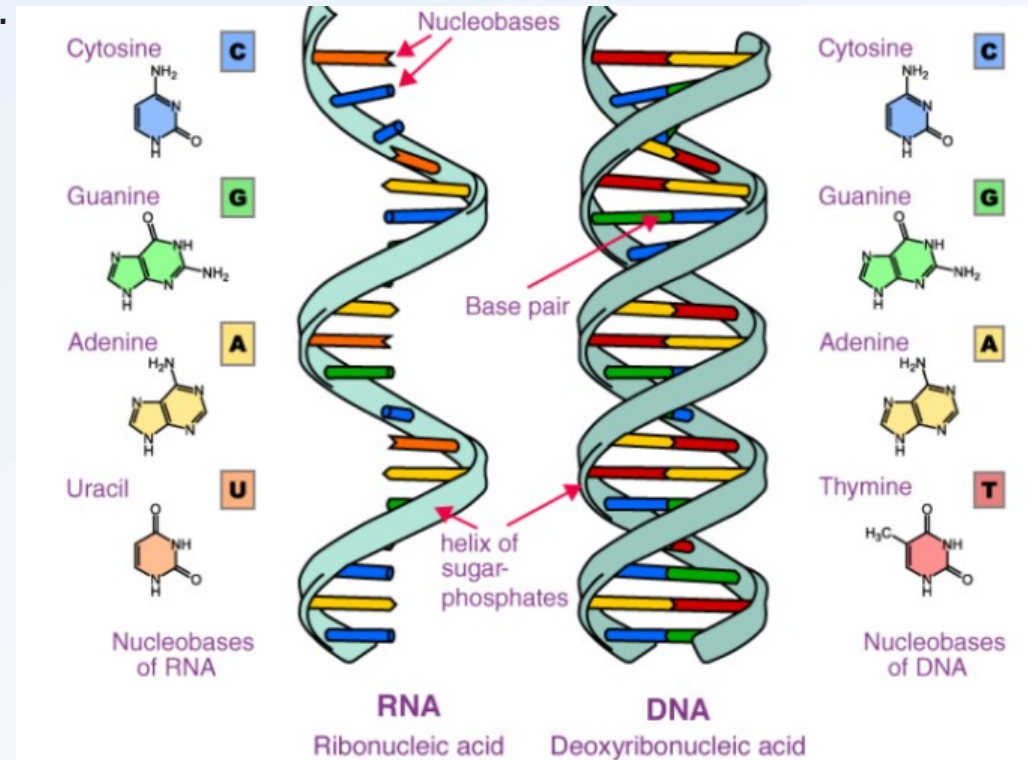


◆ Introduction of structure biology: Molecules

Nucleic Acid

Ribonucleic acid (RNA) and deoxyribonucleic acid (DNA)

Nucleic acid is an important class of macromolecules found in all cells and viruses. The functions of nucleic acids have to do with the storage and expression of genetic information. DNA encodes the information the cell needs to make proteins.

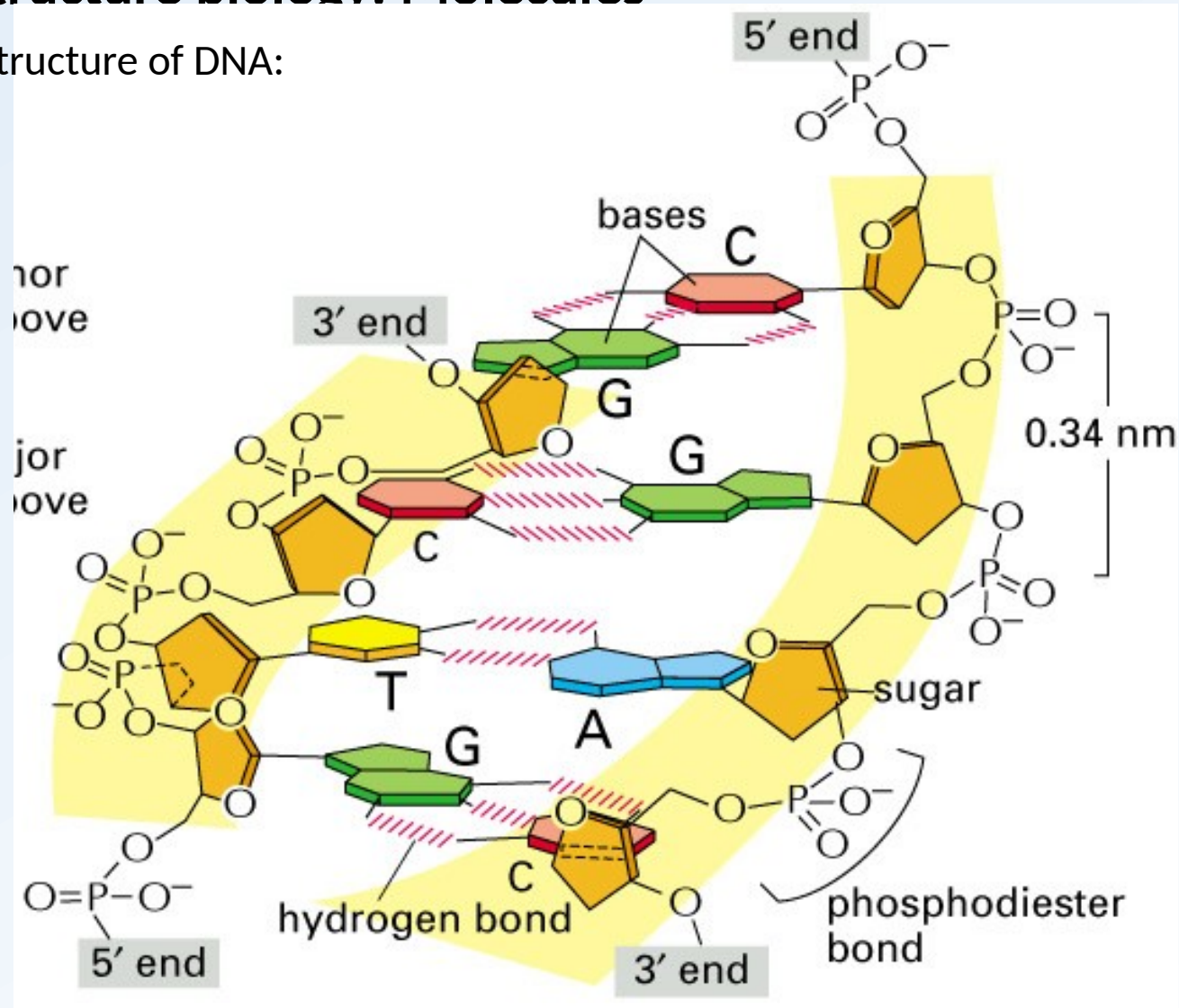


AI for biological analysis



◆ Introduction of structure biology: Molecules

Secondary structure of DNA:

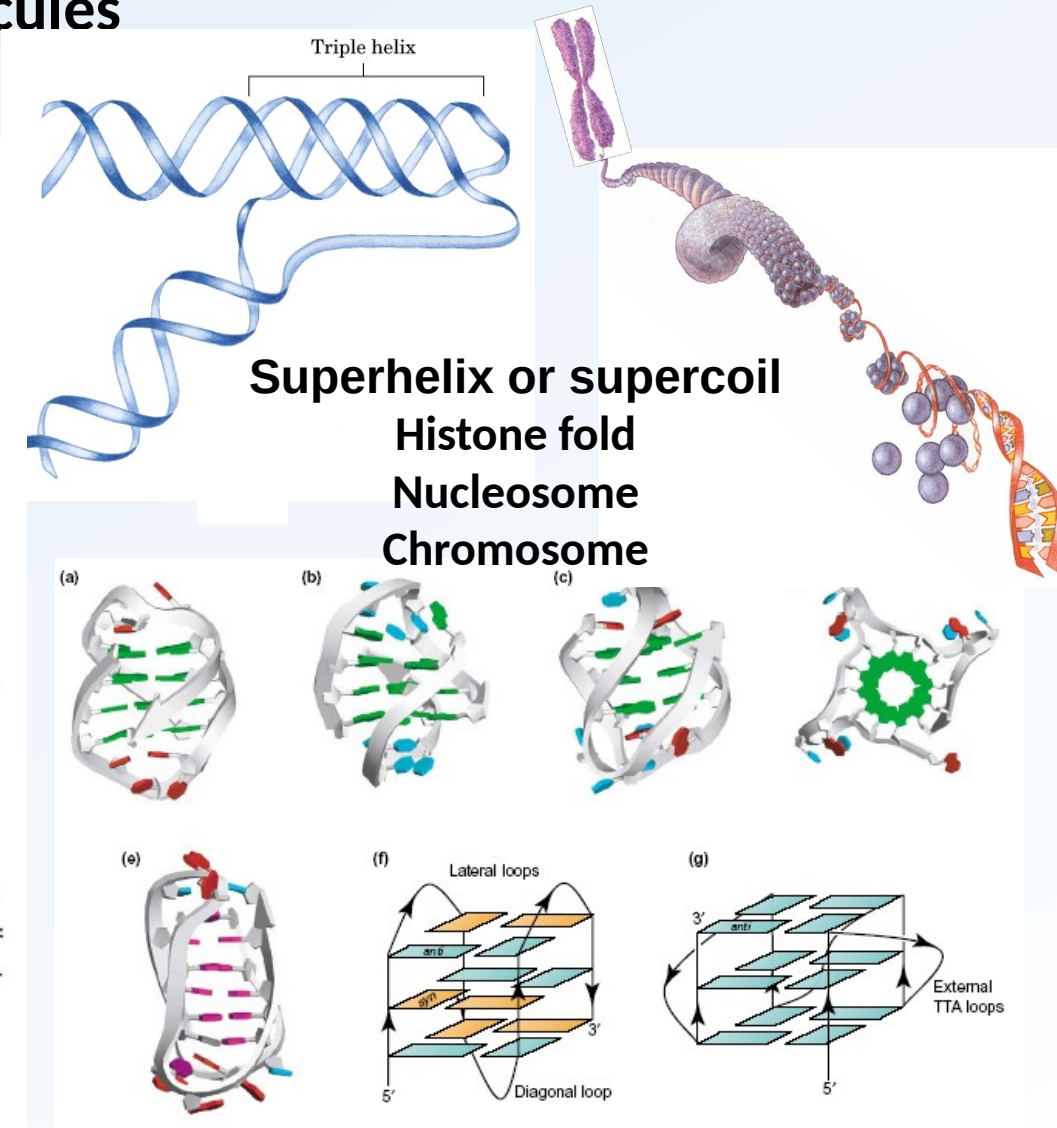
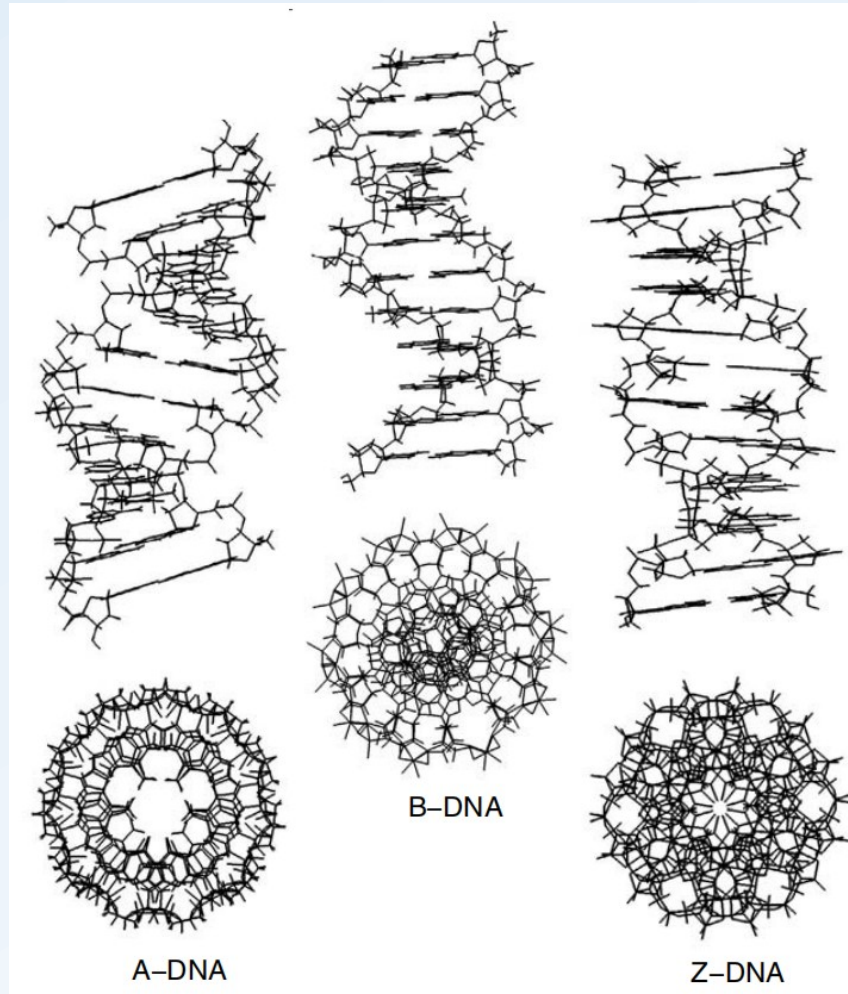


AI for biological analysis



◆ Introduction of structure biology: Molecules

Different DNA structures:



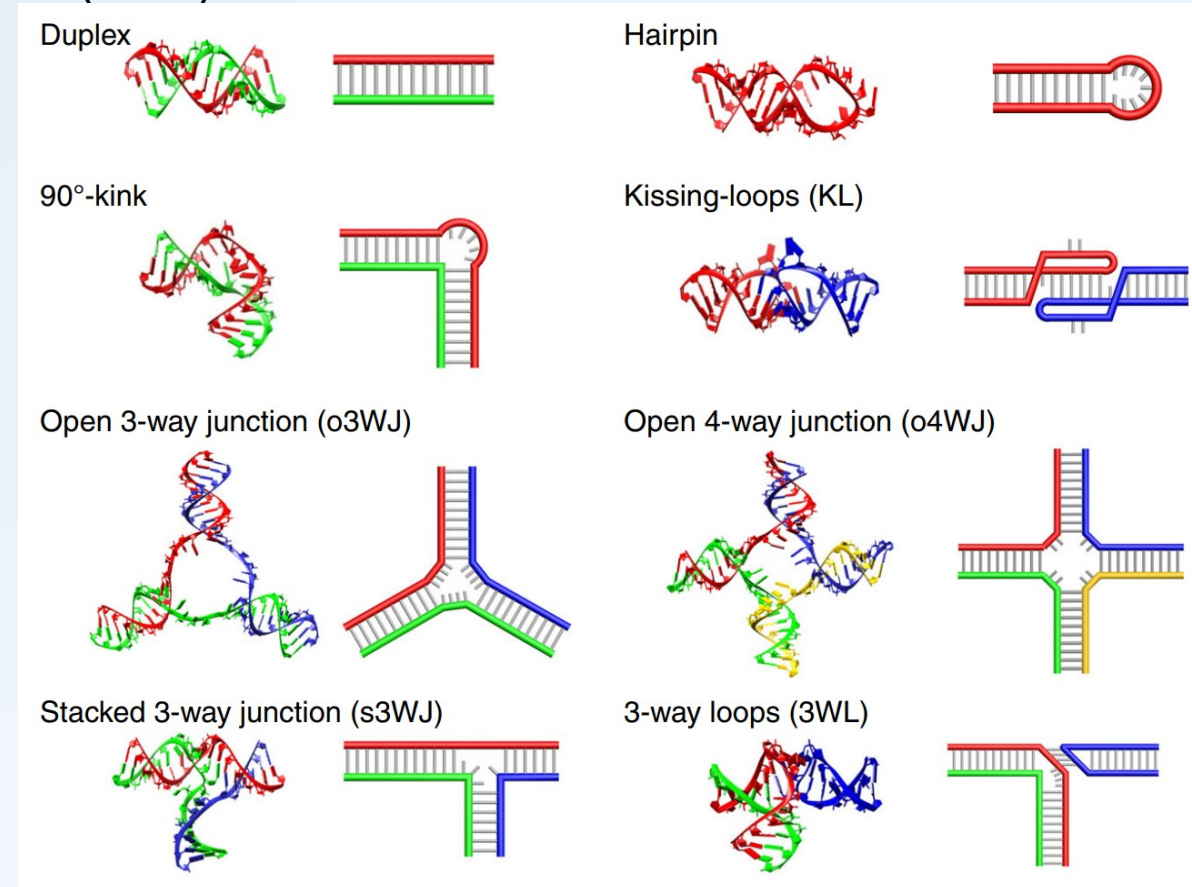
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◆ Introduction of structure biology: Molecules

Differences from DNA: Non-Watson-Crick pair...

Structure of RNA (motif):

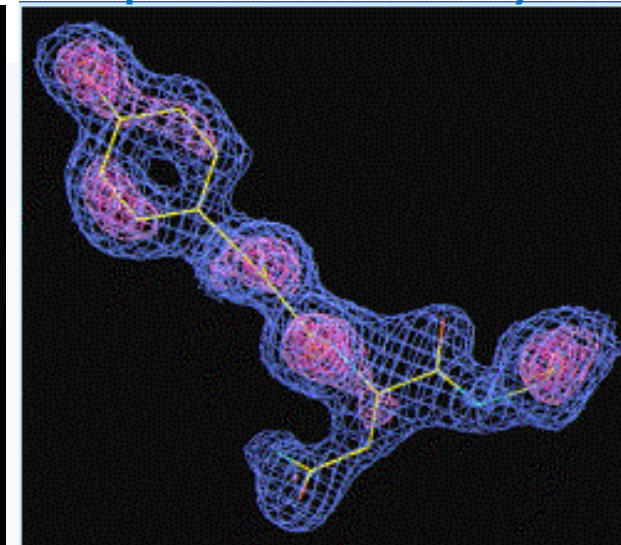
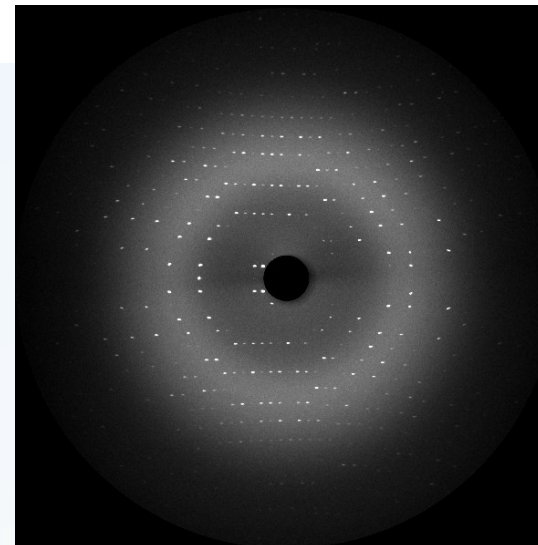
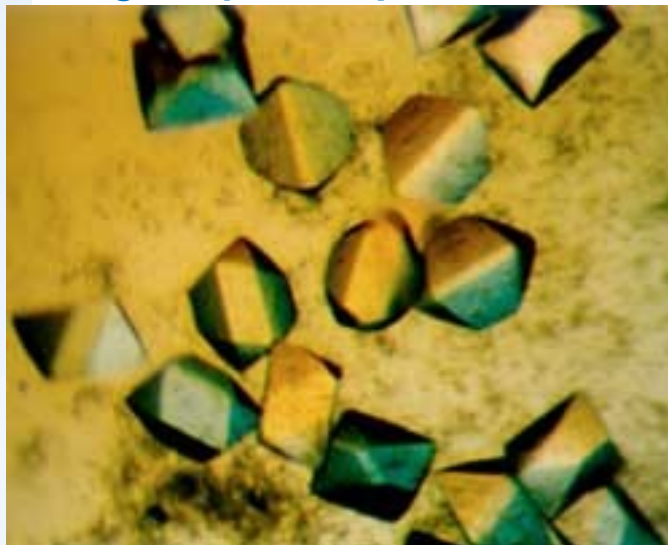
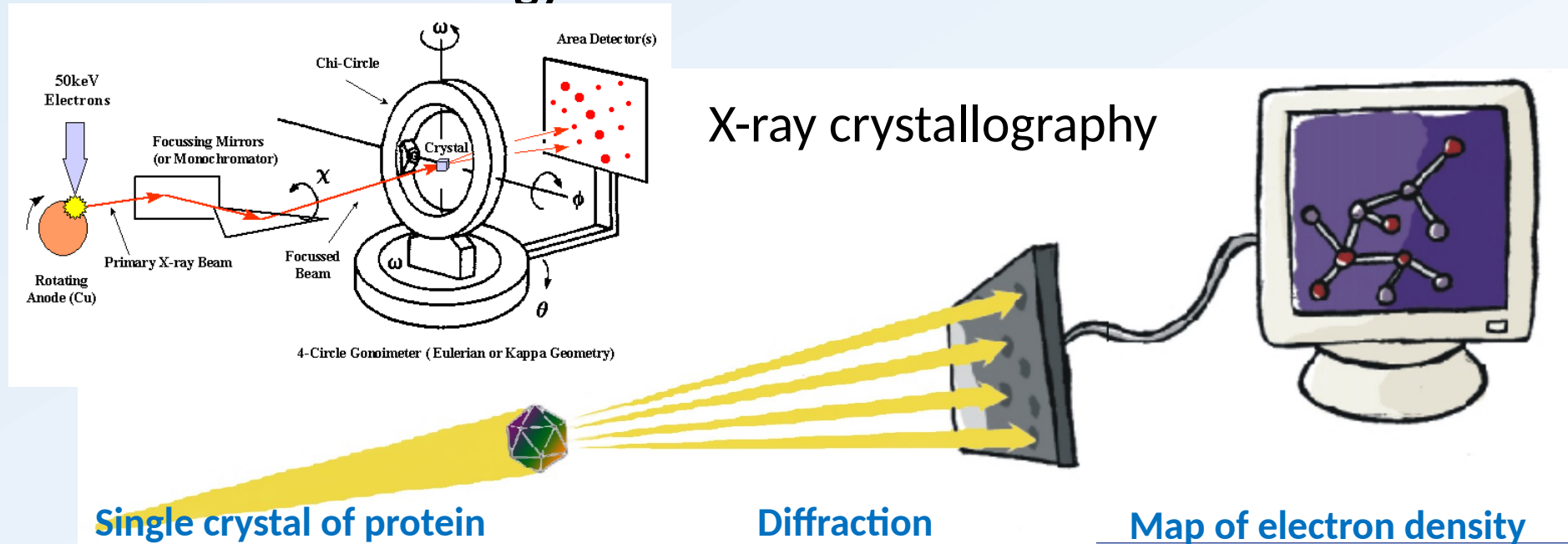


Nat Commun, **2018**, 9, 2196.

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◆ Introduction of structure biology: Methods



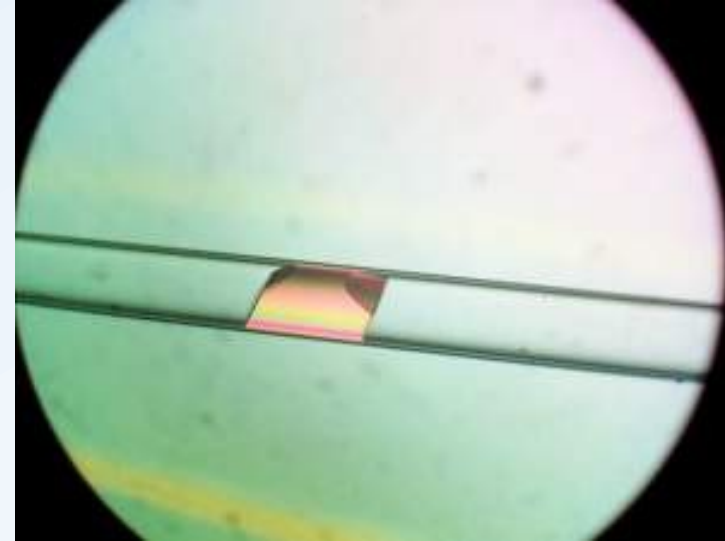
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◆ Introduction of structure biology: Methods

Factors that influence the result:

- Purity and homogeneity
- pH
- Salt and solvent
- Temperature
- Concentration 5-30mg/ml
- Ions
- Shake



Quality of Crystal needed for different instrument:

- 0.1~0.5 mm for normal XRD
- 20~40 μm for synchrotron radiation

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◆ Introduction of structure biology: Methods

Shortcomings in the early time:

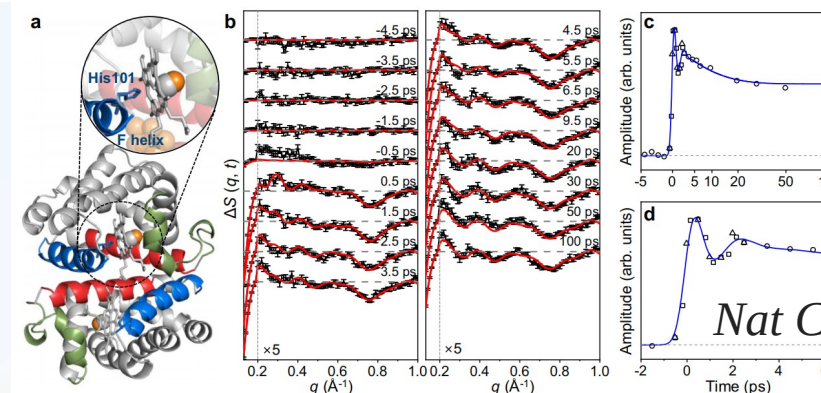
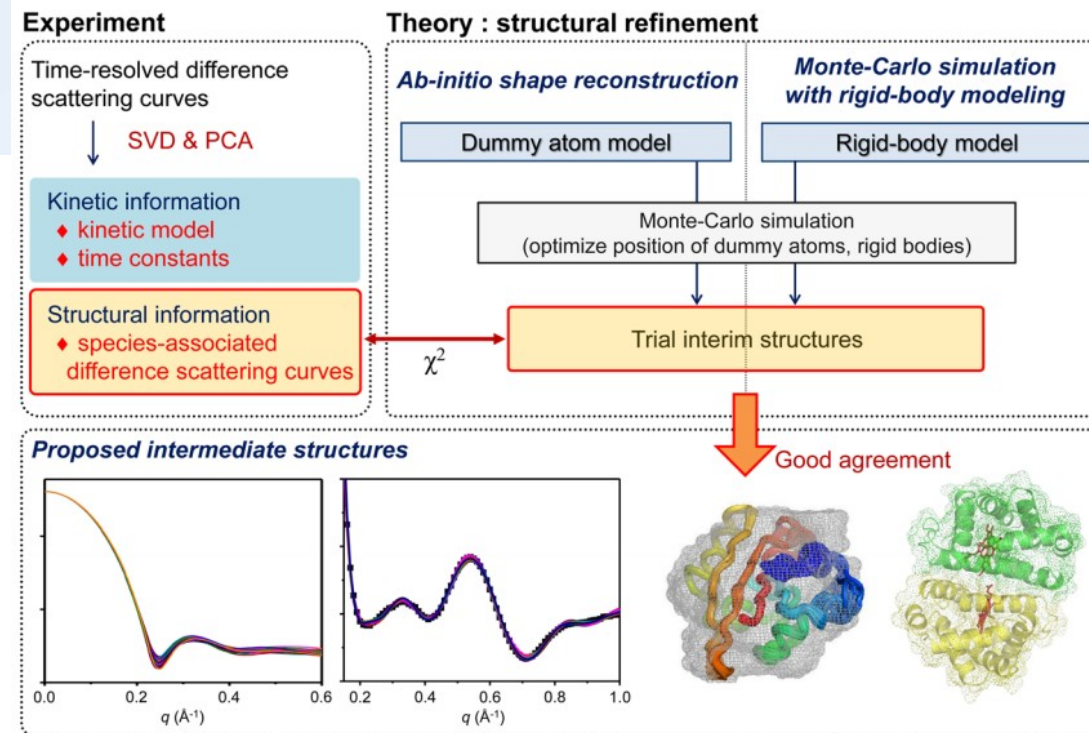
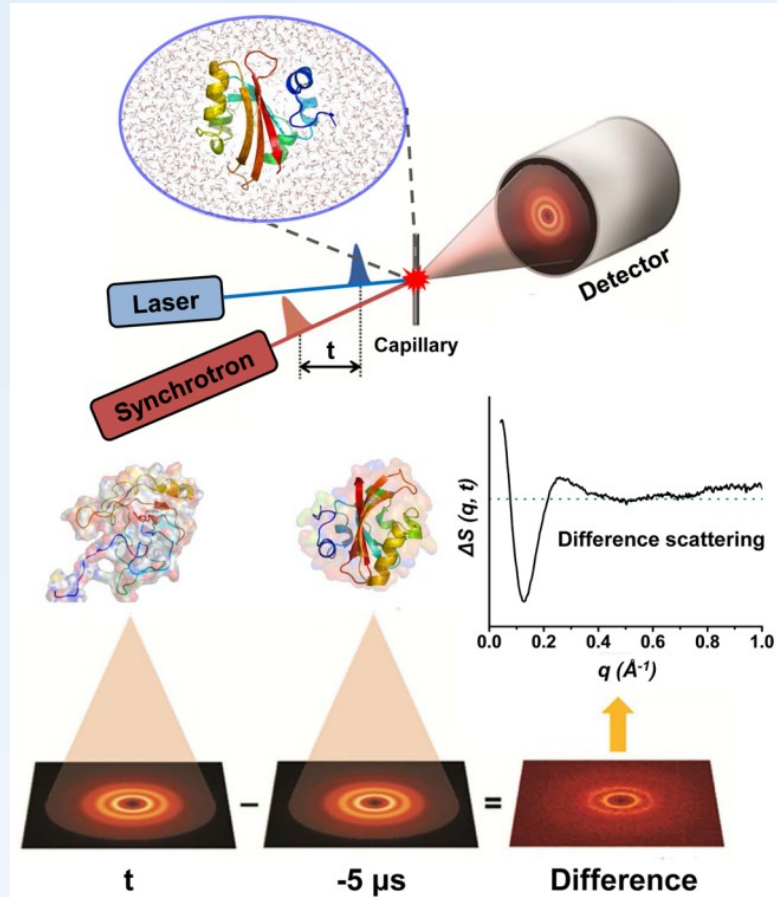
- **Difficult to prepare crystal of protein**
- Phase problem
- Weak
- Fail to reveal dynamic structure

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◆ Introduction of structure biology: Methods

Time-resolved X-ray solution scattering:



Nat Commun, 2021, 12, 3677.

Acc. Chem. Res. 2015, 48, 2200–2208

2026/9/11

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◆ Introduction of structure biology: Methods



The Nobel Prize in
Chemistry 2002

Kurt Wüthrich
h

”for his development of nuclear magnetic resonance spectroscopy for determining the three-dimensional structure of biological macromolecules in solution”.

NMR

After demonstration of x-ray crystallography, there are still several important issues to be solved:

- Dynamic
- Generality



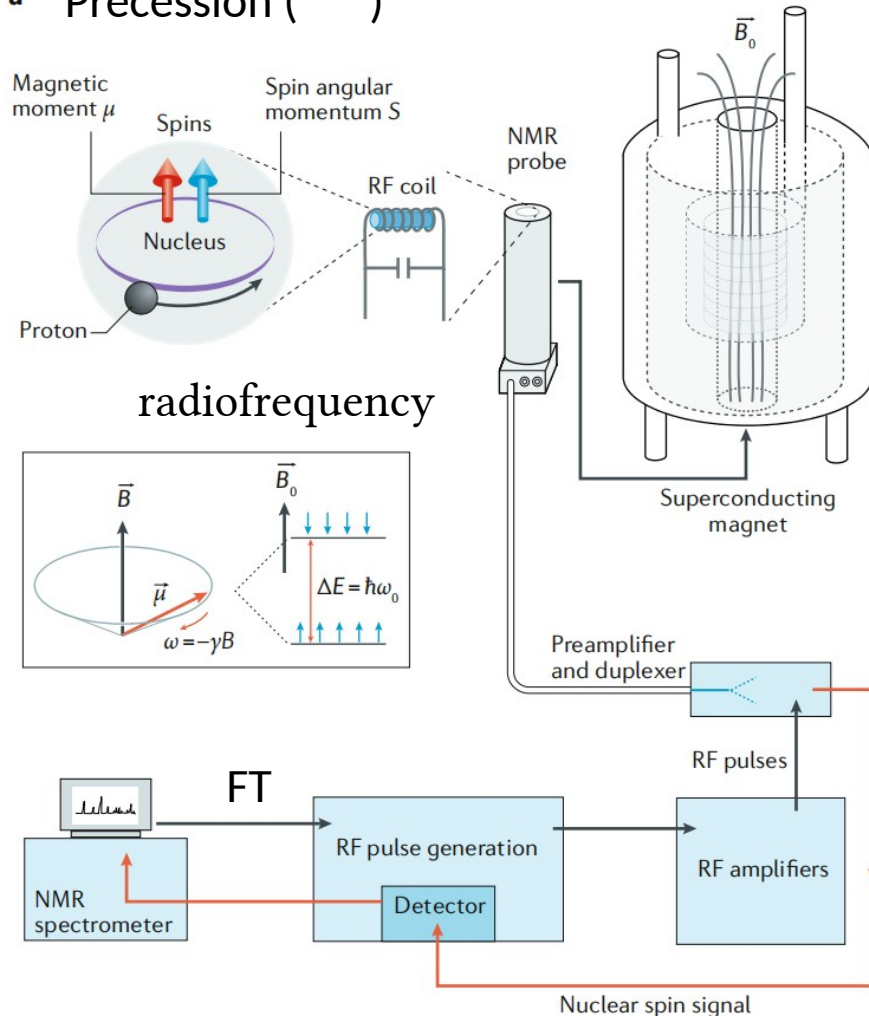
- High conc.
- “Small protein”

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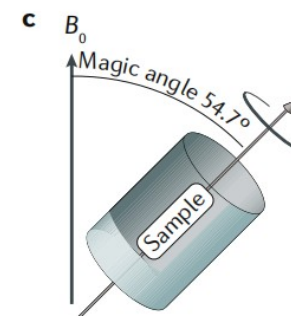
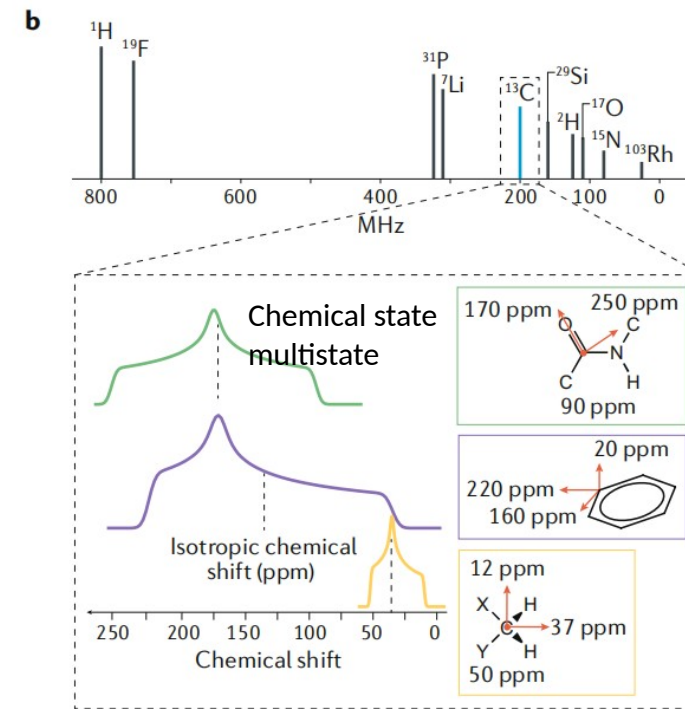


◆ Introduction of structure biology: Methods

a Precession ()



FID (Free Induction Decay,)



Anisotropic contribution to be removed by MAS

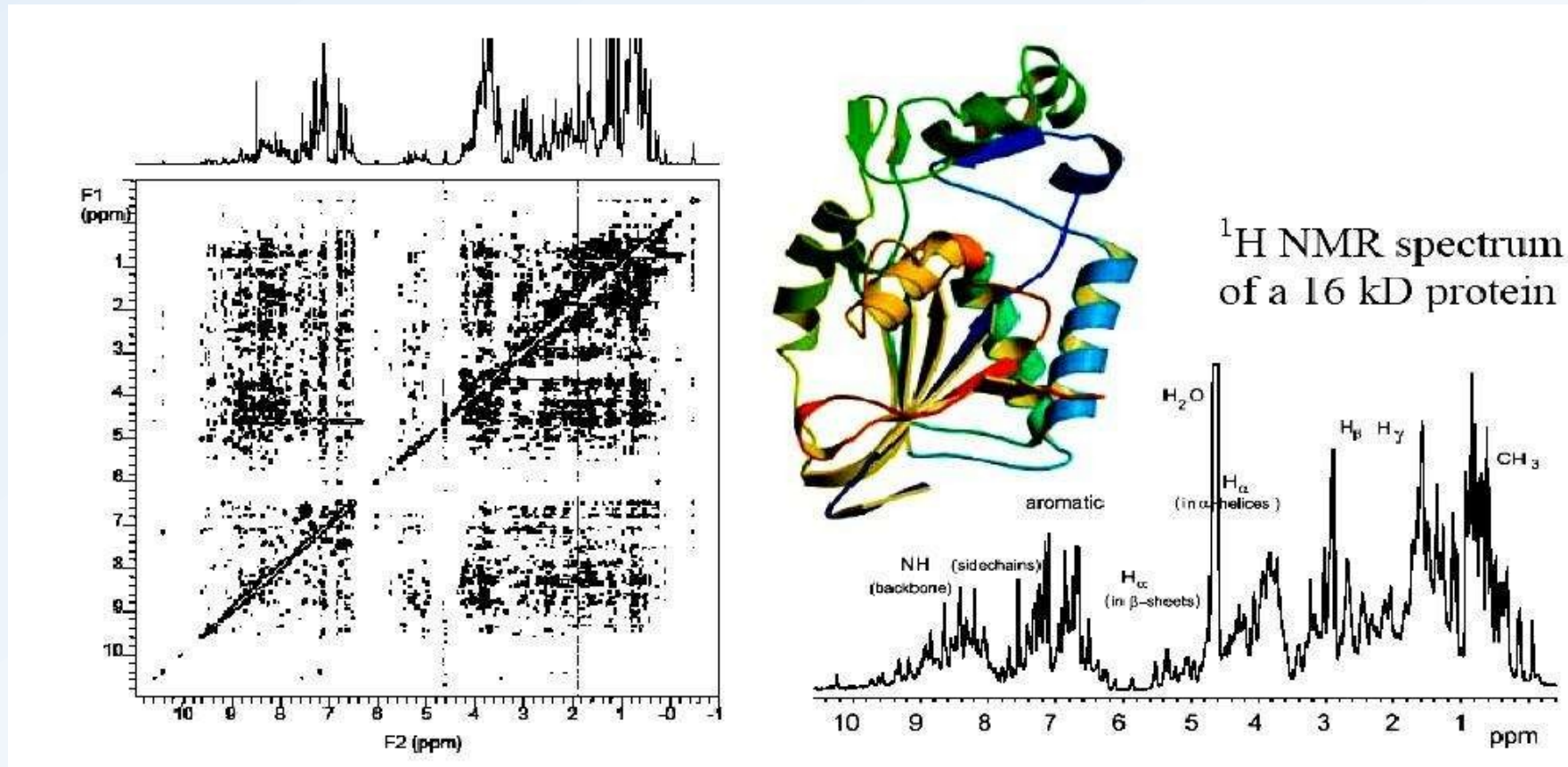
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◆ Introduction of structure biology: Methods

COSY () TOCSY ()

NOESY nuclear overhauser effect spectroscopy



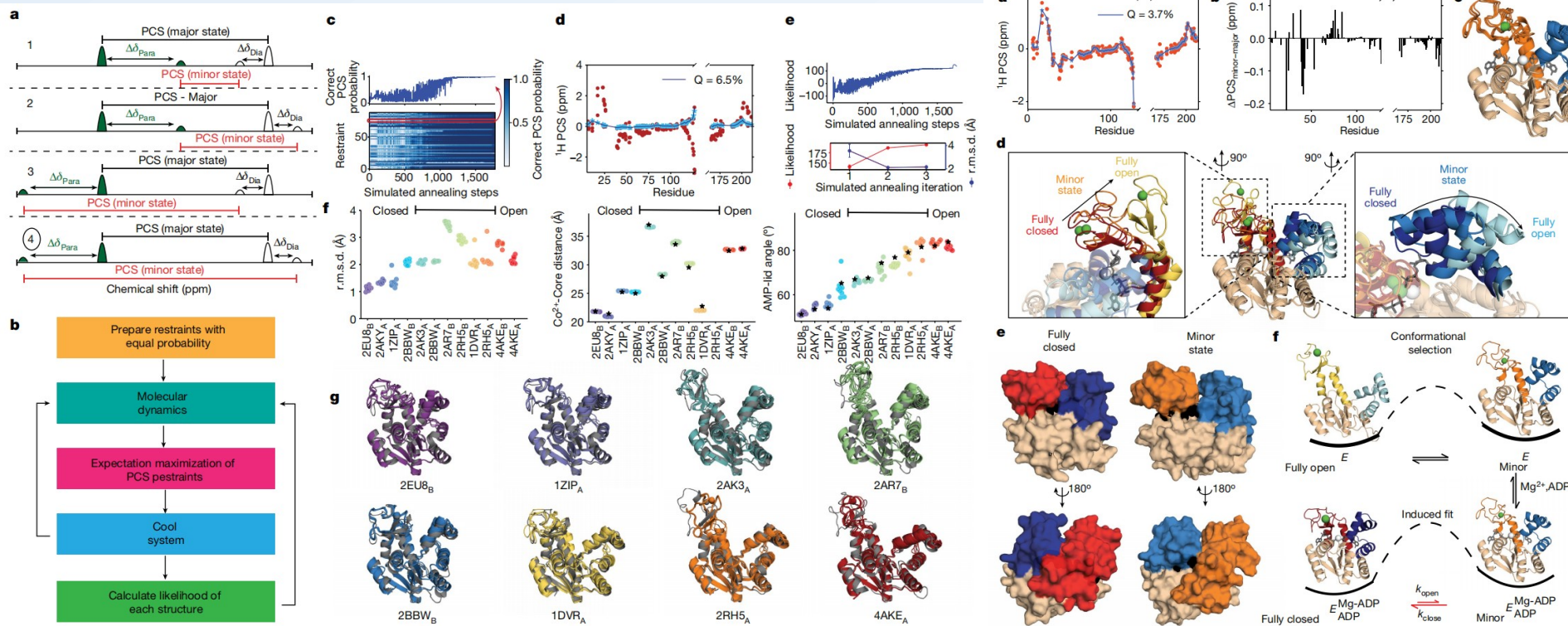
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◆ Introduction of structure biology: Methods

Article

Structure determination of high-energy states in a dynamic protein ensemble



2026/9/11



Nature **603**, 528–535 (2022)

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◆ Introduction of structure biology: Methods Cryo-TEM:



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Jacques
Dubochet

Prize share: 1/3



© Nobel Media AB.
Photo: A. Mahmoud

Joachim Frank

Prize share: 1/3



© Nobel Media AB.
Photo: A. Mahmoud

Richard
Henderson

Prize share: 1/3

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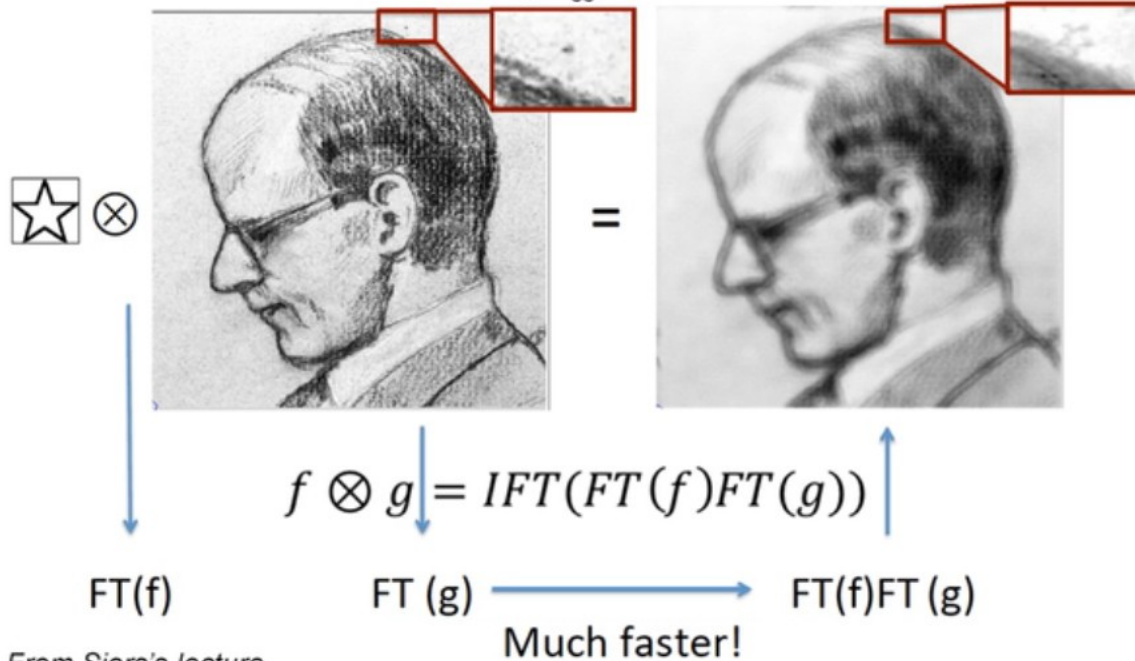


◆ Introduction of structure biology: Methods

Big protein, almost all kinds of proteins and nucleic acids

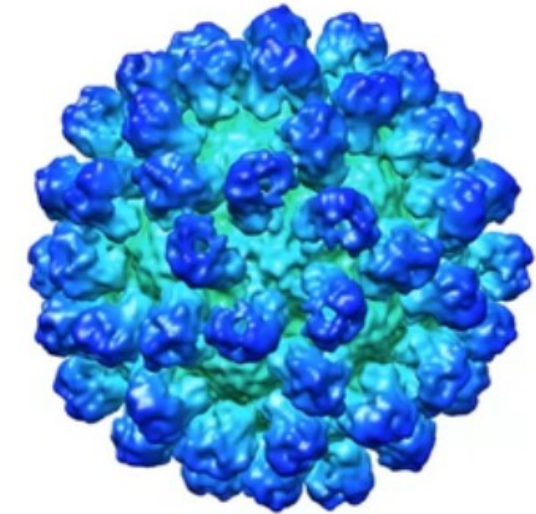
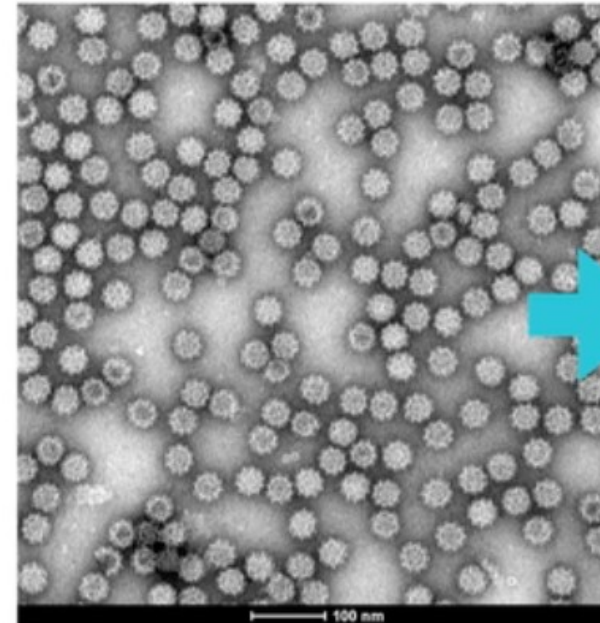
Convolution theorem

$$f \otimes g = \int_{-\infty}^{\infty} f(x)g(t-x)\partial x$$



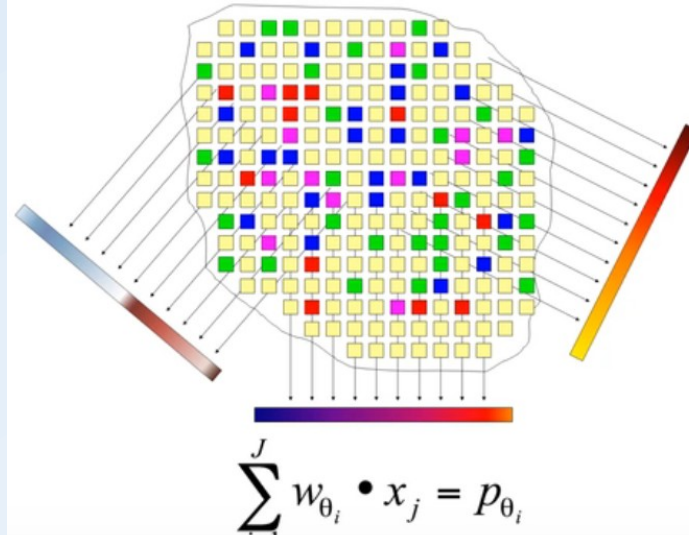
From Sjors's lecture

Three dimensional electron microscopy



3D Reconstruction

Algebraic Method



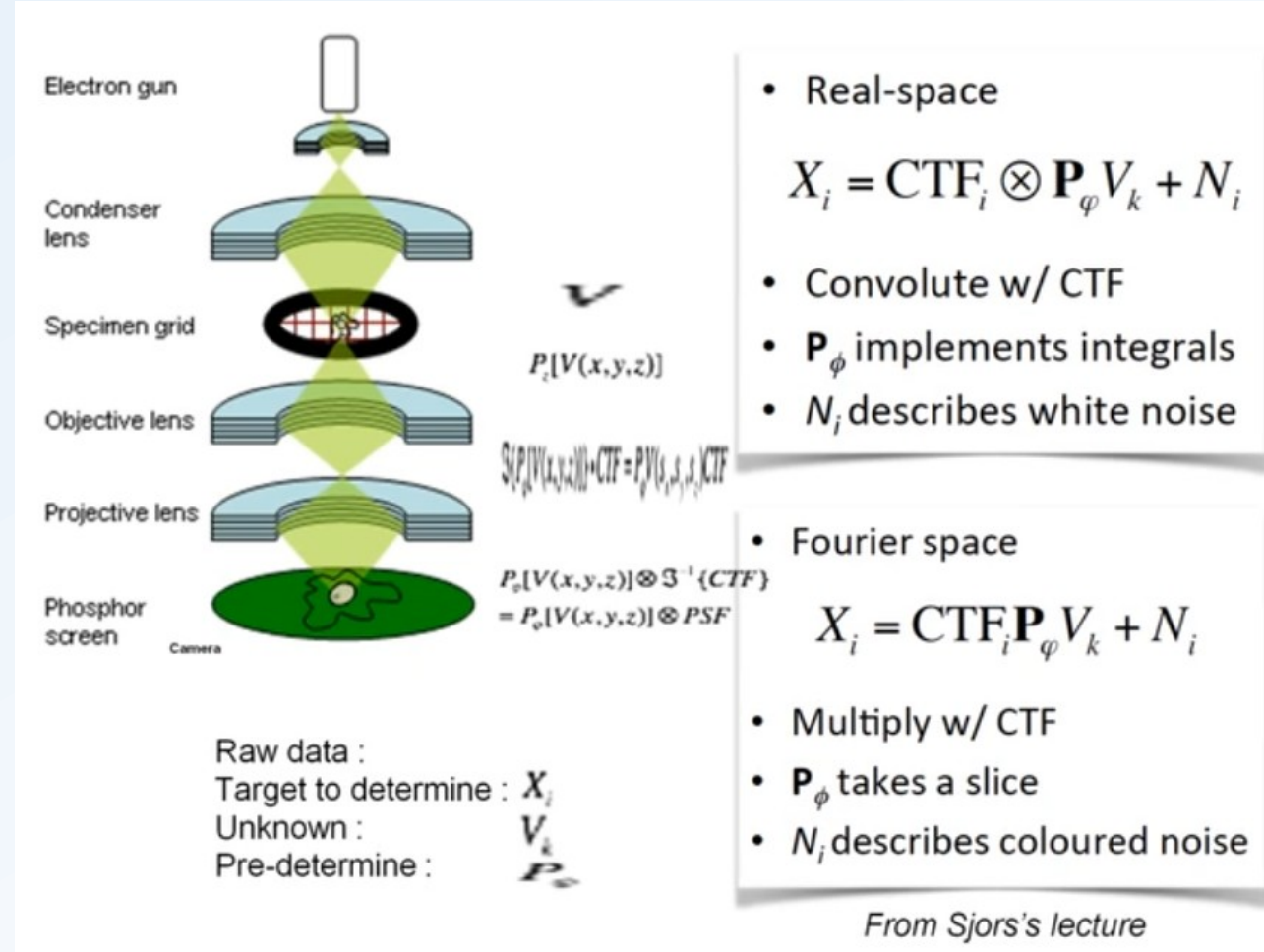
- $AX=P$

- $A = \begin{pmatrix} a_{11}, a_{12}, a_{13}, \dots, a_{1I} \\ a_{21}, a_{22}, a_{23}, \dots, a_{2I} \\ \vdots \\ a_{J1}, a_{J2}, a_{J3}, \dots, a_{JI} \end{pmatrix} = \begin{pmatrix} n1 \\ n2 \\ \vdots \\ nj \end{pmatrix}$

- A : The projection matrix

- $X = (\chi_1, \chi_2, \dots, \chi_J)^T$: The original image, J pixels in total

- $P = (p_1, p_2, \dots, p_I)^T$: The projection, I in total



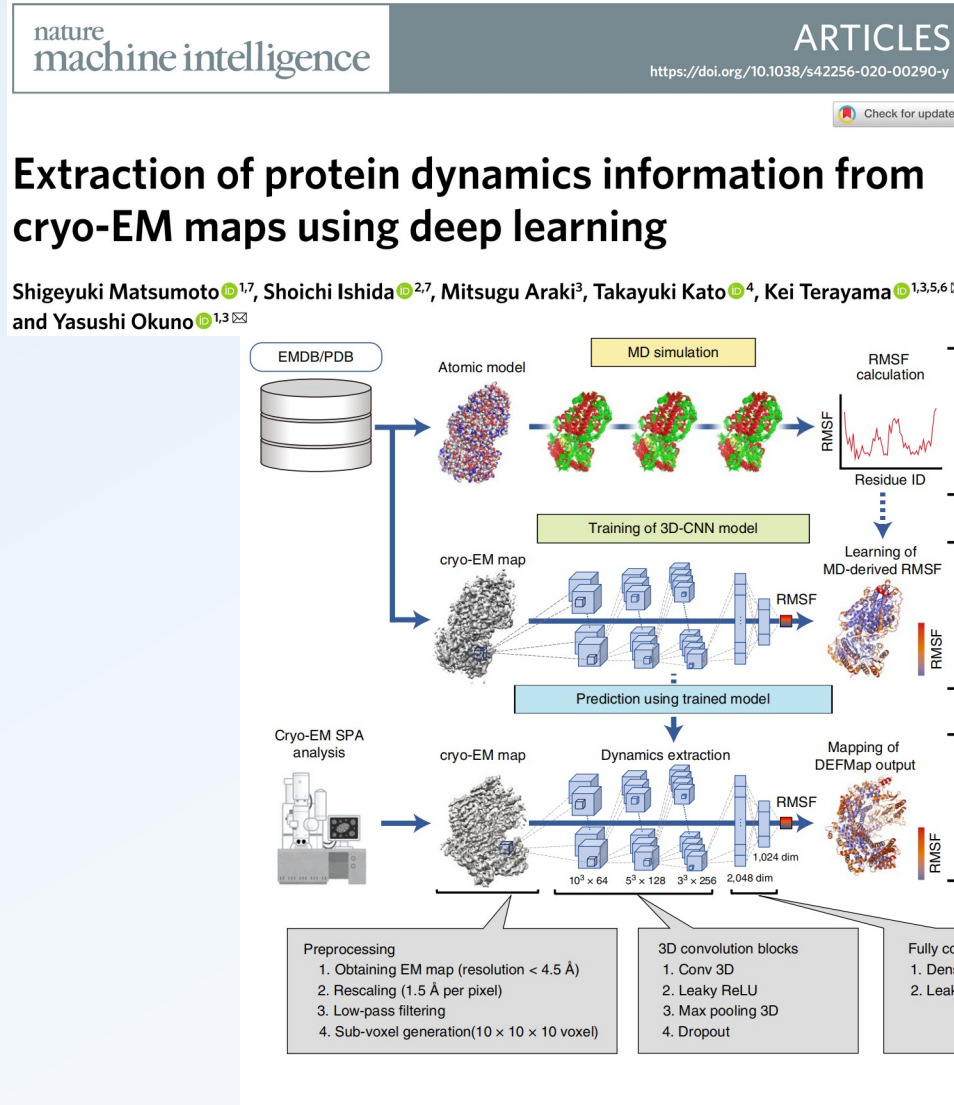
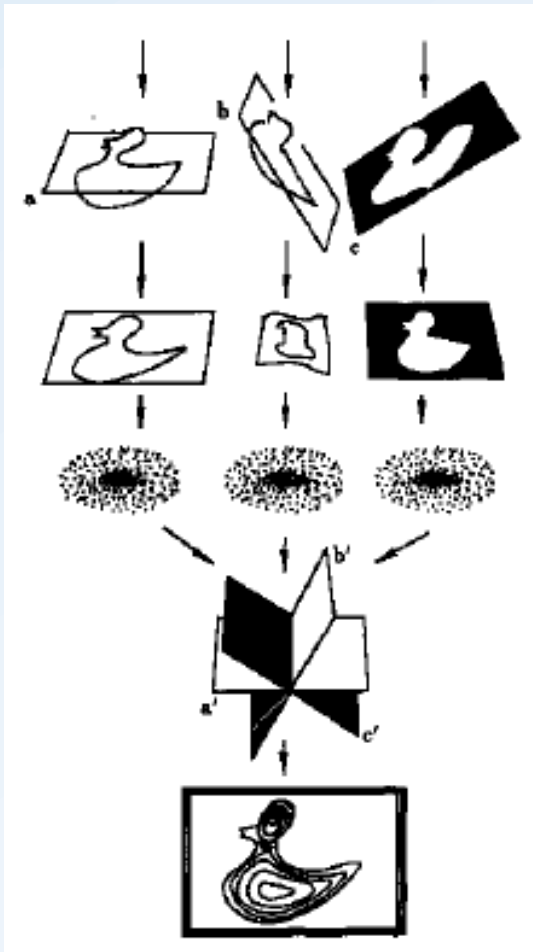
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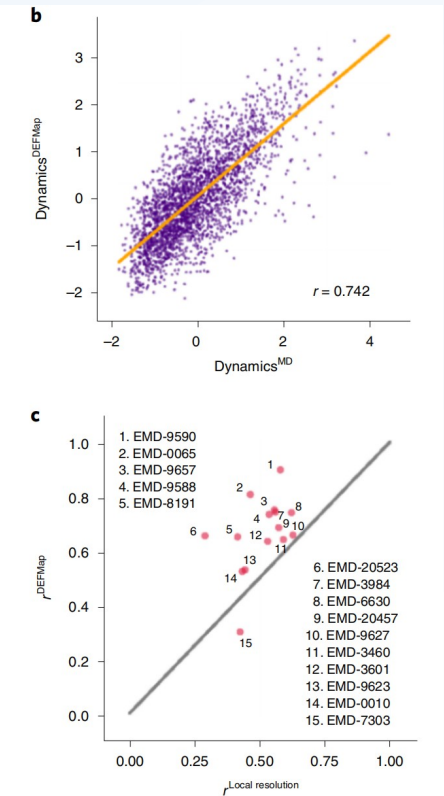
◆ Introduction of structure biology: Methods

Nat. Mach. Intell. **2021**, 3, 153–160.

Reconstruction



DL-MD assisted reconstruction



Chin. Phys. B. **2018**, 27, 066801

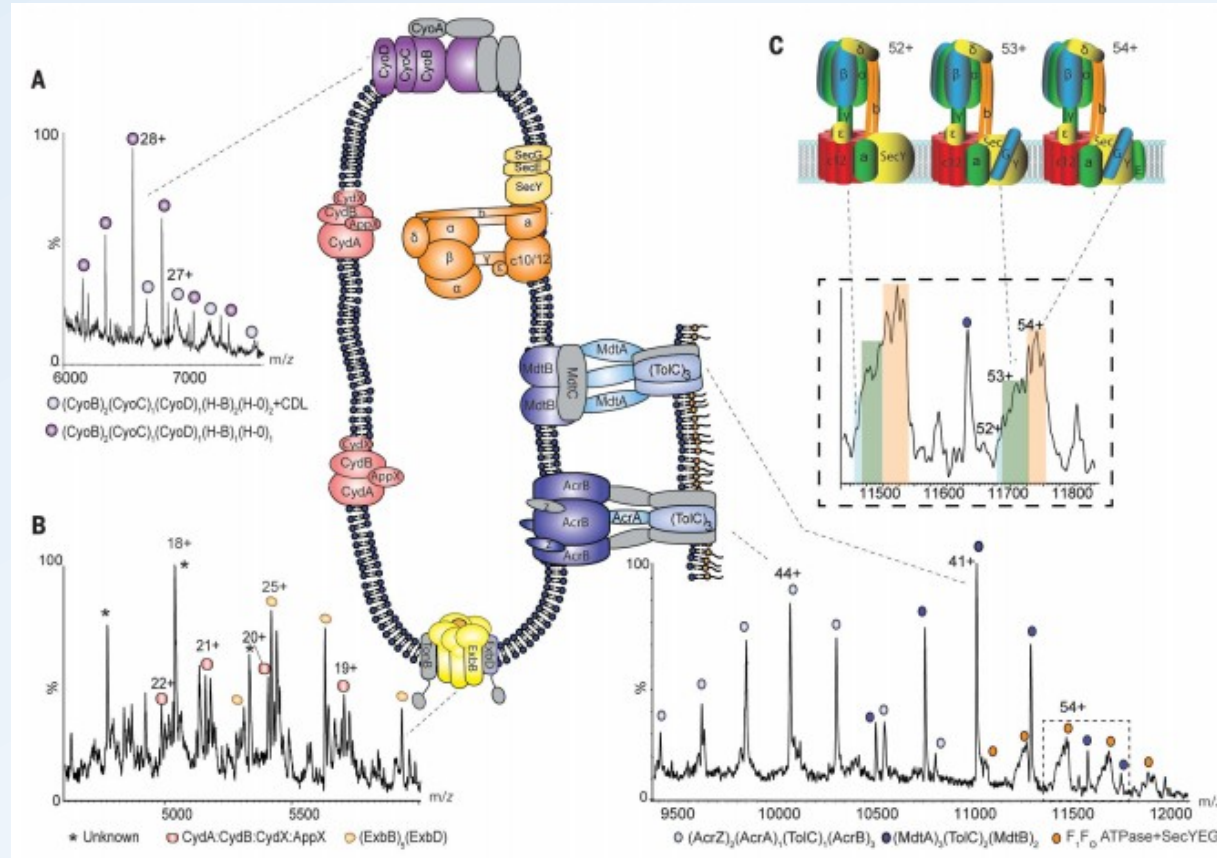
2026/9/11

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The global structure of assembled protein can be confirmed by introducing different interacting species.

Native MS



Prof. Carol V. Robinson

The 3D structure of assembled structure can be confirmed with combination of NMR or cryo-EM or X-ray crystallography.

Science, **2018**, 362, 829–834.

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Theoretical

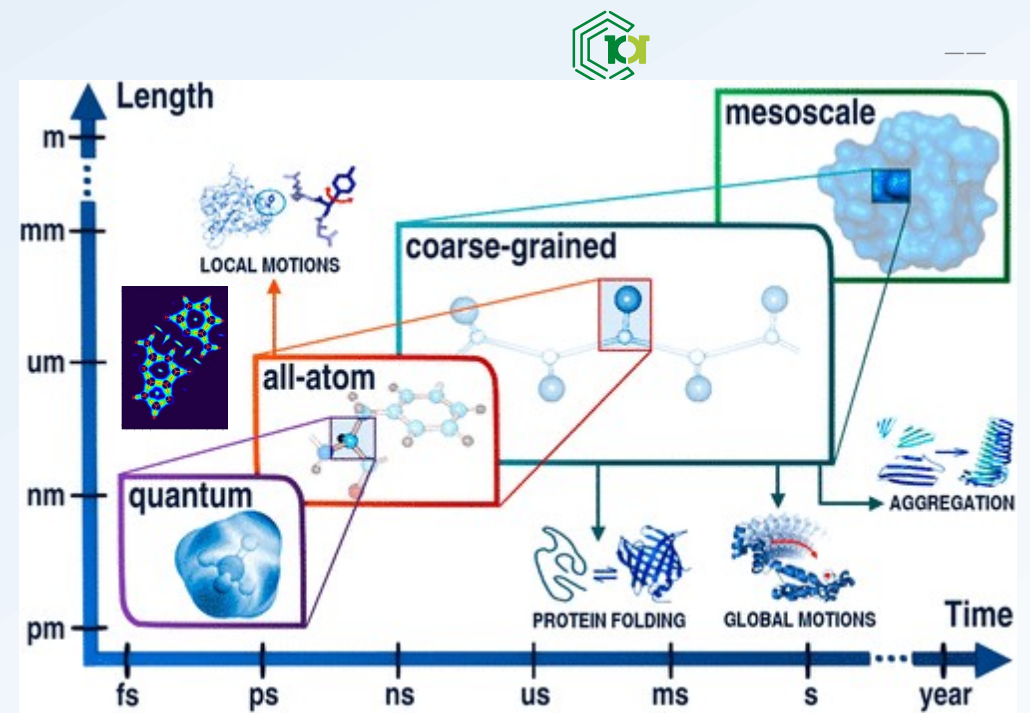
QM

The Schrödinger wave equation describes the motions of the electrons and nuclei in a molecular system from first principles. This equation can be written as

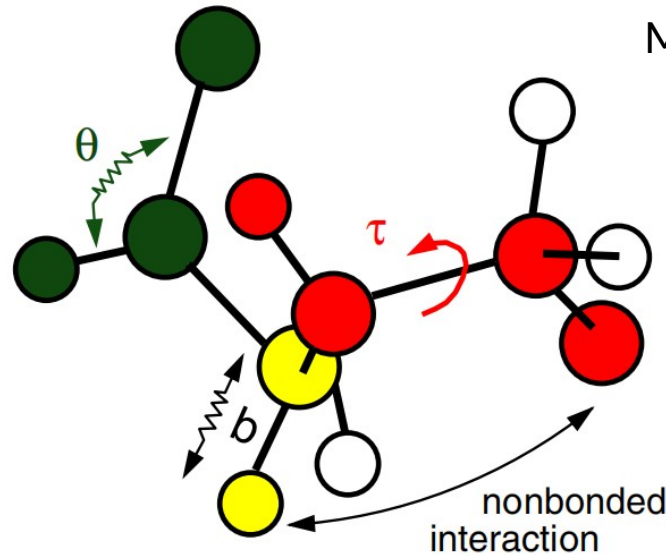
$$\hat{H}\psi_n = E_n\psi_n, \quad (7.1)$$

where the Hamiltonian operator $\hat{H}$ is the sum of the kinetic (E_k) and potential (E_p) energy of the system:

$$\hat{H}(\tilde{P}, \tilde{X}) = E_k(\tilde{P}) + E_p(\tilde{X}), \quad (7.2)$$



MM



$$E_{\text{local}}(X) = \sum_{\text{bonds } i} E_{\text{bond}}(b_i) + \sum_{\text{bond angles } i} E_{\text{bang}}(\theta_i) + \sum_{\text{dihedral angles } i} E_{\text{tor}}(\tau_i),$$

$$E_{\text{nonlocal}}(X) = \sum_{\text{nonbonded pairs } (i,j), i < j} E_r(r_{ij})$$

$$E^b(r) = \frac{1}{2} k^b (r - r_0)^2$$

$$E^a(\theta) = \frac{1}{2} k^a (\theta - \theta_0)^2$$

$$E^d(\varphi) = \sum_{i=1}^N \frac{V_i}{2} [1 + \cos(n_i \varphi - \varphi_{0,i})]$$

$$E_{AB}^{\text{ele}}(r_{AB}) = \frac{q_A q_B}{4\pi\epsilon_0 r_{AB}}$$

$$E_{AB}^{\text{vdw}}(r_{AB}) = \frac{C_{12}}{r_{AB}^{12}} - \frac{C_6}{r_{AB}^6} = 4\epsilon \left[\left(\frac{\sigma}{r_{AB}} \right)^{12} - \left(\frac{\sigma}{r_{AB}} \right)^6 \right]$$

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◆ Introduction of structure biology: Methods

If we have enough calculating resources, we can

Develop New FF and sampling methods: *Nat. Commun.* **2022.** 13, 68.

eLife, **2016**, 5, e19274

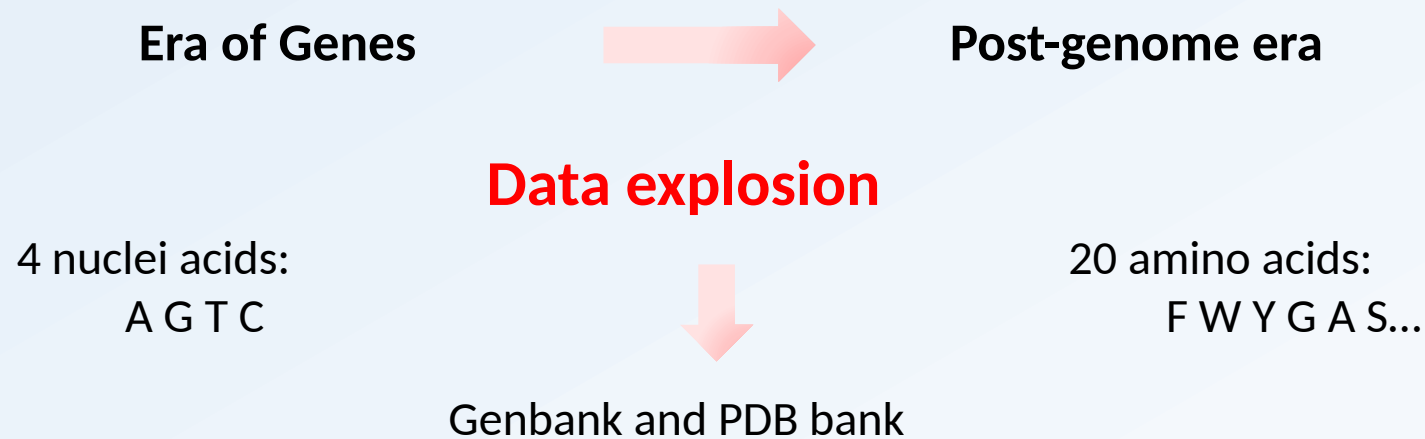
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◆ Introduction of structure biology: Methods

****A paradigm shift****



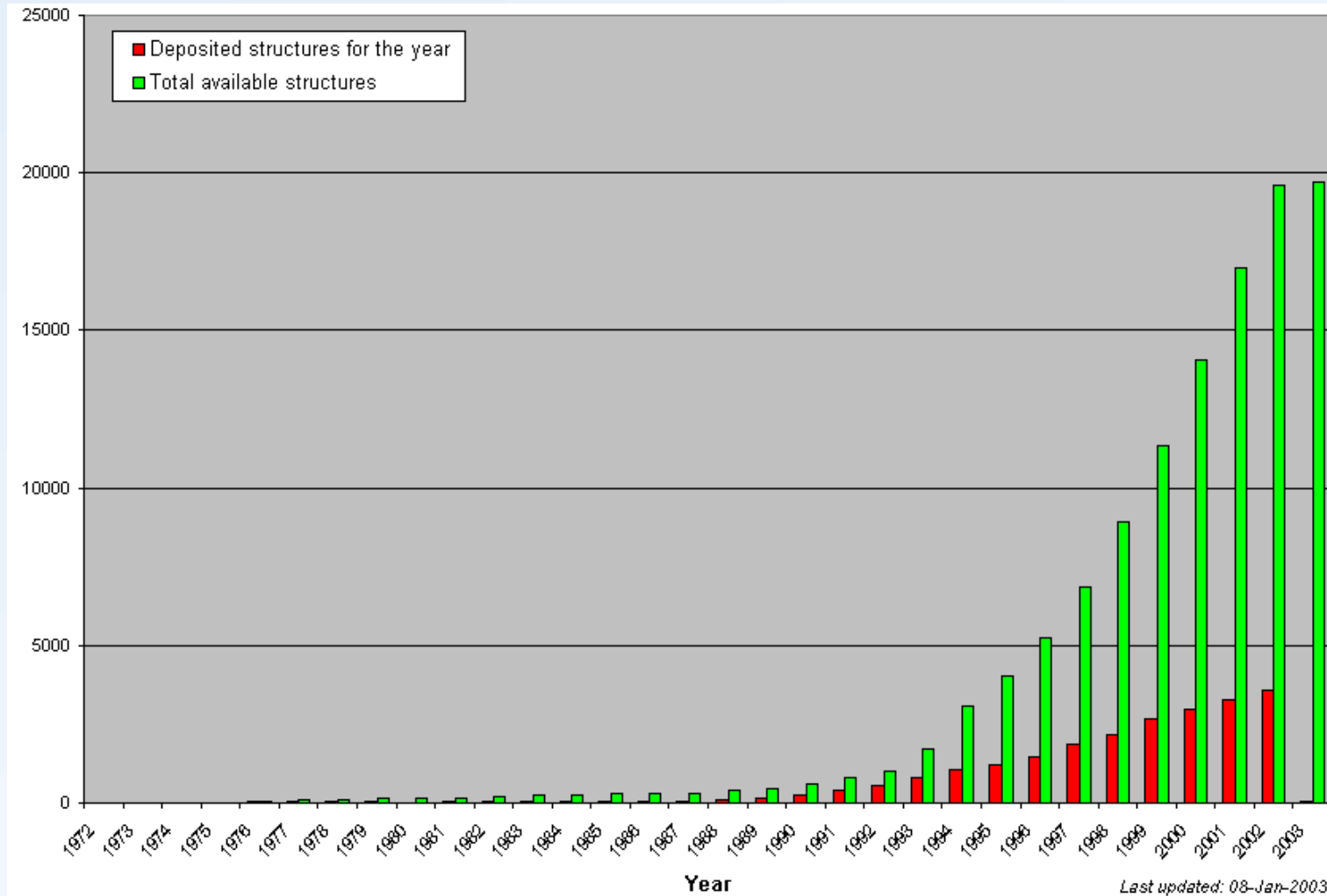
Bioinformatics is the science of the using information to understand biology. It is the discipline of obtaining information about genomic or protein sequence data. This may involve similarity searches of databases, comparing your unidentified sequence to the sequences in a database, or making predictions about the sequence based on current knowledge of similar sequences.

— —Claverie (2000)

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◆ Introduction of structure biology: Methods Bioinformatics: Data-driven method

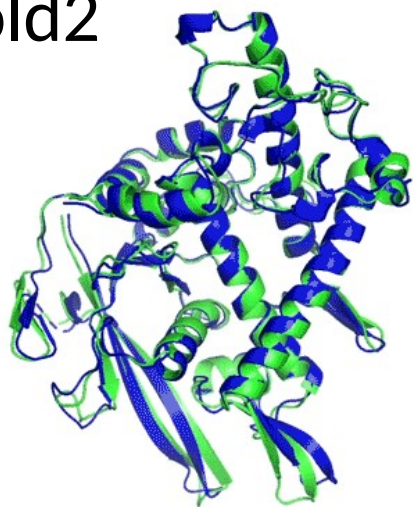


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◆ Introduction of structure biology: Methods

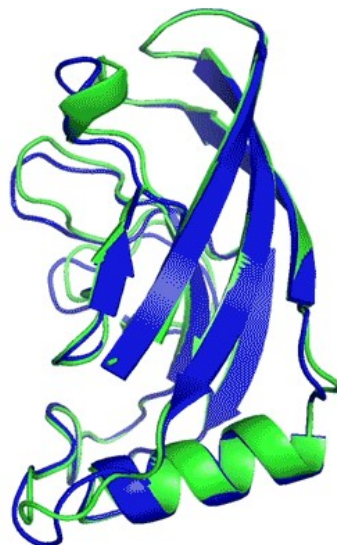
AlphaFold2



T1037 / 6vr4

90.7 GDT

(RNA polymerase domain)



T1049 / 6y4f

93.3 GDT

(adhesin tip)

● Experimental result

● Computational prediction

nature

<https://doi.org/10.1038/s41586-024-07487-w>

Accelerated Article Preview

Accurate structure prediction of biomolecular interactions with AlphaFold 3

Received: 19 December 2023

Accepted: 29 April 2024

Accelerated Article Preview

Cite this article as: Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold3. *Nature* <https://doi.org/10.1038/s41586-024-07487-w> (2024)

Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, Sebastian W. Bodenstein, David A. Evans, Chia-Chun Hung, Michael O'Neill, David Reiman, Kathryn Tunyasuvunakool, Zachary Wu, Akvilė Žemgulytė, Eirini Arvaniti, Charles Beattie, Ottavia Bertolli, Alex Bridgland, Alexey Cherepanov, Miles Congreve, Alexander I. Cowen-Rivers, Andrew Cowie, Michael Figurnov, Fabian B. Fuchs, Hannah Gladman, Rishub Jain, Yousuf A. Khan, Caroline M. R. Low, Kuba Perlin, Anna Potapenko, Pascal Savy, Sukhdeep Singh, Adrian Stecula, Ashok Thillaisundaram, Catherine Tong, Sergei Yakneen, Ellen D. Zhong, Michal Zielinski, Augustin Židek, Victor Bapst, Pushmeet Kohli, Max Jaderberg, Demis Hassabis & John M. Jumper

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Nature (2024). <https://doi.org/10.1038/s41586-024-07487-w>

Nature, **2021**, 596, 583–589

Gif from Zhihu <https://baijiahao.baidu.com/s?id=1705954669862856583&wfr=spider&for=pc>

2026/9/11

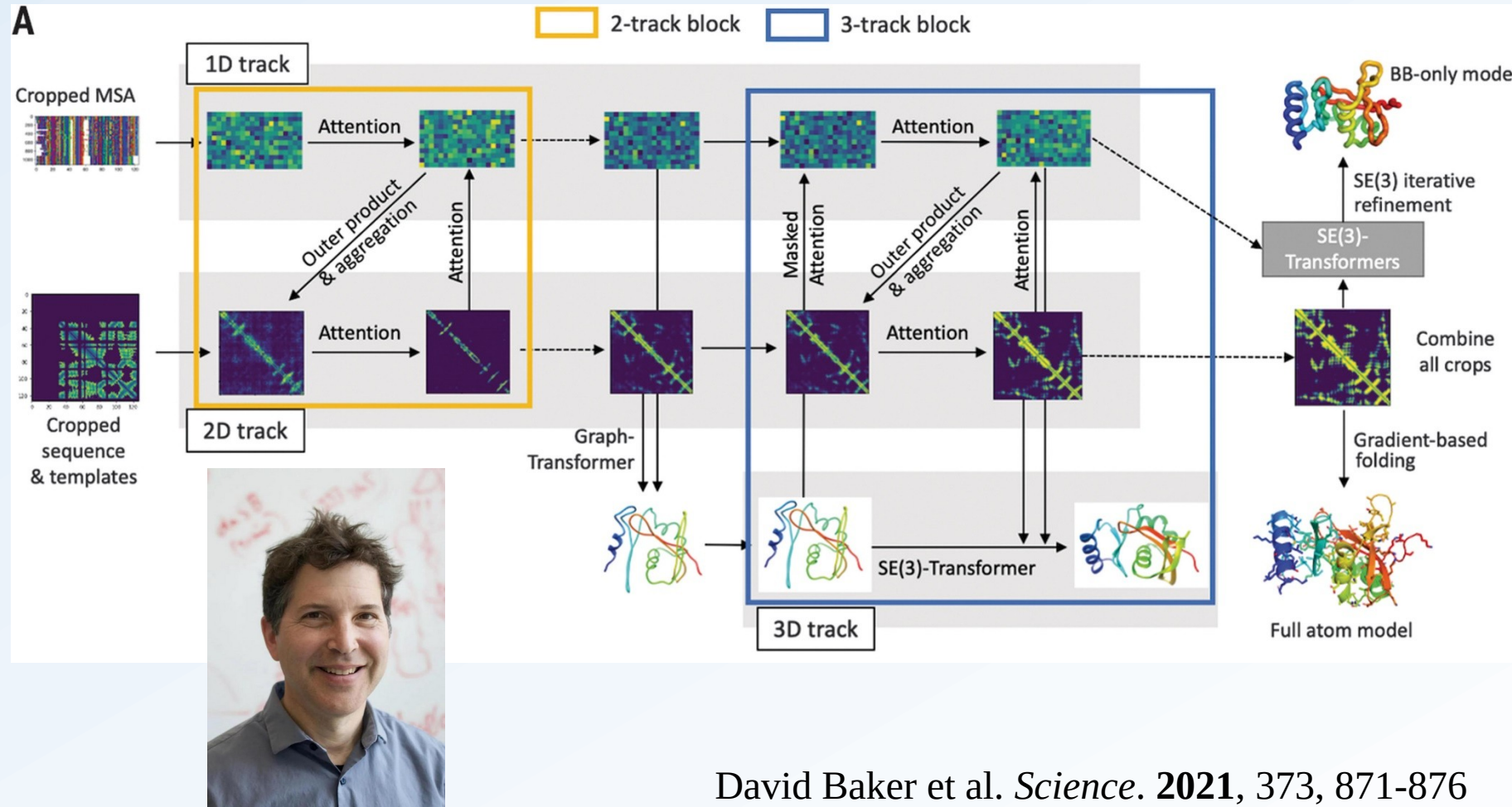


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◆ Introduction of structure biology: Methods

Rosettafold:

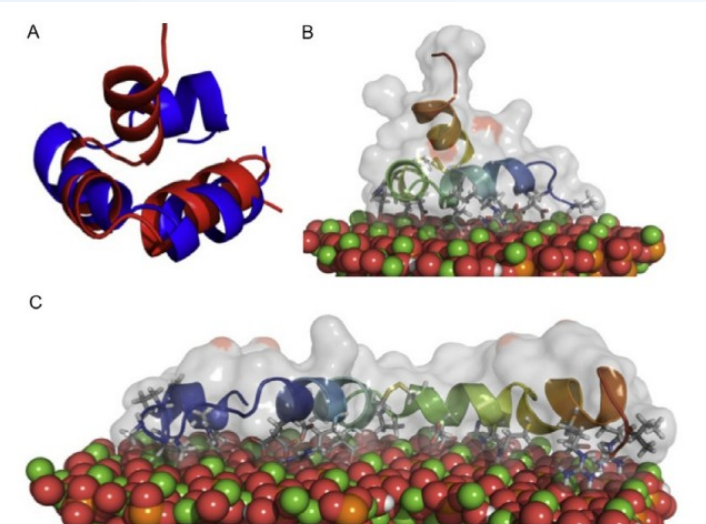
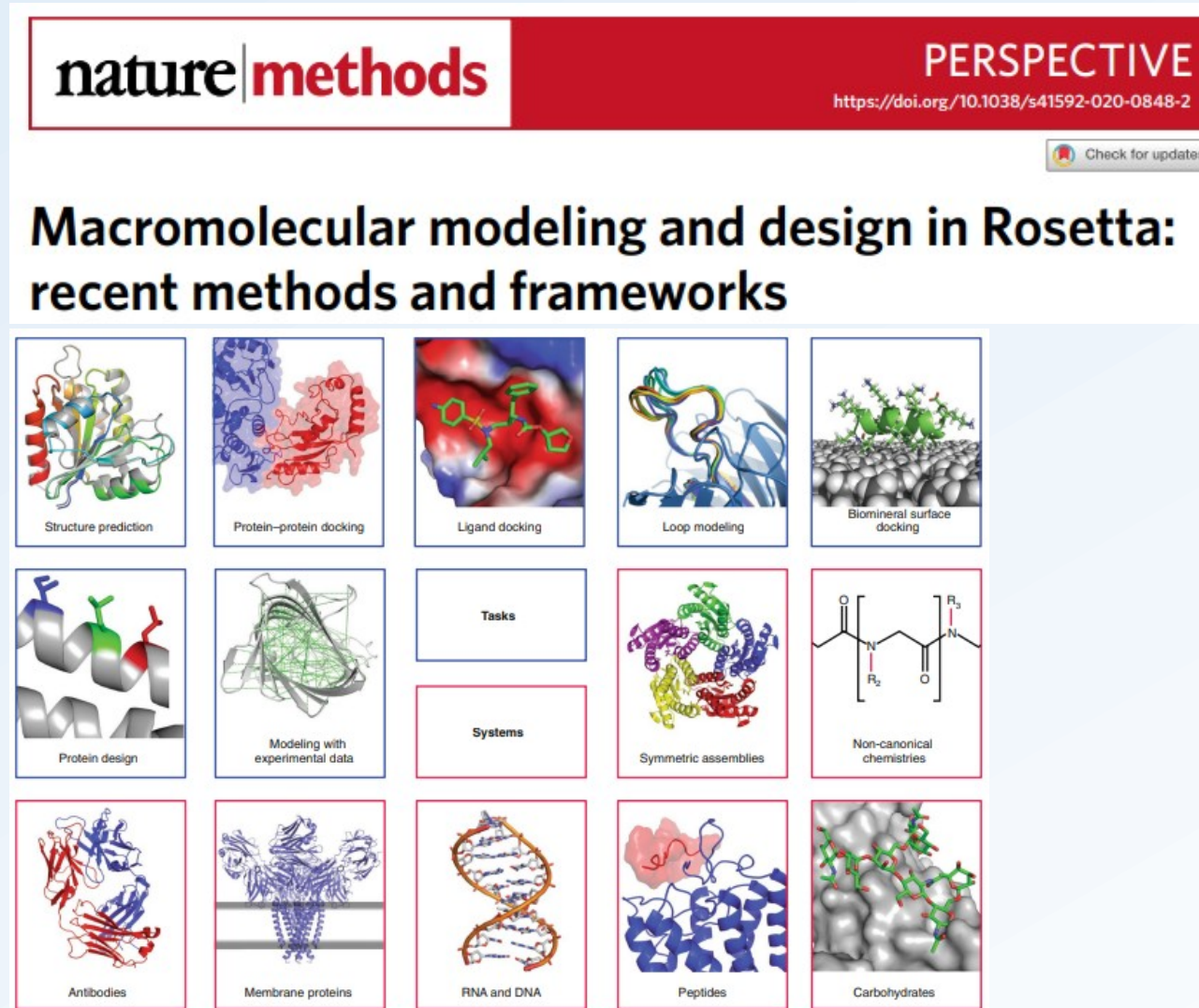


David Baker et al. *Science*. **2021**, 373, 871-876

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◆ Introduction of structure biology: Methods

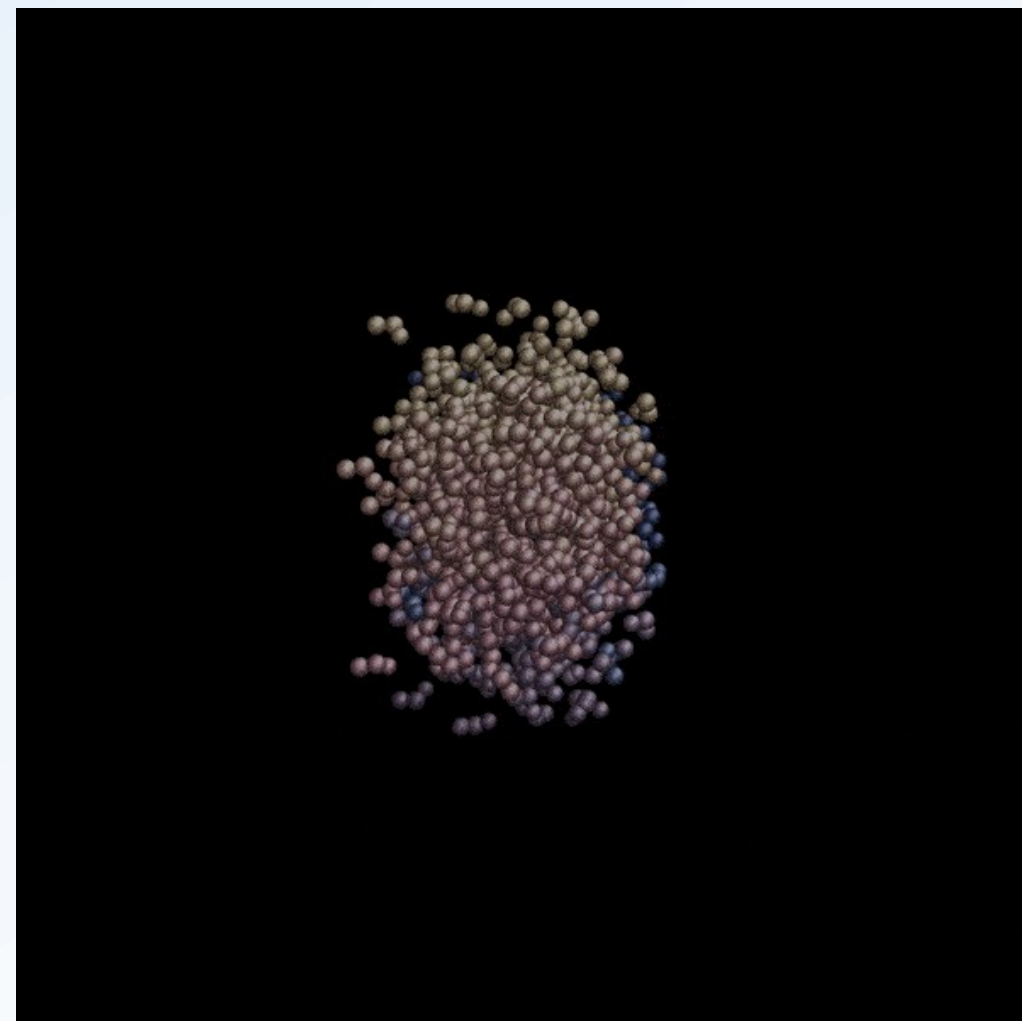


This result has been confirmed by solid-state NMR, which means we can obtain reasonable interfacial structure using PyRosetta!

Nature methods, 2020, 17, 665–680

Diffusion model

https://www.bilibili.com/video/BV1p8411777A/?spm_id_from=333.337.search-card.all.click&vd_source=d361f17018965537e5876df9d591d167

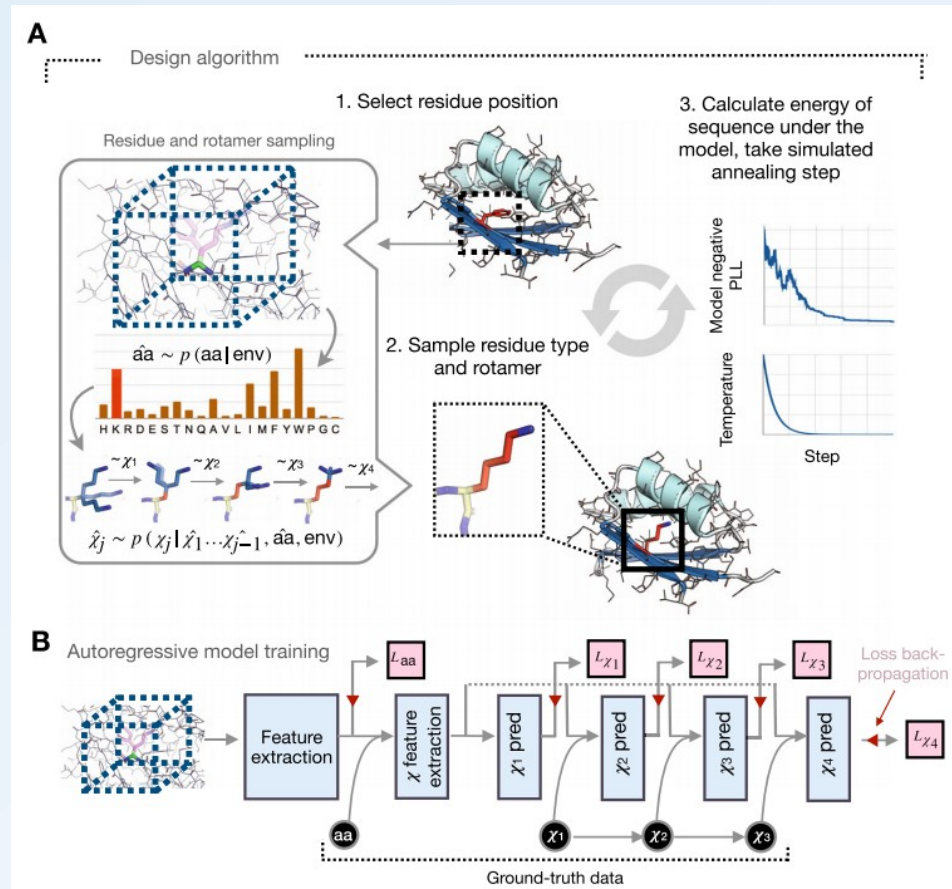


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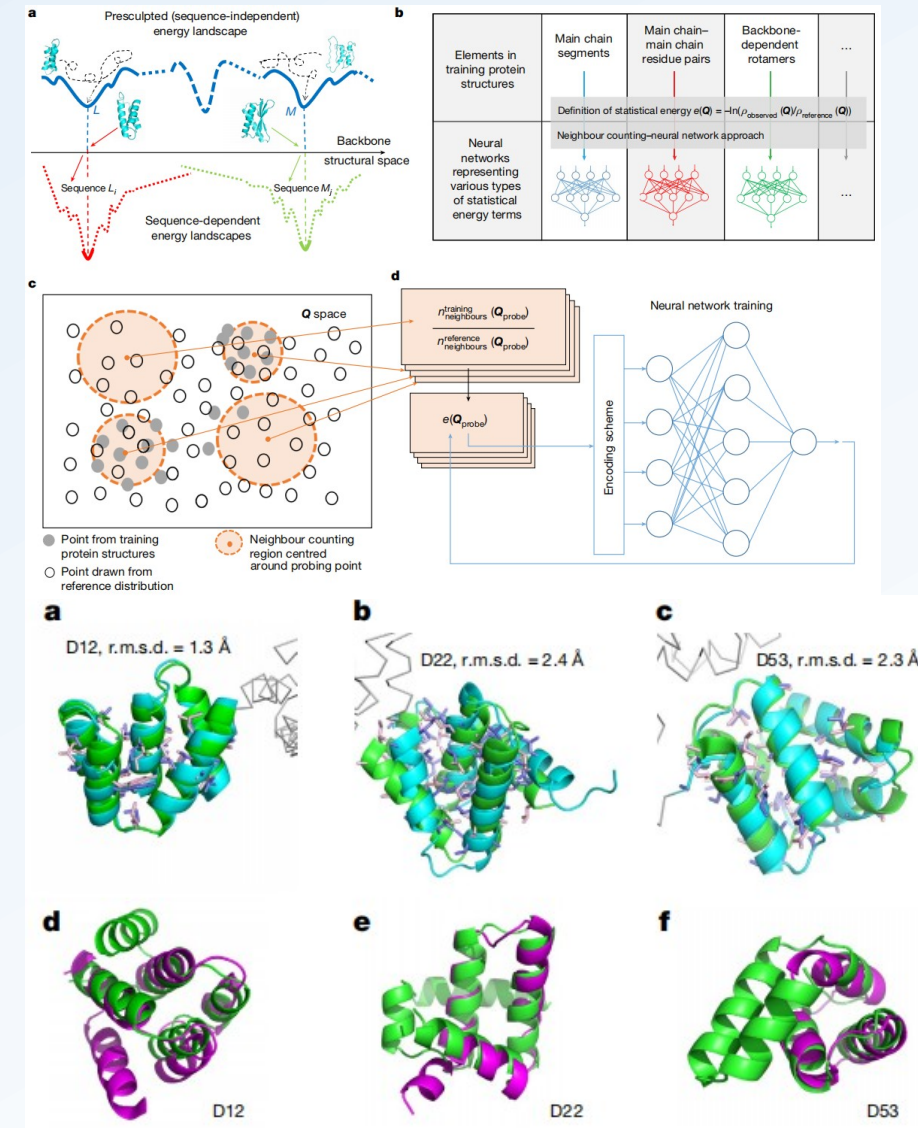


◆ Introduction of structure biology: Methods

De novo protein design ()



Nat. Commun, 2022, 13, 746



Nature, 2022, 602, 523–528

AI for biological analysis



◆ Introduction of structure biology: Methods

Cite as: R. Krishna *et al.*, *Science*
10.1126/science.adl2528 (2024).

Generalized biomolecular modeling and design with RoseTTAFold All-Atom

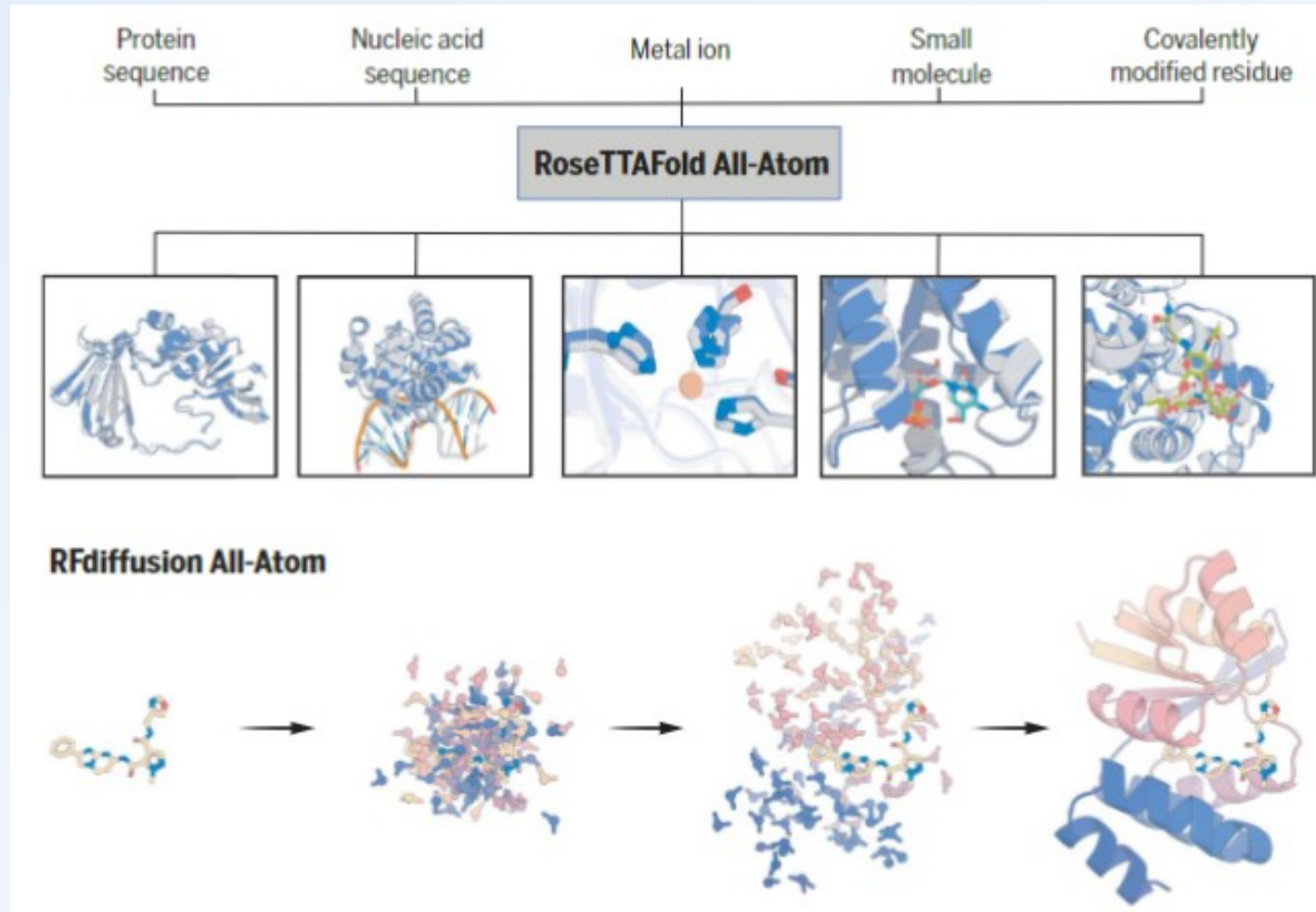
**Rohith Krishna^{1,2†}, Jue Wang^{1,2†}, Woody Ahern^{1,2,3†}, Pascal Sturmfels^{1,2,3}, Preetham Venkatesh^{1,2,4‡},
Indrek Kalvet^{1,2,5‡}, Gyu Rie Lee^{1,2,5‡}, Felix S. Morey-Burrows⁶, Ivan Anishchenko^{1,2}, Ian R. Humphreys¹,
Ryan McHugh^{1,2,4}, Dionne Vafeados^{1,2}, Xinting Li^{1,2}, George A. Sutherland⁶, Andrew Hitchcock⁶, C. Nei
Alex Kang², Evans Brackenbrough², Asim K. Bera², Minkyung Baek⁷, Frank DiMaio^{1,2}, David Baker^{1,2,5*}**



AI for biological analysis



◆ Introduction of structure biology: Methods



Review

The past, present and future of de novo protein design

<https://doi.org/10.1038/s41586-026-10328-7>

Received: 16 September 2024

Accepted: 20 February 2026

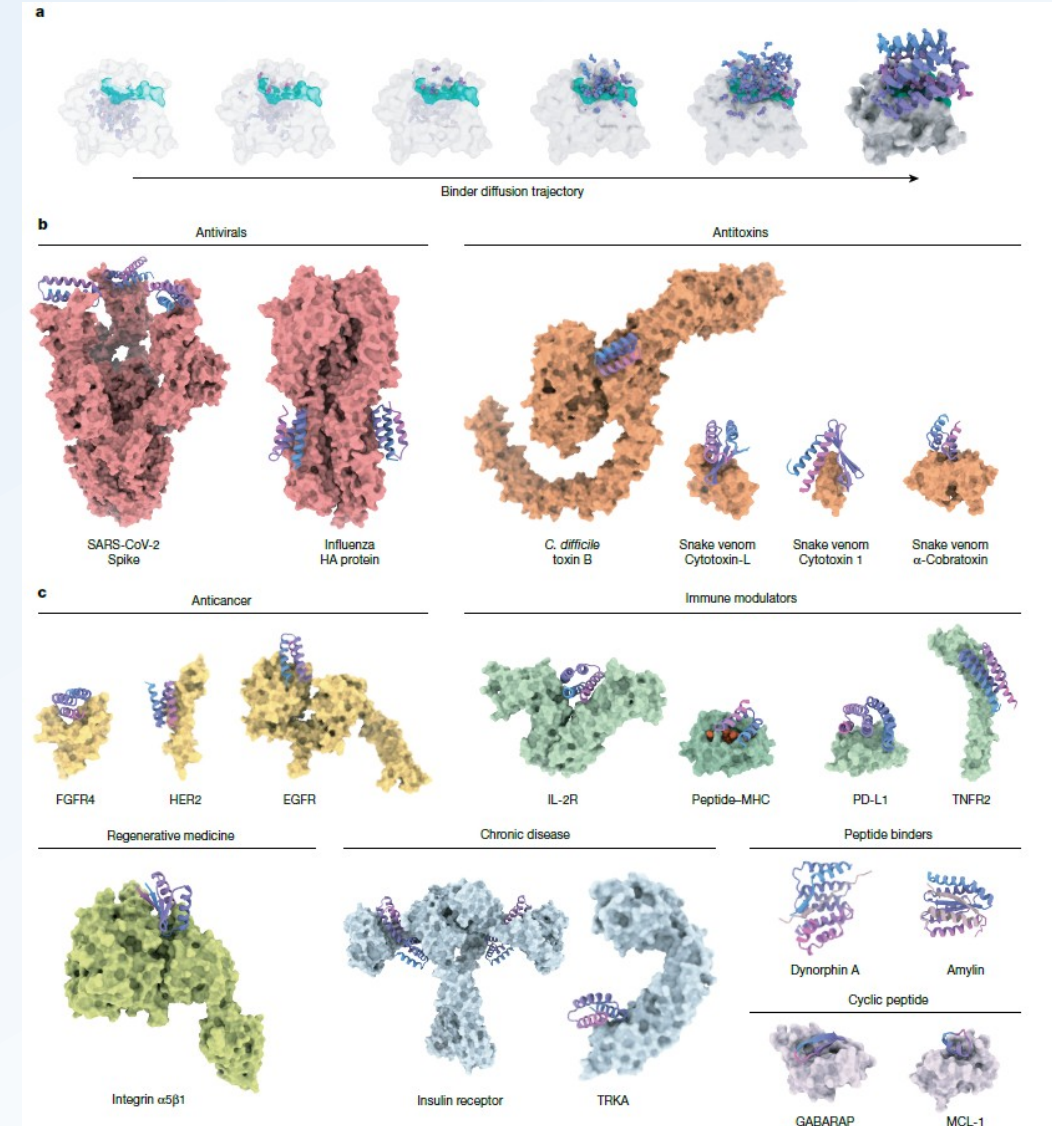
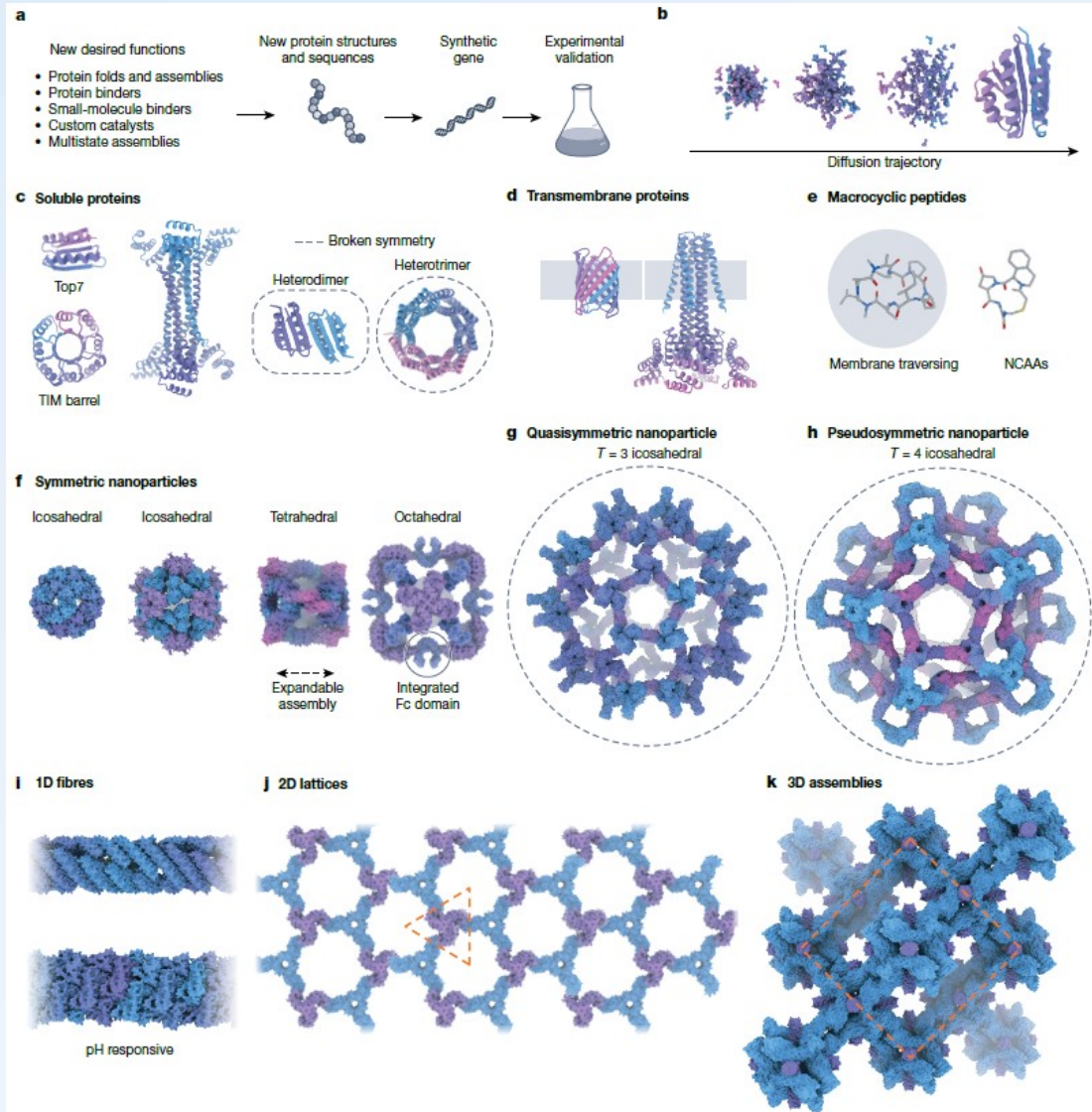
Published online: 29 April 2026



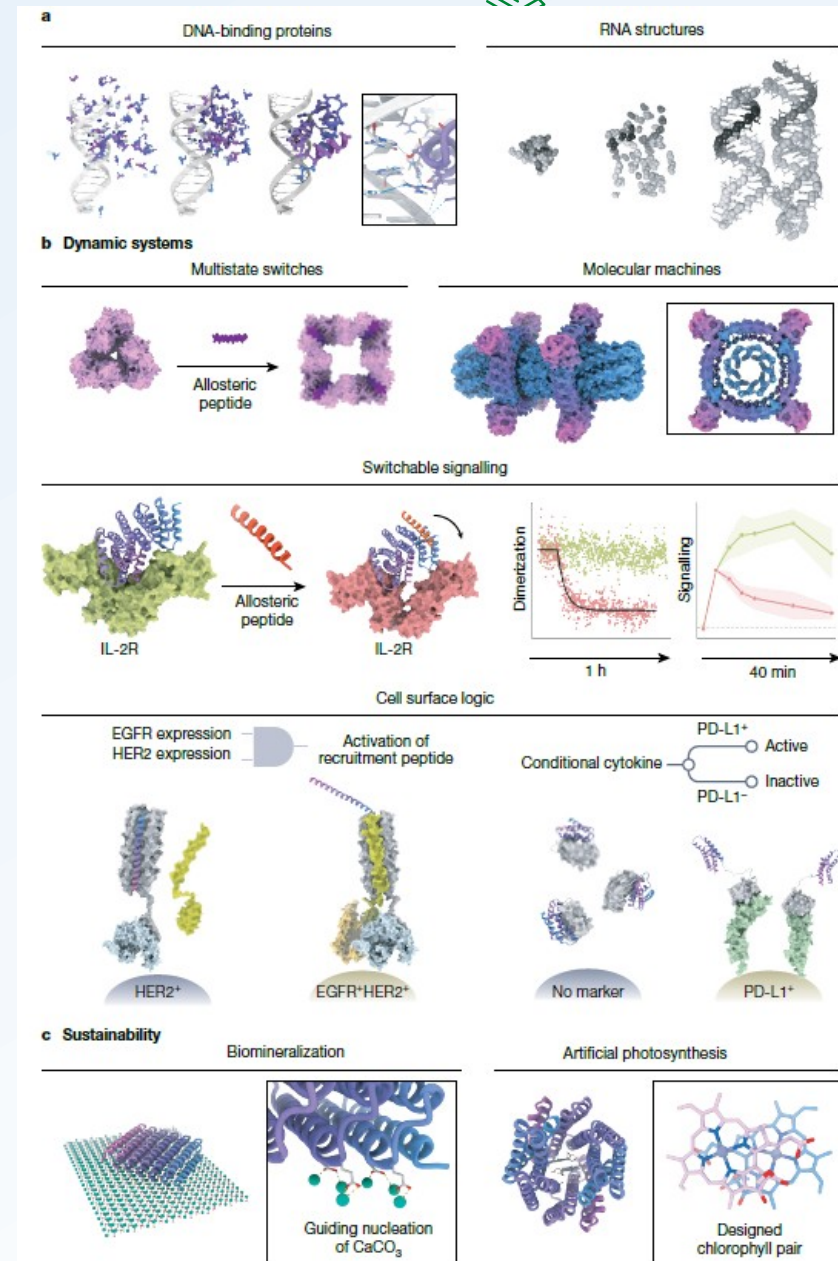
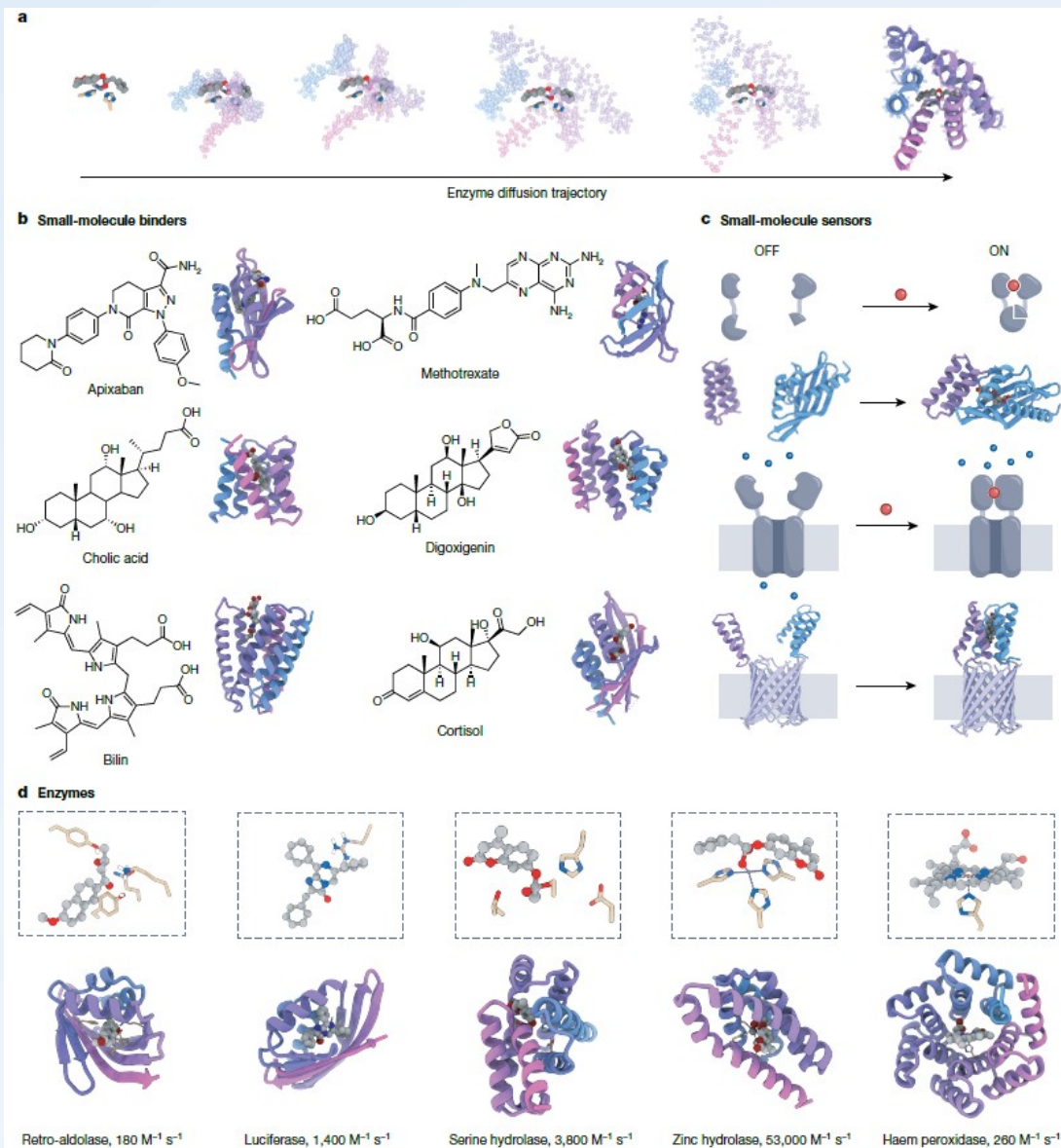
Check for updates

Wei Yang^{1,2,3,15}, Shunzhi Wang^{1,2,4,5,15}, Gyu Rie Lee^{1,2,4,6,15}, Jason Z. Zhang^{1,2,4,7,15}, Alexis Courbet^{1,2,4}, David Juergens^{1,2}, Xinru Wang^{1,2,8,9}, Thomas Schlichthaerle^{1,2,10}, Mohamad Abedi^{1,2,4,7}, Robert Ragotte^{1,2}, Linna An^{1,2}, Indrek Kalvet^{1,2,4}, Sam Pellock^{1,2}, Ljubica Mihaljevic^{1,2}, Cameron Glasscock^{1,2,11}, Arvind Pillai^{1,2}, Adam Broerman^{1,2}, Nathan Ennist^{1,2,12,13}, Ella Haefner^{1,2}, Nora McNamara-Bordewick^{1,2}, Ian Haydon^{1,2}, Lance Stewart^{1,2}, Gaurav Bhardwaj^{2,14} & David Baker^{1,2,4}✉

AI for biological analysis



AI for biological analysis



AI for biological analysis

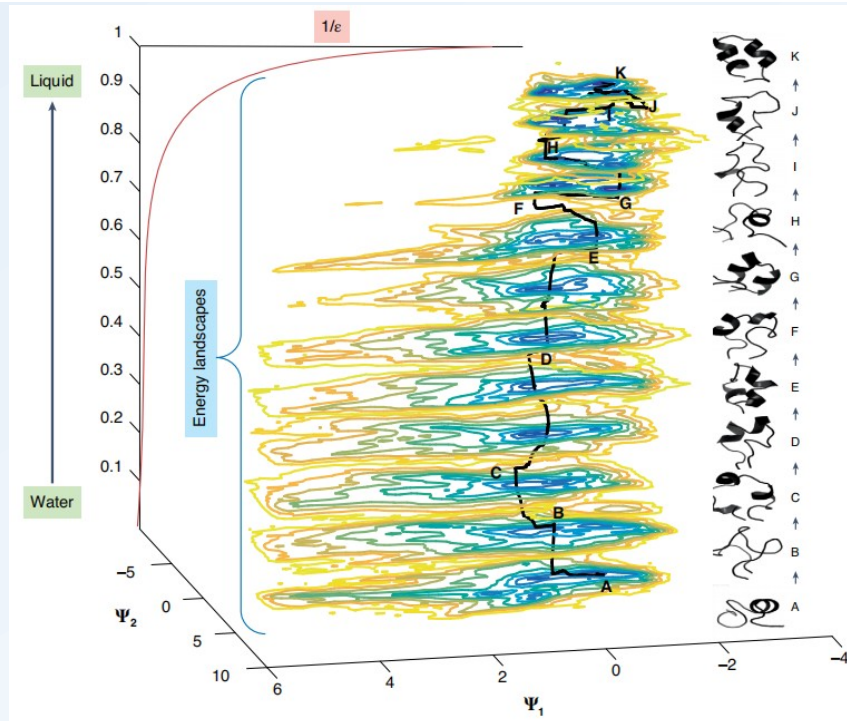


Thinking Is there any problem that AI cannot solve (well)?

Structural biology is solved — now what?

The splendid computational success of AlphaFold and RoseTTAFold in solving the 60-year-old problem of protein folding raises an obvious question: what new avenues should structural biology explore? We propose a strong pivot toward the goal of reading mechanism and function directly from the amino acid sequence. This ambitious goal will require new data analytical tools and an extensive database of the atomic-level structural trajectories traced out on energy landscapes as proteins perform their function.

Abbas Ourmazd, Keith Moffat and Eaton Edward Lattman



The next challenge involves identifying the conformational motions relevant to function. Here, the concept of an energy landscape is key, where each point corresponds to a structure of a particular energy.

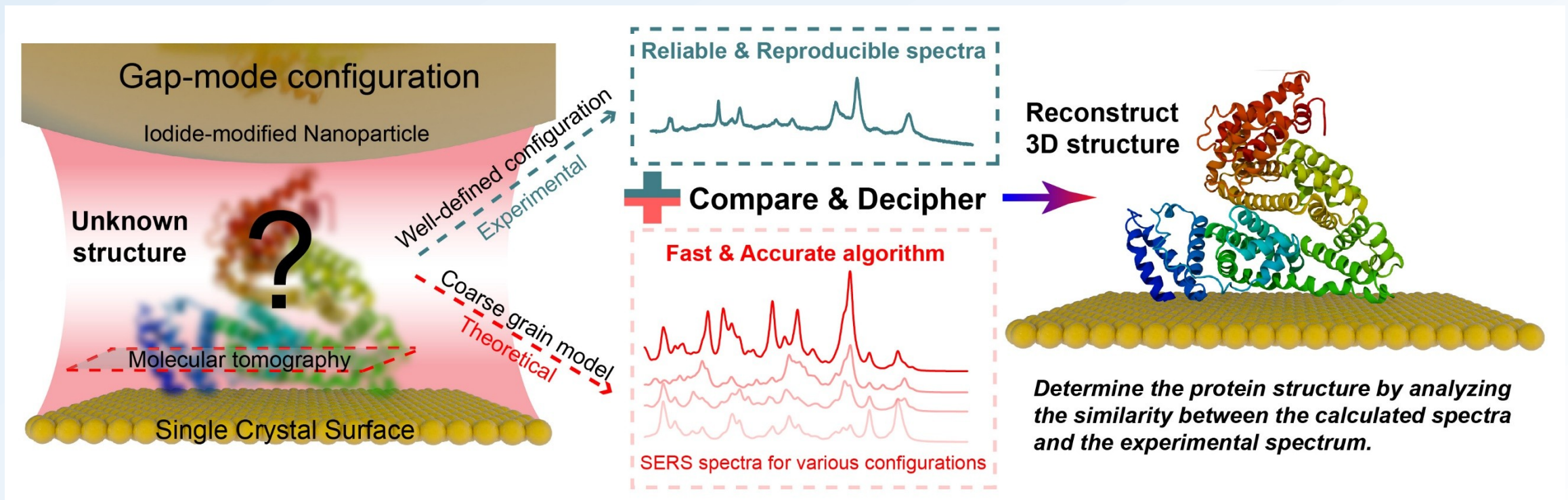
Nat Methods, 2022, 19, 24–26.

AI for biological analysis

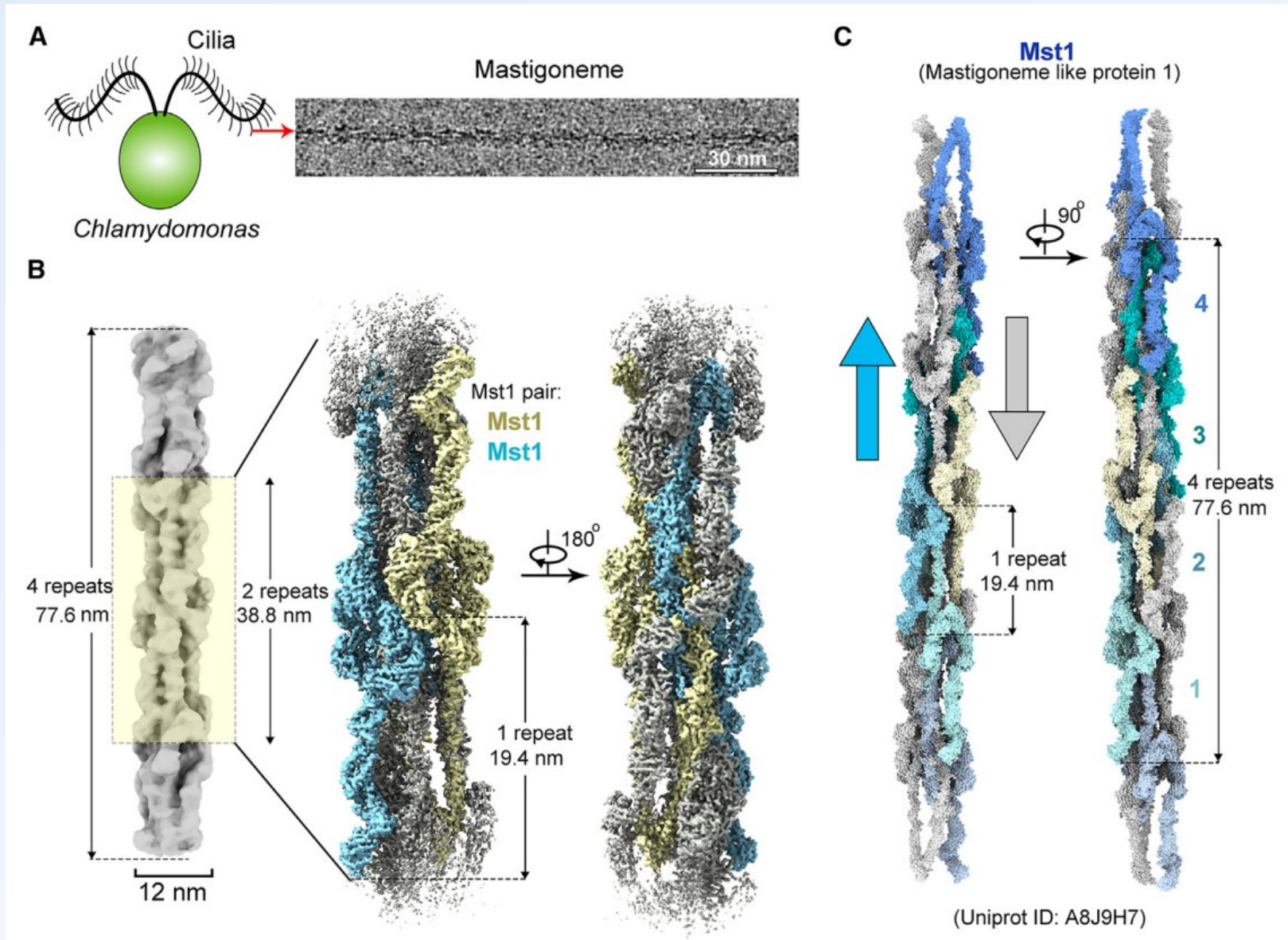


Raman & SERS:

$$S_{protein} \propto \sum_m S_{AA} = \sum_m \sum_n \frac{\omega_s^4 M_s^2 M_i^2 F_p}{c^4 (1 - e^{\hbar\omega_s/k_B T})} |E_s^L \alpha'_n(\Omega) E_i^L|^2$$



AI for biological analysis



Cell, 2024, 187, 1733–1744

Contents

1. Overview of biological analysis

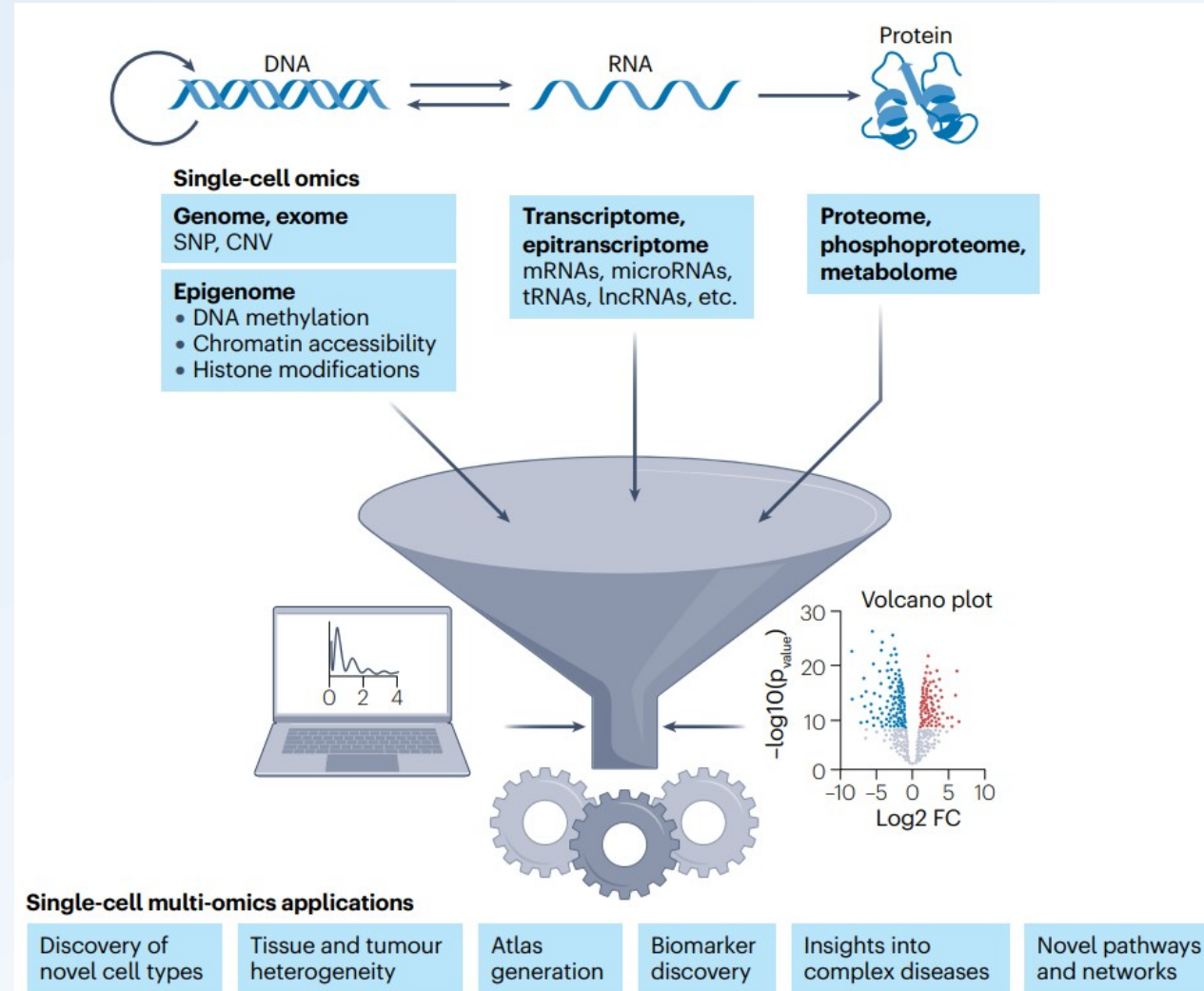
2. **AI for biological analysis**

- Nanopore
- Structure biology
- Bioinformatics
- **Single-cell**
-

AI for biological analysis



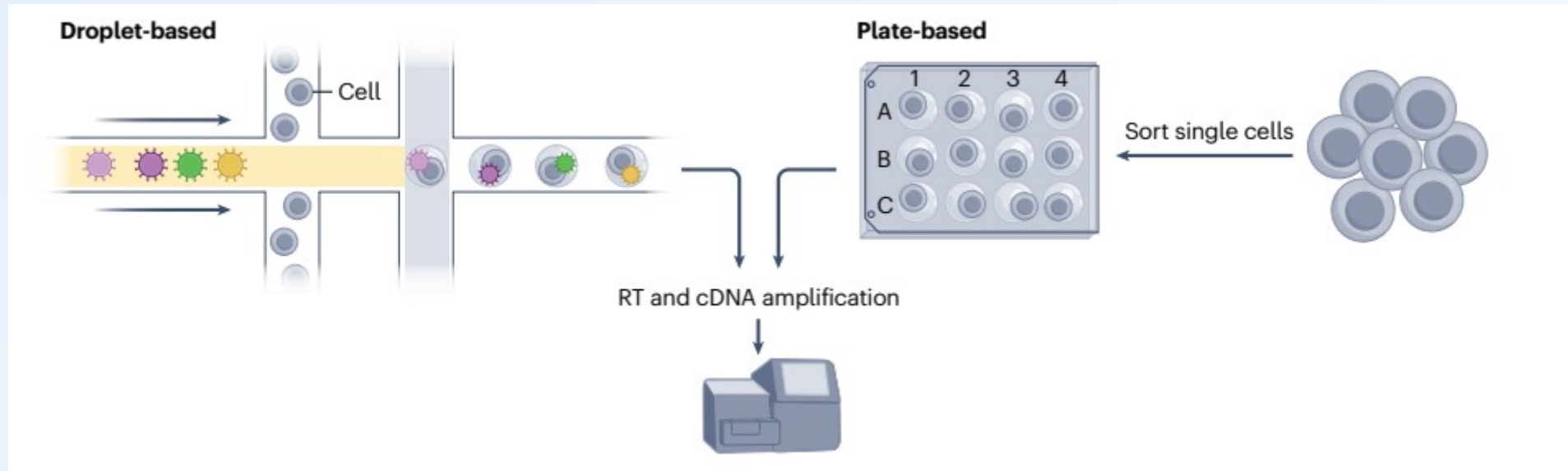
◆ Single-cell multi-omics



AI for biological analysis



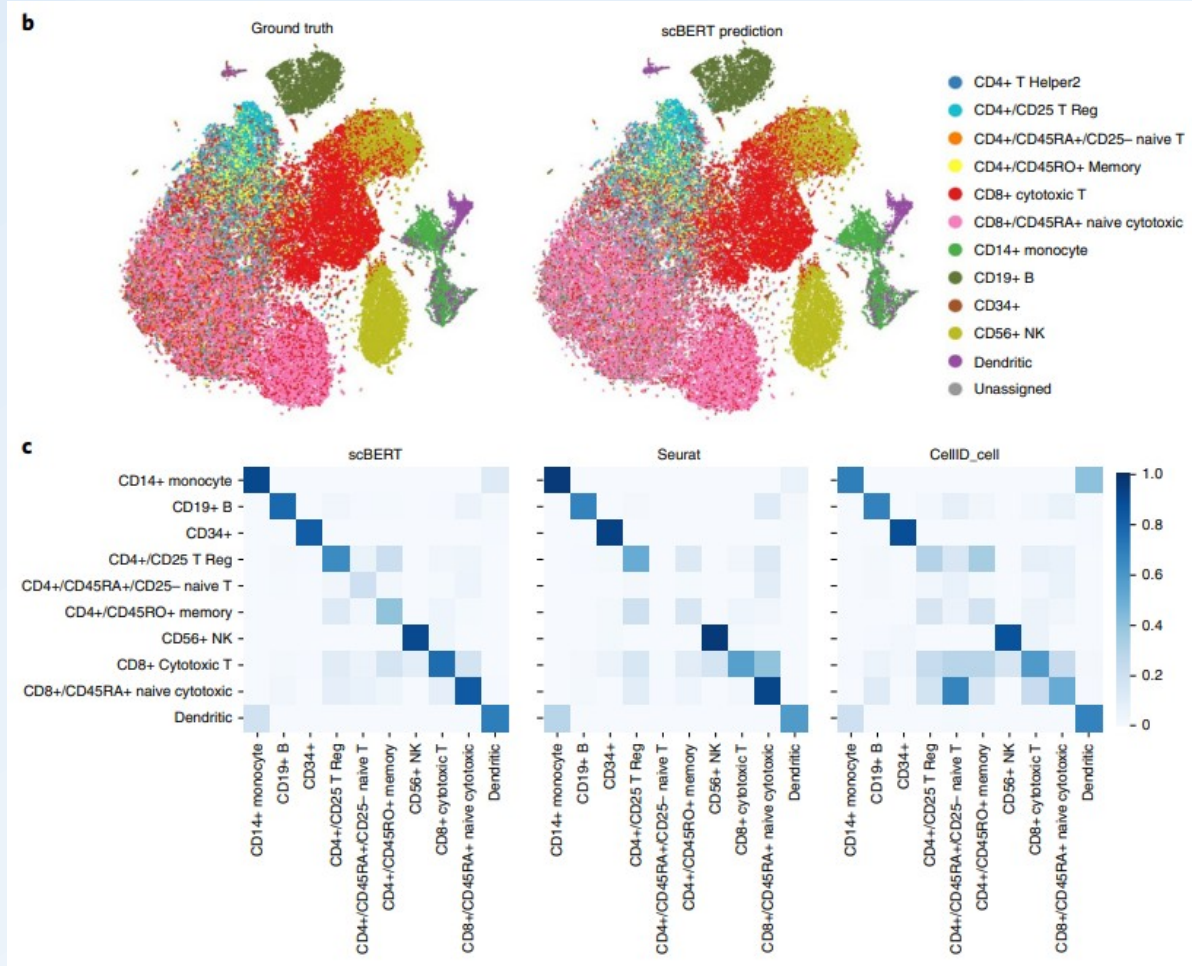
◆ Single-cell multi-omics



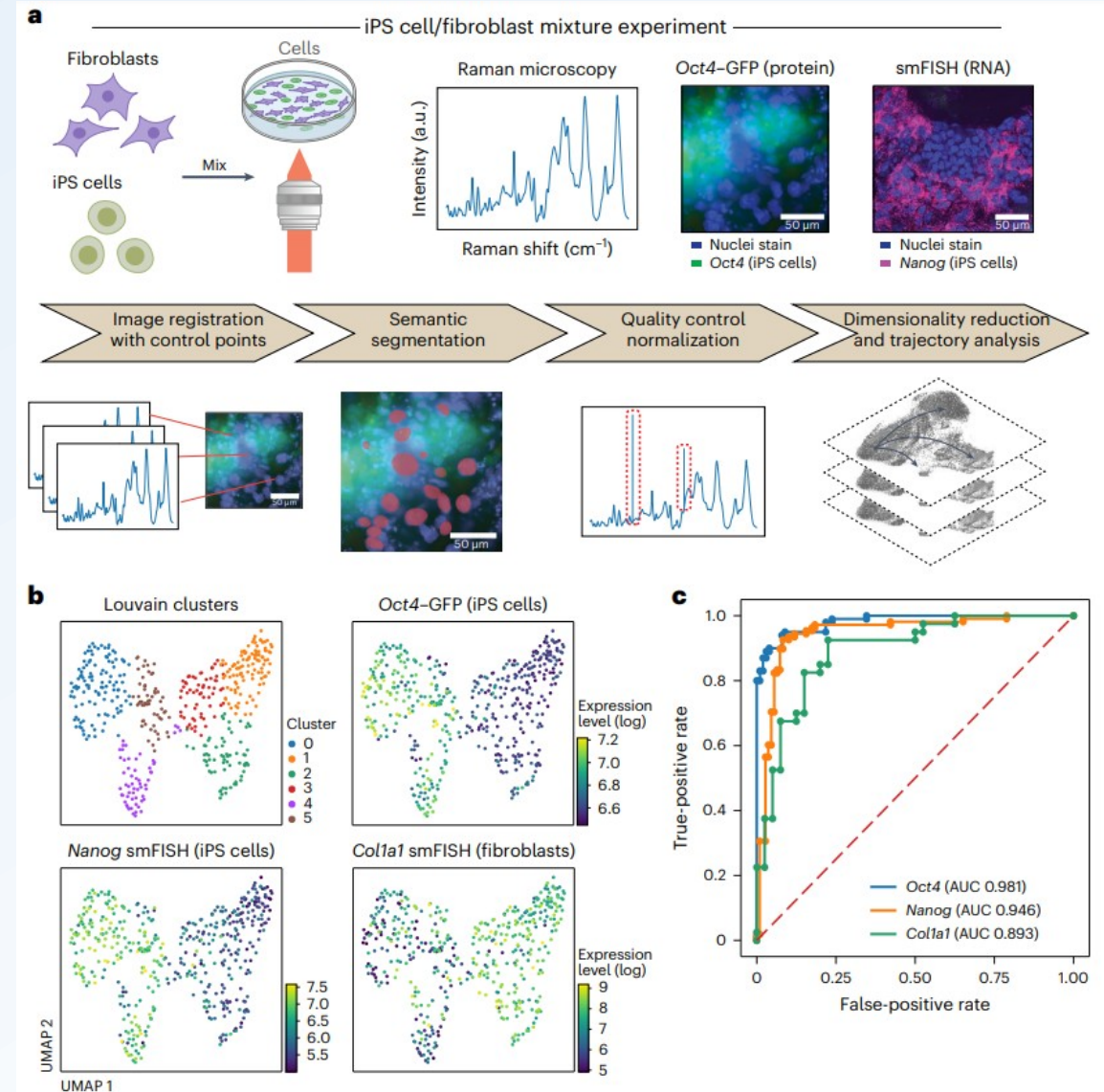
AI for biological analysis



◆ Single-cell multi-omics



Nat Mach Intell **4**, 852–866 (2022)



Nat Biotechnol (2024). <https://doi.org/10.1038/s41587-023-02082-2>